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The Carbon Dioxide Molecule

Potential, Spectroscopic, and Molecular Constants from Its Infrared Spectrum

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The potential energy function of the carbon dioxide molecule is determined from a very large set of vibrorotational data belonging to several isotopic species. Starting from this potential, theoretical values of the spectroscopic constants and the molecular constants are computed for 10 isotopic species of this molecule. These values are shown to agree with the observed values to within the limits of the mean experimental accuracy. The theoretical framework of this study (approximations and methodology), is briefly recalled. An extension is presented which results in diagonalizing infinite truncated matrices.

I. INTRODUCTION

The study of the internuclear potential has been and continues to be the purpose of a large number of studies because it gives information on the behavior of a molecule out of equilibrium. The potential energy of a polyatomic molecule is an intricate function of the displacements of the nuclei, and most other attempts have considered this function as a power series expansion with respect to coordinates describing the deformations of the molecule whose zeroth order term is harmonic. The principal reason is that the harmonicity of the zeroth order potential allows for an important simplification in the perturbation treatment of the nuclear Hamiltonian regarded from the point of view of the second quantization theory applied to bosons systems. This means that whenever possible such a "simplified" approach should be used. This is the case for a rather well-behaved molecule such as CO_2 and results were published in 1971 (*1*) which, for the first time, allowed for a correct fitting of the experimental data. At that time the adjustment of the coefficients appearing in the power series expansion of the potential up to the fourth order of approximation in the usual ordering scheme (*2*), was carried out by using only the vibrational state energies. Unfortunately, this important additional simplification resulted in several inconsistencies, and particularly in a certain lack of physical meaning due to a slower convergence of the higher orders than was presupposed by the theory.

Analyzing these results led us to the conclusion that the adjustment of the potential to the vibration–rotation energies (the spectrum itself) would overcome these difficulties. However, such an adjustment to experiment turned out to be several orders of magnitude more complicated than the adjustment to the pure vibrational state energies. With that purpose in mind, new methods were designed in order to

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systematize the algebraic approach developed in the past (*3–6*) to most of the transformations required for solving the eigenvalue problem.

In this paper, we first give the broad outlines of the resulting formalism hereafter referred to as the "algebraized contact transformation" (Section II). The results (potential, spectroscopic, and molecular constants for 10 isotopic species of CO_2) are given and compared in Section III to the presently available experimental results. As a conclusion to this section, we present a detailed study of one of the significant limitations to the assumption of the harmonicity of the zeroth order potential which occurs in the case of the carbon dioxide molecule. This extension of the algebraized contact transformation formalism consists in leaving, in the Hamiltonian, operators which are usually "destroyed" by the perturbation treatment. The infinite energy matrices which are obtained following that way are truncated and shown to converge satisfactorily for the eigenvalues of interest.

II. ALGEBRAIZED CONTACT TRANSFORMATION APPLIED TO THE ADJUSTMENT OF THE POTENTIAL OF CO_2 TO EXPERIMENT

A. Generalities on the Method

The carbon dioxide molecule (a linear molecule) is a relatively "rigid" molecule, which means that the displacements of the nuclei from the equilibrium configuration when the molecule vibrates or rotates are small compared to the internuclear distances. This physical fact is at the origin of the theoretical model chosen for representing the potential: a power series expansion in the nuclear displacements whose zeroth order term is harmonic. Such a representation may be expected to be quite good in the lower and medium spectrum and progressively worse in the upper spectrum where the leading role of the zeroth approximation may become questionable.

(a) The Darling and Dennison (*7*) Hamiltonian from which we start is expanded in orders of magnitude, up to the fourth order, following the usual ordering scheme (*2*). This expansion is put under the form of a sum of tensorial operators $O_i^{A_1}$ totally symmetric in the point group $C_{\infty v}$ (we have designed the method for a general linear triatomic molecule):

$$H = H_0 + H_1 + \cdots + H_4 = \sum_i O_i^{A_1}.$$

In that case, the index i runs from 1 to 471.

(b) Each of these operators is a linear combination of products of the elementary operators (normal coordinates, conjugate momenta, components of the total angular momentum) Hermitian not only with respect to the vibrational operators but also with respect to the components of the total angular momentum. Then each tensor is a sum of monomials, half of them being ordered in a definite way, half in the opposite way. There are about 1700 monomials in the complete expansion.

(c) The Hamiltonian is then subjected to two successive canonical transformations of type ($i = \sqrt{-1}$):

$$H' = T_1 H T_1^{-1}, \qquad T_1 = \exp\{iS_1\}$$

and

$$H'' = T_2 H' T_2^{-1}, \qquad T_2 = \exp\{iS_2\} \tag{1}$$

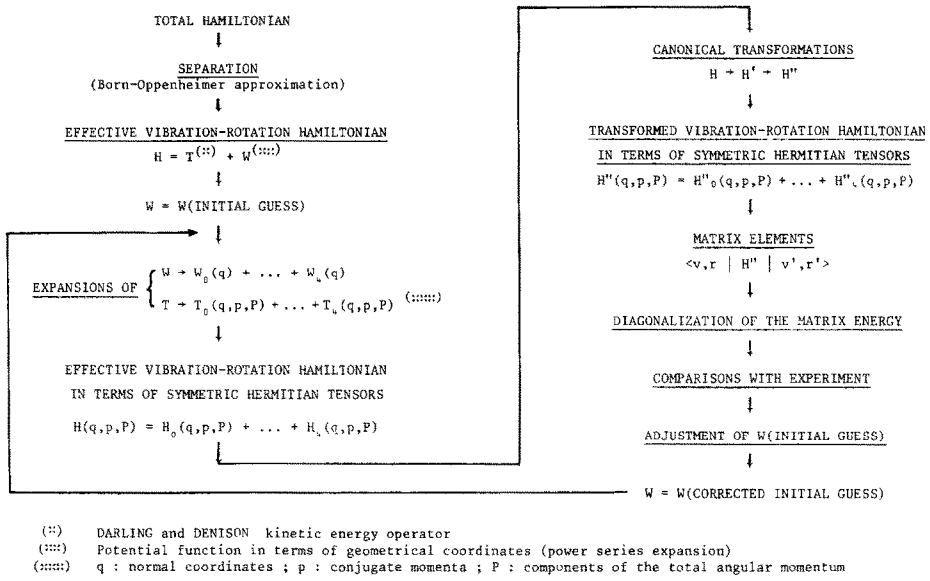


FIG. 1. The iterative process for the adjustment of the Born–Oppenheimer potential to experiment. (It is in fact divided into two parts: the algebraic part carried out once and for all, the numeric part).

where S_1 and S_2 are two Hermitian operators. It is well known that the aim of these transformations is to diagonalize, or to partly diagonalize (to make block-diagonal) the Hamiltonian step by step. This "decoupling process" is a particular case of the general effective Hamiltonian theory (*8–11*).

(d) The two generators S_1 and S_2 for the two steps of the Hamiltonian transformation are automatically determined and this work is made much easier owing to a systematic use of the creation and annihilation operators (*3, 4*):

$$a_{s\sigma} = (1/2^{1/2})(q_{s\sigma} + ip_{s\sigma}/\hbar)$$
$$a^{\dagger}_{s\sigma} = (1/2^{1/2})(q_{s\sigma} - ip_{s\sigma}/\hbar) \tag{2}$$

associated with the operators $q_{s\sigma}$ (normal coordinate) and $p_{s\sigma}$ (conjugate momentum). Introducing the operators $a_{s\sigma}$ and $a^{\dagger}_{s\sigma}$ leads to a greater flexibility in the choice of the couplings that one does not want to be removed by the perturbation treatment. Examples of this flexibility in decoupling subblocks from the "remainder" are given in Section III.F. It is one of the important characteristics of the algebraized contact transformation. In the same way, the use of creation and annihilation operators allows for a "visualization" of the matrix elements of the Hamiltonian. The algebraized contact transformation enters the frame of the second quantization theory.

(e) None of the formulas necessary to the computation of the Hamiltonian eigenvalues are derived by hand. These formulas are automatically generated and coded under algebraic form once and for all. This is based upon the definition of an algebra which may be either commutative or noncommutative (*5, 6*) and consists in determining its multiplication table.

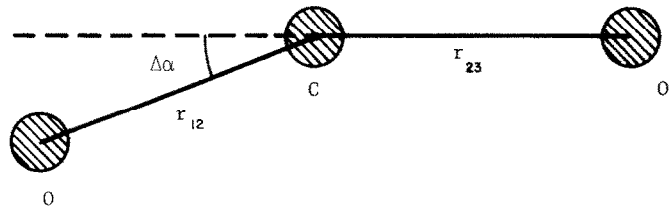


FIG. 2. The carbon dioxide molecule out of equilibrium.

(f) The algebraized contact transformation is an *inverse* method. Indeed, the preceding only concerns the direct computation of the eigenvalues of the Hamiltonian starting from a given solution for the potential. If one wants to adjust the original potential (initial guess) to experiment it is obvious that, in addition to the direct computation, the reverse should be implemented in the whole process. Indeed, after each transformation, the Jacobian of the system transformed quantities/untransformed quantities is derived for each step of the computation. The final Jacobian is obtained as the product of all partial Jacobians. This procedure makes it possible to "connect" the potential to the observed energy levels owing to an iterative cycle which goes through a damped least squares method (*12*). All these operations are summarized in Fig. 1. They are detailed in the next paragraphs.

B. Coordinates

Figure 2 shows the nuclei of a linear triatomic molecule out of equilibrium. Let us call r_{12} and r_{23} the two instantaneous bond lengths, r_{12}^0 and r_{23}^0 the two corresponding equilibrium lengths, and $\Delta\alpha$ the angle of the two vectors 1–2 and 2–3. The potential may be expanded with respect to the three quantities (often called "exact internal coordinates"):

$$r_{12}-r_{12}^0,\ r_{23}-r_{23}^0, \text{ and } \Delta\alpha \tag{3}$$

and, from the assumed symmetry properties of the potential, this expansion must be even with respect to $\Delta\alpha$. Moreover, for a symmetric linear molecule like CO_2 this expansion must be invariant while changing simultaneously $r_{12}-r_{12}^0$ to $r_{23}-r_{23}^0$ and $r_{23}-r_{23}^0$ to $r_{12}-r_{12}^0$.

In the following we shall make use, in addition to Eq. (3), of other sets of coordinates: the ξ's defined by:

$$\xi_1 = (r_{12}-r_{12}^0)/r_{12}^0, \qquad \xi_2 = \Delta\alpha, \qquad \xi_3 = (r_{23}-r_{23}^0)/r_{23}^0 \tag{4}$$

which are dimensionless; the ξ^*'s defined by:

$$\xi_1^* = (r_{12}-r_{12}^0)/r_{12}, \qquad \xi_2^* = \Delta\alpha, \qquad \xi_3^* = (r_{23}-r_{23}^0)/r_{23} \tag{5}$$

identical to the ξ's except that the instantaneous bond lengths appear in the denominators instead of the equilibrium bond lengths (*13*); the R's defined by:

$$R_1 = a_1(\xi_1 + \xi_3), \qquad R_2 = a_2\xi_1, \qquad R_3 = a_3(\xi_1 - \xi_3) \tag{6}$$

which are first symmetric (R_1 is of symmetry $\Sigma_{(g)}^+$ and R_3 is of symmetry $\Sigma_{(u)}^+$) and second, with a correct choice for the numbers a_1, a_2, and a_3, quasi-normal internal

coordinates (*3*). We have adopted the values $a_1 = 20.294$, $(a_2)^2 = 58.75$, and $a_3 = 14.248$. We shall also make use of the R^*'s defined by:

$$R_1^* = a_1(\xi_1^* + \xi_3^*), \qquad R_2^* = a_2\xi_2^*, \qquad R_3^* = a_3(\xi_1^* - \xi_3^*). \tag{7}$$

C. Strong Interactions in the Case of a Triatomic Linear Molecule; Application to CO_2

For the carbon dioxide molecule, two types of interactions cannot be removed by the successive canonical transformations on the Hamiltonian. First, the so-called Fermi interaction which couples states differing in their principal quantum numbers by:

$$\Delta v_1 = \pm 1; \qquad \Delta v_2 = \mp 2; \qquad \Delta v_3 = 0 \tag{8}$$

and, second, the so-called Coriolis interaction which couples states differing in their principal quantum numbers by:

$$\Delta v_1 = \mp 1, \qquad \Delta v_2 = \mp 1, \qquad \Delta v_3 = \pm 1, \tag{9}$$

the latter being nondiagonal in the secondary quantum number l:$\Delta l_2 = \pm 1$. In addition to these two types of interactions, one has to consider the l-type doubling which is diagonal in the principal quantum numbers v_1, v_2, and v_3 but nondiagonal in l_2:

$$\Delta v_1 = 0, \qquad \Delta v_2 = 0, \qquad \Delta v_3 = 0, \qquad \Delta l_2 = \pm 2. \tag{10}$$

The definition of the transformation operators must take into account the fact that the couplings (8) and (9) cannot be removed from the Hamiltonian by the perturbation treatment. The transformation operator S_1 takes account of the first order coupling (8), while S_2 takes account of the second order coupling (9). As a consequence, the matrix energy associated with the transformed Hamiltonian will not be diagonal and will precisely include the two couplings (8) and (9) in addition to the coupling (10) which cannot be removed from the Hamiltonian either. In addition to these couplings, two higher order interactions are included in this matrix: a Fermi-type interaction of the fourth order which couples states differing in their principal quantum numbers by:

$$\Delta v_1 = \pm 2, \qquad \Delta v_2 = \mp 4, \qquad \Delta v_3 = 0 \tag{11}$$

and $\Delta l_2 = 0$, and a mixing of the Fermi interaction and of the l-type interaction:

$$\Delta v_1 = \pm 1, \qquad \Delta v_2 = \mp 2, \qquad \Delta v_3 = 0, \qquad \Delta l_2 = \pm 2 \tag{12}$$

which is of the third order. Actually the method is more general and allows for the treatment of a general linear triatomic molecule like N_2O, OCS, etc. Thirteen[1] types of matrix elements may be taken into account and included in the energy matrix; they are given in Table I. In this table, the first four columns give the differences in the quantum numbers corresponding to each matrix element, an identification number being given in the fifth column. The sixth column gives the square root part of the matrix element $\langle v_1 v_2 v_3 l_2 | H'' | v_1 + \Delta v_1 v_2 + \Delta v_2 v_3 + \Delta v_3 l_2 + \Delta l_2 \rangle$:

[1] Additional interactions may be easily included when necessary.

no square root part if the column is empty. The last column indicates YES or NO, respectively, for the presence or the absence of the matrix element in the energy matrix when the case of CO_2 is considered. From this table it is seen that only eight interactions are considered in the latter case. The polynomial part of the matrix element associated with a given interaction is computed by summing the contributions of all the tensors in the twice transformed Hamiltonian. This sum appears as a polynomial in the quantum numbers, the numerical coefficients of which are usually called the spectroscopic constants.

D. Vibrorotational Spectroscopic Constants of CO_2

Eight types of matrix elements are considered in the case of CO_2 and the polynomials thus obtained have the following form:

Matrix element number 1 (diagonal):

$$\begin{aligned}P_1(v_1,v_2,v_3,l_2,J) &= E_{00^00} + \omega_1^0 v_1 + \omega_2^0 v_2 + \omega_3^0 v_3 + x_{11}^0 v_1^2 + x_{12}^0 v_1 v_2 + x_{13}^0 v_1 v_3 + x_{22}^0 v_2^2 \\ &+ x_{23}^0 v_2 v_3 + x_{33}^0 v_3^2 + x_{ll} l_2^2 + y_{111} v_1^3 + y_{112} v_1^2 v_2 + y_{113} v_1^2 v_3 + y_{122} v_1 v_2^2 + y_{123} v_1 v_2 v_3 \\ &+ y_{133} v_1 v_3^2 + y_{222} v_2^3 + y_{223} v_2^2 v_3 + y_{233} v_2 v_3^2 + y_{333} v_3^3 + y_{1ll} v_1 l_2^2 + y_{2ll} v_2 l_2^2 \\ &+ y_{3ll} v_3 l_2^2 + (B_0 - \alpha_1^0 v_1 - \alpha_2^0 v_2 - \alpha_3^0 v_3 + \gamma_{11} v_1^2 + \gamma_{12} v_1 v_2 + \gamma_{13} v_1 v_3 + \gamma_{22} v_2^2 \\ &+ \gamma_{23} v_2 v_3 + \gamma_{33} v_3^2 + \gamma_{ll} l_2^2)[J(J+1) - l_2^2] \\ &- (D_0 + \beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3)[J(J+1) - l_2^2]^2 + H[J(J+1) - l_2^2]^3 \\ &+ [a_l + a_{1ll} v_1 + a_{3ll} v_3 + a_{Jll}[J(J+1) - l_2^2]] l_2 K \quad (13)\end{aligned}$$

where[2] most of the symbols have their usual meanings. In this formula, E_{00^00} is the energy of the fundamental level with respect to the minimum of the potential. The last factor, $l_2 K$, in the last term should in fact be written as l_2^2 since for CO_2 we have $l_2 = K$. We have maintained the quantum number K explicitly because this term comes from those tensors in the transformed Hamiltonian where P_z appears. These (relatively small) contributions are in fact contributions to $x_{l_2 l_2}$, y_{1ll}, y_{3ll}, and γ_{ll}.

Matrix element number 2 (Fermi interaction):

$$P_2(v_1,v_2,v_3,l_2,J) = F_0 + F_1 v_1 + F_2 v_2 + F_3 v_3 + F_J[J(J+1) - l_2^2] \quad (14)$$

where the coefficients F_0, F_1, F_2, F_3, and F_J (F for Fermi) are related to the usual, but sometimes very confusing, W_{122}, λ_1, λ_2, λ_3, and δ.

Matrix element number 5 (l-type doubling):

$$P_5(v_1,v_2,v_3,l_2,J) = L_0 + L_1 v_1 + L_2 v_2 + L_3 v_3 + L_J[J(J+1) - l(l+2)], \quad (15)$$

where the coefficients L_0, L_1, L_2, L_3, and L_J (L for l-type) are related to the usual, but also very confusing, q_0, f_1, f_2, f_3, and f_J.

Matrix element number 6 (Coriolis interaction):

$$P_6(v_1,v_2,v_3,l_2,J) = C_0 + C_1 v_1 + C_2 v_2 + C_3 v_3 + C_l l_2 + C_J[J(J+1) - l^2]. \quad (16)$$

[2] We have followed the usual definitions for the α's, for D_0, and for the β's always preceded by a minus sign.

TABLE I

Matrix Elements which May Be Taken into Account in Constructing the Energy Matrix Associated with the Transformed Hamiltonian of a Linear Triatomic Molecule

Δv_1	Δv_2	Δv_3	$\Delta \ell_2$	Nb.	Square root part of $\langle v_1 v_2 \ell_2 v_3 \lvert H'' \rvert v_1+\Delta v_1\ v_2+\Delta v_2\ \ell_2+\Delta\ell_2\ v_3+\Delta v_3 \rangle$ [a]	CO_2
0	0	0	0	1		YES
-1	2	0	0	2	$v_1(v_2+\ell_2+2)(v_2-\ell_2+2)$	YES
-2	0	1	0	3	$v_1(v_1-1)(v_3+1)$	NO
-1	-2	1	0	4	$v_1(v_2-\ell_2)(v_2+\ell_2)(v_3+1)$	NO
0	0	0	2	5	$(v_2+\ell_2+2)(v_2-\ell_2)[f-K(K+1)][f-(K+1)(K+2)]$	YES
-1	-1	1	1	6	$v_1(v_2-\ell_2)(v_3+1)[f-K(K+1)]$	YES
-1	-1	1	-1	7	$v_1(v_2+\ell_2)(v_3+1)[f-K(K-1)]$	YES
-1	2	0	2	8	$v_1(v_2+\ell_2+2)(v_2+\ell_2+4)[f-K(K+1)][f-(K+1)(K+2)]$	YES
-1	2	0	-2	9	$v_1(v_2-\ell_2+2)(v_2-\ell_2+4)[f-K(K-1)][f-(K-1)(K-2)]$	YES
0	-4	1	0	10	$(v_2-\ell_2-2)(v_2-\ell_2)(v_2+\ell_2-2)(v_2+\ell_2)(v_3+1)$	NO
-2	4	0	0	11	$v_1(v_1-1)(v_2+\ell_2+2)(v_2+\ell_2+4)(v_2-\ell_2+2)(v_2-\ell_2+4)$	NO
-4	0	2	0	12	$v_1(v_1-1)(v_1-2)(v_1-3)(v_3+1)(v_3+2)$	YES
-3	2	1	0	13	$v_1(v_1-1)(v_1-2)(v_2+\ell_2+2)(v_2-\ell_2+2)(v_3+1)$	NO

[a] Square root part of the matrix element ; f stands for $J(J+1)$ and $K=\ell_2$.

No previous symbols have been clearly defined for these coefficients except perhaps for the first one (C_0) related to a symbol sometimes written as Z. The polynomial for the coupling number 7 is simply obtained by changing the sign of C_l.

Matrix element number 8 (Fermi + l-type):

$$P_8(v_1,v_2,v_3,l_2,J) = F^L. \tag{17}$$

Here the polynomial reduces to a constant. It is exactly the same for the coupling number 9.

Matrix element number 12 (Fermi-type interaction):

$$P_{12}(v_1,v_2,v_3,l_2,J) = F^{IV} \tag{18}$$

which is also a constant.

Altogether there are 62 spectroscopic constants for each isotopic species and 58 provide independent algebraic contributions to the energy [the four a's of Eq. (13) are contributions to other constants]. This number of constants is to be compared to the number of potential constants: 28 in the case of CO_2. The difference, which is 30, gives the number of relationships between the spectroscopic constants. Moreover, the same potential makes it possible to compute with fairly good accuracy (see the next section) all the isotopic species of the molecule at the same time. This means that the potential "contains" as many times 58 spectroscopic constants as there are isotopic species investigated.

E. From the Spectroscopic Constants to the Energy Levels

A program has been written for computing the energy levels from the spectroscopic constants. This program first constructs the energy matrix, diagonalizes it, and allows for the computation of either the energies of the vibrorotational levels or the molecular constants associated with each vibrational state (the band center ν_0, the inertia constant B, the centrifugal distortion constant D, and the fourth order constant H) or the energies of the lines of any given transition. This program was written by Cihla (*14*).

The energy matrix is first constructed on the basis of the zeroth order wavefunctions: $\psi^0(v_1,v_2,v_3,l_2,J,M)$ of the zeroth order Hamiltonian and then symmetrized owing to the transformation

$$\psi_{|l|}^{\text{sym}} = (1/2^{1/2})\{\psi^0(v_1, v_2, v_3, +l_2, J, M) + \psi^0(v_1, v_2, v_3, -l_2, J, M)\},$$

$$\psi_{|l|}^{\text{anti}} = (1/2^{1/2})\{\psi^0(v_1, v_2, v_3, +l_2, J, M) - \psi^0(v_1, v_2, v_3, -l_2, J, M)\}. \quad (19)$$

The effect of this transformation is to split each subblock of the original matrix into two blocks, the first one allowing for the computation of the energies of the so-called "c-levels" (including the levels with a zero l_2 value) and the second one allowing for the computation of the energies of the so-called "d-levels".[3]

From the energies of the levels, one can compute the energies corresponding to the lines of any transition. This procedure, however, requires the diagonalization of the energy matrix as many times as there are J values considered. This is, in general, hardly possible if one takes into account the sizes of the subblocks which constitute the energy matrix. Another procedure, much less time consuming, consists in diagonalizing the energy matrix for a few values of J and deducing the molecular constants associated with each vibrational state following a simple least-squares method. For instance, if the energy matrix has been diagonalized for the J values $J_1,J_2, \ldots ,J_p$, we may, in most cases, represent the eigenvalues for the p values of J by:

$$E_{J_1} = \nu_0 + BJ_1(J_1 + 1) - DJ_1^2(J_1 + 1)^2 + HJ_1^3(J_1 + 1)^3,$$

$$E_{J_p} = \nu_0 + BJ_p(J_p + 1) - DJ_p^2(J_p + 1)^2 + HJ_p^3(J_p + 1)^3. \quad (20)$$

If $p = 4$, this linear system is exactly determined. It is overdetermined if $p > 4$. Its resolution leads to the values of ν_0, B, D, and H which are the molecular constants associated with the vibrational state considered. It is well known that the term ν_0 is not the vibrational energy value since it includes rotational contributions like $-Bl_2^2$. The constants B, D, and H are the effective values of, respectively, the inertia constant, the centrifugal distortion constant, and the fourth order diagonal rotational contribution to the energy. In certain cases and especially when two vibrational states coupled by the Coriolis interaction are sufficiently close to each other, the two corresponding rotational series of levels are distorted and the ex-

[3] The "c-levels," in the notations of the interesting work by Bunker and Papousek (*15*), are the Σ^+ levels for J even and the Σ^- levels for J odd. It is the reverse for the "d-levels." In addition, they are (u) or (g) according to the symmetry of the vibronic state. The c and d labels, respectively, correspond to the e and f labels of Ref. (*16*).

TABLE II

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to the Quasi-Normal Internal Coordinates of Eq. (6) (in cm^{-1}) in Two Cases

F [a]	W(R) FROM PURE VIBRATION	W(R) FROM VIBRATION-ROTATION
F_{11}	0.7098943E+03	0.7101243E+03
F_{22}	0.3355588E+03	0.3354951E+03
F_{33}	0.1230881E+04	0.1231576E+04
F_{111}	-0.4643927E+02	-0.4890794E+02
F_{122}	-0.2922401E+02	-0.2925546E+02
F_{133}	-0.2662095E+03	-0.2698750E+03
F_{1111}	0.1749718E+01	0.2010455E+01
F_{1122}	0.9732242E+00	0.1326540E+01
F_{1133}	0.2106288E+02	0.2322218E+02
F_{2222}	0.8123874E+00	0.8234544E+00
F_{2233}	-0.1383630E+01	-0.1307093E+01
F_{3333}	0.7069187E+01	0.7506295E+01
F_{11111}	-0.3100623E+01	-0.7573783E-01
F_{11122}	-0.6713903E-01	-0.3753167E-01
F_{11133}	0.5931097E+01	-0.5491342E+00
F_{12222}	-0.8896714E-01	-0.1081372E+00
F_{12233}	0.1294839E+00	-0.3072761E-01
F_{13333}	-0.3622097E+01	-0.1013888E+01
F_{111111}	0.6938491E+00	0.1584061E-01
F_{111122}	0.4428989E+00	-0.6867379E-01
F_{111133}	0.9442407E+00	-0.1466336E+00
F_{112222}	0.3168358E-02	-0.4834831E-02
F_{112233}	-0.5674895E+00	-0.3709000E-01
F_{113333}	-0.1547290E+01	-0.2496887E+00
F_{222222}	0.7378560E-02	0.6894737E-02
F_{222233}	-0.2053079E-01	-0.3964920E-02
F_{223333}	0.8552808E-01	0.1161854E-02
F_{333333}	0.2828982E+00	-0.1724850E-01

[a] For example F_{12233} is the coefficient of $R_1R_2^2R_3^2$.

pansions (20) are no longer valid. The values of ν_0, B, D, and H that one would obtain by the above method would be meaningless. In such cases it is necessary to diagonalize the energy matrix for all the values of J.

III. PRESENTATION OF THE RESULTS

We report in this section the results that we have obtained by iterating the potential energy function of CO_2 from the molecular constants of a large set of vibrational

TABLE III

Convergence of the Potential Function Expressed in Terms of Quasi-normal Internal Coordinates; Comparison between the Mean Absolute Values of the Coefficients Appearing in a Given Order of Magnitude in Two Cases

a	Potential from pure vibration	Potential from vibration rotation
$\overline{\lvert V_0 \rvert}$	759	760
$\overline{\lvert V_1 \rvert}$	114	116
$\overline{\lvert V_2 \rvert}$	6	6
$\overline{\lvert V_3 \rvert}$	2.2	.3
$\overline{\lvert V_4 \rvert}$	.5	.05

in cm^{-1}

[a] For example, $\overline{|V_2|}$ denotes the mean absolute value of the coefficients appearing in the second order of magnitude part of the potential expansion with respect to the quasi normal internal coordinates of Eq. (6).

states belonging to several isotopic species. The constants ν_0, B, D, and H are computed from the linear system of Eq. (20) where $p = 4$. With this value of p, due only to consideration of computation time, the system (20) is exactly determined and gives accurate values for ν_0, B, and D, but not for H. This may be simply explained: the four J values we use, which are not necessarily the same for different subblocks, range from 10 to 60. For $J = 60$, the contribution of H (mean order of magnitude: 1.0×10^{-13} cm^{-1}) is about 0.005 cm^{-1} and is too small to allow for a good determination of this constant. However, when for a given vibrational state there is an experimental value for H, we may extend the range for the values of J and increase the value of p to go through an overdetermined linear system. We could have iterated the potential directly from the observed transition energies. The expected difference in the determination of the potential is probably small; we should say very small compared to the considerable changes which have been observed between former purely vibrational approaches (*1*, *17*, *18*) and the present vibrorotational approach to the potential. The real change is to adjust the potential either to pure vibrational experimental data or to vibrorotational data.

The results we present here have been obtained at the end of the iterative process (end of convergence) and concern the potential itself, the spectroscopic constants and the molecular constants for a large set of vibrational states belonging to 10 isotopic species. We also give some indications about the results of a number of comparisons between computed and experimental transitions.

A. The Potential Energy Function

In Table II are given the 28 coefficients which appear in the expansion of the potential energy function with respect to the mass-independent quasi-normal

TABLE IV

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to Dimensionless Normal Coordinates for Various Symmetric Species of CO_2 (in cm^{-1})

Ident. [a]	$^{12}C^{16}O_2$	$^{13}C^{16}O_2$	$^{12}C^{18}O_2$	$^{13}C^{18}O_2$	$^{14}C^{16}O_2$
11	+0.676899873E+03	+0.676899872E+03	+0.638091265E+03	+0.638091261E+03	+0.676899872E+03
22	+0.336446287E+03	+0.326870662E+03	+0.331296957E+03	+0.321568033E+03	+0.318464851E+03
33	+0.119826611E+04	+0.116416216E+04	+0.117992655E+04	+0.114527665E+04	+0.113422455E+04
111	-0.455153434E+02	-0.455153434E+02	-0.416583564E+02	-0.416583564E+02	-0.455150434E+02
122	+0.732544227E+02	+0.731126306E+02	+0.719475691E+02	+0.698347444E+02	+0.712324357E+02
133	-0.256357852E+03	-0.249061630E+03	-0.245092894E+03	-0.237895454E+03	-0.242656757E+03
1111	+0.182666773E+01	+0.182666773E+01	+0.162326384E+01	+0.162326385E+01	+0.182666773E+01
1122	-0.117704177E+02	-0.114354189E+02	-0.109259287E+02	-0.106050760E+02	-0.111413455E+02
1133	+0.215366834E+02	+0.209237261E+02	+0.199914967E+02	+0.194044230E+02	+0.203856513E+02
2222	+0.245594300E+01	+0.259705394E+01	+0.257541186E+01	+0.242637284E+01	+0.237976904E+01
2233	-0.233222204E+02	-0.267330304E+02	-0.274619082E+02	-0.258726983E+02	-0.253757467E+02
3333	+0.710574164E+01	+0.679702334E+01	+0.688989854E+01	+0.649118029E+01	+0.636650295E+01
11111	-0.671850000E-01	-0.671849000E-01	-0.579674000E-01	-0.579674000E-01	-0.671850000E-01
11122	+0.844467000E+00	+0.823423000E+00	+0.761982000E+00	+0.738732000E+00	+0.799334000E+00
11133	-0.497217000E+00	-0.483063000E+00	-0.448120000E+00	-0.434961000E+00	-0.470643000E+00
12222	-0.106256000E+01	-0.100393000E+01	-0.100041000E+01	-0.942521000E+00	-0.952109000E+00
12233	+0.573744000E+01	+0.541559000E+01	+0.540137000E+01	+0.508880000E+01	+0.514056000E+01
13333	-0.937056000E+00	-0.884476000E+00	-0.882168000E+00	-0.831117000E+00	-0.839570000E+00
111111	+0.137190000E-01	+0.137190000E-01	+0.114925000E-01	+0.114926000E-01	+0.137190000E-01
111122	-0.113598000E+00	-0.113744000E+00	-0.967748000E-01	-0.933349000E-01	-0.104695000E+00
111133	-0.129625000E+00	-0.125436000E+00	-0.113428000E+00	-0.110098000E+00	-0.122698000E+00
112222	+0.128344000E+00	+0.121143000E+00	+0.117312300E+00	+0.110523000E+00	+0.114992000E+00
112233	-0.436670000E+00	-0.412173000E+00	-0.399142000E+00	-0.376043000E+00	-0.391246000E+00
113333	-0.224530300E+00	-0.212660300E+00	-0.205937000E+00	-0.194019000E+00	-0.201863000E+00
222224	-0.272593000E-01	-0.249971300E-01	-0.260264000E-01	-0.239003000E-01	-0.231179000E-01
222233	+0.314740000E+00	+0.288623000E+00	+0.300508000E+00	+0.274804000E+00	+0.266925000E+00
223333	-0.177643000E+00	-0.162932700E+00	-0.169610000E+00	-0.155102000E+00	-0.150655000E+00
333333	-0.138864000E-01	-0.145682300E-01	-0.151681000E-01	-0.139707000E-01	-0.134729000E-01

[a] Identification of the constants: 1 corresponds to q_1; 22 to $(q_{21}^2 + q_{22}^2)$; 3 to q_3. For example, "133" stands for k_{133} with the usual notations.

internal coordinates of Eq. (6) in two cases: the result of the last (unpublished) pure vibrational approach (*18*) and the result of the present vibrorotational approach. The majority of the differences and the most important appear in the third and fourth orders, which is not surprising. Moreover, it is immediately seen that the convergence of the "new potential" (in the following we shall use this expression to designate the potential adjusted to vibrorotational data, the "old potential" being the one which was adjusted to pure vibrational data) is much faster than that of the "old potential." An illustration of this fact is given in Table III. One may simply conclude that both vibration (by the x's) and rotation (by the α's) are necessary to fix the potential up to the second order and again both vibration (by the y's) and rotation (by the γ's) are necessary to fix the potential up to the fourth order. To simplify we have not mentioned the role of strong interactions, which is also important. However, they do not participate directly in the preceding arguments.

In Tables IV and V we give, for 10 isotopic species, the coefficients of the expansion of the potential with respect to the dimensionless normal coordinates $q_{s\sigma}$ ($s\sigma$ = 1, 21, 22, 3). In Table VI we give the coefficients of the expansion of the potential

TABLE V

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to Dimensionless Normal Coordinates for Various Nonsymmetric Species of CO_2 (in cm^{-1})

Ident.[a]	$16_0 12_C 18_0$	$16_0 12_C 17_0$	$16_0 13_C 18_0$	$16_0 13_C 17_0$	$17_0 12_C 18_0$
11	+0.637345882E+03	+0.666698202E+03	+0.657307389E+02	+0.666687466E+03	0.647911288D+03
22	+0.133881534E+03	+0.335088372E+03	+0.324230188E+02	+0.325472490E+03	0.332512858D+03
33	+0.118336995E+04	+0.119349568E+04	+0.115502516E+04	+0.115925758E+04	0.118430965D+04
111	-0.436003140E+02	-0.445031687E+02	-0.436043539E+02	-0.445043765E+02	-0.425735852D+02
113	+0.472678014E+01	+0.255592809E+01	+0.507247371E+01	+0.274440465E+01	0.217959495D+01
122	+0.735273426E+02	+0.743690259E+02	+0.713882047E+02	+0.722271705E+02	0.727167879D+02
133	-0.253845430E+03	-0.253302049E+03	-0.243090299E+03	-0.246014325E+03	-0.247693957D+03
223	+0.652662804E+01	+0.346832023E+01	+0.668869163E+01	+0.355529009E+01	0.305963508D+01
333	-0.899631615E+01	-0.432925039E+01	-0.843919016E+01	-0.451538744E+01	-0.377234522D+01
1111	+0.172733371E+01	+0.177342147E+01	+0.172799363E+01	+0.177361630E+01	0.167149976D+01
1113	-0.517907079E+00	-0.292087388E+00	-0.555741062E+00	-0.302882178E+00	-0.237033586D+00
1122	-0.113291175E+02	-0.115423084E+02	-0.109938928E+02	-0.112102905E+02	-0.111213663D+02
1133	+0.207119465E+02	+0.211142513E+02	+0.201040636E+02	+0.205057190E+02	0.203454269D+02
1223	-0.361929420E+00	-0.458769862E+00	-0.967154915E+00	-0.461443814E+00	-0.403962081D+00
1333	+0.899924254E+00	+0.452668155E+00	+0.875286956E+00	+0.472099100E+00	0.388749762D+00
2222	+0.262124575E+01	+0.263622624E+01	+0.247253969E+01	+0.248727926E+01	0.259557141D+01
2233	-0.279362030E+02	-0.281064351E+02	-0.263506111E+02	-0.265182379E+02	-0.276735929D+02
3333	+0.701420056E+01	+0.705314496E+01	+0.661734625E+01	+0.665496167E+01	0.694414208D+01
11111	-0.625004000E-01	-0.647022000E-01	-0.625023128E-01	-0.647029544E-01	-0.600969685D-01
11113	+0.677732000E-02	+0.371454000E-02	+0.727373000E-02	+0.399857525E-02	0.307533522D-02
11122	+0.801211000E+00	+0.821911300E+00	+0.777891000E+00	+0.798280733E+00	0.780319954D+00
11133	-0.472341000E+00	-0.484117000E+00	-0.458714000E+00	-0.470221552E+00	-0.459585276D+00
11223	-0.178888000E-02	-0.264197000E-02	-0.719379000E-02	-0.454194794E-02	-0.120222707D-02
11333	+0.191510000E-01	+0.134012300E-01	+0.199532870E-01	+0.108471154E-01	0.880223496D-02
12222	-0.133969000E+01	-0.134595000E+01	-0.971796000E+00	-0.986737280E+00	-0.101489856D+01
12233	+0.556500000E+01	+0.564729000E+01	+0.524702000E+01	+0.532757424E+01	0.547960521D+01
13333	-0.907837000E+00	-0.922031000E+00	-0.855800351E+00	-0.869784270E+00	-0.894714669D+00
22223	-0.144591000E+00	-0.769946000E-01	-0.143386000E+00	-0.763750994E-01	-0.675715714D-01
22333	+0.262614000E+00	+0.140461000E+00	+0.263624000E+00	+0.141089352E+00	0.122202661D+00
33333	-0.293681000E-01	-0.157636000E-01	-0.297266000E-01	-0.159664826E-01	-0.136298457D-01
[b]					
111111	+0.125042000E-01	+0.130907000E-01	+0.124908000E-01	+0.130868316E-01	0.119853149D-01
111122	-0.113435000E+00	-0.106837000E+00	-0.100423000E+00	-0.103765009E+00	-0.999406960D-01
111133	-0.120787000E+00	-0.125089000E+00	-0.117185000E+00	-0.121467989E+00	-0.117046656D+00
112222	+0.122512000E+00	+0.125326000E+00	+0.115485000E+00	+0.118223000E+00	0.119832548D+00
112233	-0.417073000E+00	-0.426475000E+00	-0.393199000E+00	-0.402315000E+00	-0.407768040D+00
113313	-0.215027000E+00	-0.219993000E+00	-0.202682000E+00	-0.207521000E+00	-0.210353286D+00
222222	-0.267676000E-01	-0.269659000E-01	-0.245273000E-01	-0.247145000E-01	-0.263422697D-01
222233	+0.308888000E+00	+0.311307000E+00	+0.283038000E+00	+0.285317000E+00	0.304113593D+00
223333	-0.174434000E+00	-0.175733000E+00	-0.159863000E+00	-0.161070000E+00	-0.171664883D+00
333333	-0.157331000E-01	-0.157540000E-01	-0.144429000E-01	-0.144464000E-01	-0.153807657D-01

[a] See footnote of Table IV.

[b] Off-diagonal fourth order operators are not considered except if they participate in a strong interaction.

with respect to the internal coordinates of Eq. (4), the ξ's. In the same table, we give the coefficients of the expansion of the potential with respect to the internal coordinates given by Eq. (5) and introduced by Simons *et al.* (*13*). Table VII gives the coefficients of the expansion of the potential with respect to the quasi-normal internal coordinates of Eq. (7), the R*'s. This representation of the potential, $W(R^*)$, may be interesting for studying the effects of truncation at a given order in the potential function expansion and to compare these effects to the ones which

TABLE VI

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to the Internal Coordinates of Eqs. (4) and (5) (in cm^{-1})

f [a]	W(ξ)	W(ξ*)
f_{11}	0.5424846E+06	0.5424846E+06
f_{13}	0.8487462E+05	0.8487462E+05
f_{22}	0.1971034E+05	0.1971034E+05
f_{111}	-0.1520628E+07	-9.4356592E+06
f_{113}	-0.1144548E+06	-0.2958022E+05
f_{122}	-0.3488041E+05	-0.3488041E+05
f_{1111}	0.2491949E+07	-0.3424827E+06
f_{1113}	0.1265733E+06	-0.1746171E+05
f_{1122}	0.1650723E+05	-0.1837318E+05
f_{1133}	0.1902179E+05	-0.1250132E+06
f_{1223}	0.9537275E+05	0.9537275E+05
f_{2222}	0.2844204E+04	0.2842204E+04
f_{11111}	-0.2040452E+07	0.1373510E+07
f_{11113}	0.3087399E+06	0.4299700E+06
f_{11122}	-0.2586662E+05	-0.2773257E+05
f_{11133}	-0.2439542E+07	-0.2533415E+07
f_{11223}	-0.4785009E+05	0.4752266E+05
f_{12222}	-0.7574559E+04	-0.7574559E+04
f_{111111}	-0.8325041E+07	-0.5101677E+07
f_{111113}	0.5883079E+07	0.7504534E+07
f_{111122}	-0.8637048E+06	-0.9266633E+06
f_{111133}	0.2372085E+08	0.1677479E+08
f_{111223}	-0.2748575E+07	-0.2748902E+07
f_{111333}	0.2826158E+08	0.1845971E+08
f_{112222}	-0.9650980E+04	-0.1722554E+05
f_{112233}	-0.3724728E+07	-0.3725055E+07
f_{122223}	-0.8189004E+04	-0.8189004E+04
f_{222222}	0.1398110E+04	0.1389110E+04

[a] For example, f_{111223} is the coefficient of the operator $(\xi_1^3\xi_3 + \xi_1\xi_3^3)\xi_2^2$.

are observed when truncating the representation $W(R)$ given in Table II. This work is in progress. However, some preliminary indications may be obtained from Table VIII, which gives a comparison between the mean of the absolute values of the coefficients appearing in a given order in the expansion of the potential function with respect to the internal coordinates of either Eq. (6), the R's, or Eq. (7), the R^*'s. Although the argument is only classical it is immediately seen that the convergence of the first terms (the zeroth order terms are obviously the same) is much faster when using the R^*'s than when using the R's. Another illustration is given in Figs. 3 and 4 which show constant energy contour maps of $W(R)$ (Fig. 3) and $W(R^*)$ (Fig. 4) for R_1 and R_3 (R_1^* and R_3^*) varied separately with $R_2 = 0$. In Fig. 4 the general appearance of the curves and the values of the dissociation ener-

TABLE VII

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to the Quasi-normal Coordinates of Eq. (7) (in cm^{-1})

F^{**} [a]	$W(R^{**})$
F^{**}_{11}	+0.7101241003+03
F^{**}_{22}	+0.335494900E+03
F^{**}_{33}	+0.123157700E+04
F^{**}_{111}	-0.139160500E+02
F^{**}_{122}	-0.292554200E+02
F^{**}_{133}	-0.775139000E+02
F^{**}_{1111}	-0.311328700E+00
F^{**}_{1122}	+0.605746900E+00
F^{**}_{1133}	-0.288527900E+01
F^{**}_{2222}	+0.823452800E+00
F^{**}_{2233}	-0.276934900E+01
F^{**}_{3333}	-0.117536600E+01
F^{**}_{11111}	-0.132534700E-01
F^{**}_{11122}	+0.100757900E-01
F^{**}_{11133}	+0.724243800E+00
F^{**}_{12222}	-0.108136900E+00
F^{**}_{12233}	-0.135016900E+00
F^{**}_{13333}	+0.227484000E+00
F^{**}_{111111}	+0.127081500E-01
F^{**}_{111122}	-0.694695600E-01
F^{**}_{111133}	-0.757489900E-01
F^{**}_{112222}	-0.749907800E-02
F^{**}_{112233}	-0.466945000E-01
F^{**}_{113333}	-0.189878100E+00
F^{**}_{222222}	+0.689471500E-02
F^{**}_{222233}	-0.936985200E-02
F^{**}_{223333}	-0.207992900E-02
F^{**}_{333333}	-0.189038700E-01

[a] See footnote of Table II.

gies of CO_2 to OC + O or O + CO ($\simeq$40 000 cm^{-1} instead of the correct value of $\simeq$44 000 cm^{-1}) are much closer to the reality than in Fig. 3. In fact $W(R)$ and $W(R^*)$ give exactly the same expansions with respect to normal coordinates through fourth order terms and lead to same eigenvalues of the Hamiltonian. The main problem is to determine whether or not the behavior of a truncated expansion of type $W(R^*)$ is superior to that of a truncated expansion of type $W(R)$. The answer right now seems to be yes. For more details see Carney *et al.* (*19*).

The most important result of the present study is the considerable change brought to the potential function expansion by the introduction of the rotation in the adjustment process to experiment. The convergence and, as a consequence, the physical meaning of this new potential are much more satisfactory. We shall see that this potential also gives a good agreement with experiment.

TABLE VIII

Convergence of the Potential Function Expressed in Terms of Quasi-normal Internal Coordinates; Comparison between the Mean Absolute Values of the Coefficients Appearing in a Given Order of Magnitude in Two Cases

a	Potential from vibration-rotation	
	in terms of the R	in terms of the R*
$\lvert\overline{V_0}\rvert$	760	760
$\lvert\overline{V_1}\rvert$	116	39
$\lvert\overline{V_2}\rvert$	6	1.5
$\lvert\overline{V_3}\rvert$	0.3	0.2
$\lvert\overline{V_4}\rvert$	.05	.04

(in cm^{-1})

[a] For instance, $\lvert\overline{V_2}\rvert$ denotes the mean absolute value of the potential coefficients appearing in the second order of magnitude part of the potential expansion.

B. The Spectroscopic Constants

We give in Table IX the spectroscopic constants for 10 isotopic species of CO_2: five symmetric species and five nonsymmetric species. In this table, the first column gives the vibrational diagonal spectroscopic constants. Since the computer output does not allow for Greek letters we have identified, in all of the columns, the spectroscopic constants by their dependences in the quantum numbers. In the first columns, for example, v_1 means ω_1^0, $v_2{:}\omega_2^0$, $v_{1l}{:}y_{1ll}$, $v_3v_3v_3{:}y_{333}$, etc. The second column gives the rotational diagonal spectroscopic constants. Here, JJ should read $[J(J+1) - l^2]$. So the first constant is B_0; v_1JJ, v_2JJ, and v_3JJ are the α^0's and $JJJJ$ is D_0. The following constants are the $\gamma_{ss'}$'s, the β_s's, and $JJJJJJ$ is H_0. The four remaining constants come from operators where P_z appears explicitly (see Section II. D, where they are denoted by a_{ll}, a_{1ll}, a_{3ll}, and a_{Jll}) and are, in fact, contributions to x_{ll}, y_{1ll}, y_{3ll}, and γ_{ll}. The third column gives the vibrational off-diagonal spectroscopic constants: F_0, F_1, F_2, F_3, F_J (coupling number 2; see Table I), and F^{IV} (coupling number 12). The last column gives the rotational off-diagonal spectroscopic constants: L_0, L_1, L_2, L_3, L_J (coupling number 5), C_0, C_1, C_2, C_3, C_J, C_l (coupling number 6), and F^L (coupling number 8).

C. The Molecular Constants

From the potential corresponding to the present end of the adjustment process (see Tables II, IV, and V) we have computed the molecular constants (the ν_0's, the B's, and the D's) for 10 isotopic species of CO_2 ($^{12}C^{16}O_2$, $^{13}C^{16}O_2$, $^{12}C^{18}O_2$, $^{13}C^{18}O_2$, and $^{14}C^{16}O_2$ for the symmetric species; $^{16}O^{12}C^{18}O$, $^{16}O^{12}C^{17}O$, $^{16}O^{13}C^{18}O$, $^{16}O^{13}C^{17}O$, and $^{17}O^{12}C^{18}O$ for the nonsymmetric species). For each species, the molecular constants of 564 vibrational states have been obtained. Tables X and

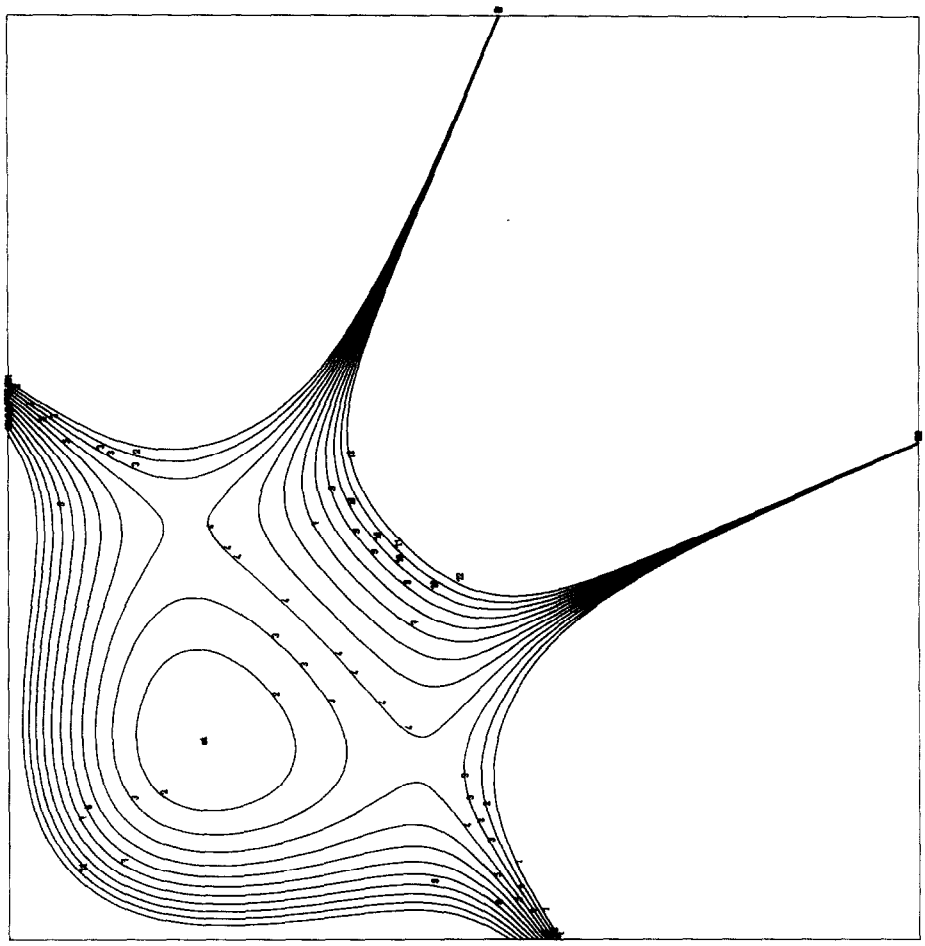


FIG. 3. Constant energy contour maps for CO_2 using the coordinates R for the expansion of the potential energy. The valence angle is fixed at the equilibrium value (0°). Energy contours correspond to $n \times 6000$ cm^{-1} ($n = 0, \ldots, 12$) above the equilibrium configuration which is indicated by a dot.

XI give the most important part of these results. In these tables, the largest subblocks are only given in part: the molecular constants for the vibrational states corresponding to the highest values of l_2 are not always given. In these tables, the first column gives the identification of the species and the vibrational assignment following the notation $(v_1v_2v_3l_2)i$ where the index i ($i = 1, 2, \ldots$) numbers the states within a Fermi polyad: "1" for the highest energy, "2" for the following, etc. In addition a letter, "C" or "D", indicates, respectively, the symmetric states (c-levels) and the antisymmetric states (d-levels). The second column gives the theoretical energy value of the "band center" and eventually the difference between the observed value (or one of the observed values) and the calculated value. The third column gives the theoretical value of the inertia constant and eventually the difference between an observed value and the calculated value. The last column gives the theoretical value of the centrifugal distortion constant

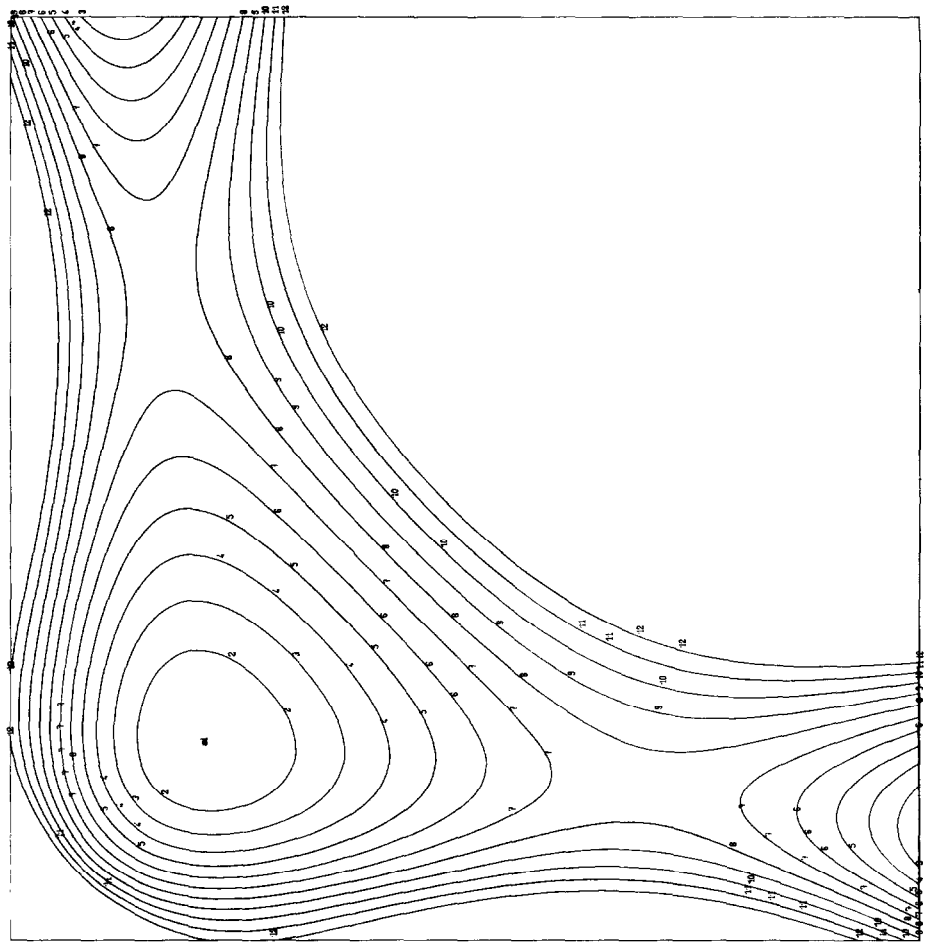


FIG. 4. Constant energy contour maps for CO_2 using the coordinates R^* for the expansion of the potential energy. The valence angle is fixed at the equilibrium value (0^0). Energy contours correspond to $n \times 6000$ cm^{-1} ($n = 0, \ldots, 12$) above the equilibrium configuration which is indicated by a dot.

(multiplied by 10^{+7}) and eventually the difference between the observed value and the calculated value. All the differences are in units of the last digit given. From time to time one will notice anomalous values for either the B's or for the D's or for both.[4] This is typical of a strong Coriolis interaction. In such a case the molecular constants of the two states involved in the interaction are meaningless.

The complete experimental data set included in Tables X and XI is formed from about 800 informations (see Appendix A). Only the first four species participated in the iterative process, which represents about 700 informations (≃85% of the complete set of data). The ratio number of informations (≅700)/number of un-

[4] These relatively numerous anomalies are indicated by an asterisk. The manifestation of the Coriolis interaction on the molecular constants depends upon the values of J used for their determination. As a consequence, other strong interactions may exist which have not been detected. A few additional anomalies are due to accidental inversion of eigenvalues (see, for example, $^{12}C^{16}O_2$, subblock 60^00).

TABLE IX

Spectroscopic Constants of the Molecule $^{16}O^{12}C^{16}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
ν_1	1	1.33567915E+03	JJ	1	3.90221417E-01	C	2	-2.56904266E+01	C	5	-1.53342000E-04
ν_2	1	6.67204346E+02	ν_1 JJ	1	1.20297656E-03	ν_1	2	2.70593643E-01	ν_1	5	5.32967420E-07
ν_3	1	2.36164697E+03	ν_2 JJ	1	7.32528279E-04	ν_2	2	3.23432207E-01	ν_2	5	-4.35577306E-07
$\nu_1\nu_1$	1	-2.99261665E+00	ν_3 JJ	1	3.11362580E-03	ν_3	2	1.38327360E-01	ν_3	5	-3.41206373E-07
$\nu_1\nu_2$	1	-5.27637768E+00	JJJJ	1	1.33311175E-07	JJ	2	1.25486703E-04	JJ	5	1.88001323E-10
$\nu_1\nu_3$	1	-1.91404419E+01	$\nu_1\nu_1$ JJ	1	-1.63962886E-06	C	12	-1.91577822E-02	C	6	-3.66601050E-02
L L	1	-1.01428127E+00	$\nu_1\nu_2$ JJ	1	1.19123788E-05				ν_1	6	3.18510691E-04
$\nu_2\nu_2$	1	1.58003044E+00	$\nu_1\nu_3$ JJ	1	1.47130513E-05				ν_2	6	5.23148803E-04
$\nu_2\nu_3$	1	-1.25418367E+01	L L JJ	1	5.85632370E-06				ν_3	6	1.16898300E-04
$\nu_3\nu_3$	1	-1.25033007E+01	$\nu_2\nu_2$ JJ	1	-9.99300482E-06				JJ	6	-3.55724650E-09
$\nu_1\nu_1\nu_1$	1	2.42212862E-02	$\nu_2\nu_3$ JJ	1	1.03141792E-05				L	6	5.42742433E-04
$\nu_1\nu_1\nu_2$	1	8.15915316E-03	$\nu_3\nu_3$ JJ	1	8.87217811E-06				C	8	-9.76831325E-06
$\nu_1\nu_1\nu_3$	1	-7.73569942E-02	ν_1 JJJJ	1	9.66982605E-11						
ν_1 L L	1	6.31570226E-02	ν_2 JJJJ	1	2.30092101E-09						
$\nu_1\nu_2\nu_2$	1	-5.16645583E-02	ν_3 JJJJ	1	3.24831495E-10						
$\nu_1\nu_2\nu_3$	1	9.56051946E-02	JJJJJJ	1	8.98219487E-15						
$\nu_1\nu_3\nu_3$	1	6.14239648E-02	L K	1	-2.13933716E-04						
ν_2 L L	1	7.02122226E-03	ν_1 L K	1	0.0						
$\nu_2\nu_2\nu_2$	1	-4.70886379E-03	ν_1 L K	1	-4.27867519E-04						
ν_2 L L	1	2.56735977E-02	ν_3 K JJ	1	1.06395760E-07						
$\nu_2\nu_2\nu_3$	1	-2.05716769E-02									
$\nu_2\nu_3\nu_3$	1	1.83352269E-02									
$\nu_3\nu_3\nu_3$	1	6.31367043E-03									

(1) Identification of the constant (2) Coupling number

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{13}C^{16}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.33628711E+03	JJ	1	3.90239716E-01	C	2	-2.51654356E+01	C	5	-1.57829665E-04
V_2	1	6.48341797E+02	V_1 JJ	1	1.20363291E-03	V_1	2	2.57700443E-01	V_1	5	4.59267369E-07
V_3	1	2.29523677E+03	V_2 JJ	1	6.94377115E-04	V_2	2	3.02010000E-01	V_2	5	-4.18775244E-07
$V_1 V_1$	1	-3.00017262E+00	V_3 JJ	1	3.00089781E-03	V_3	2	1.59508165E-01	V_3	5	-3.12059001E-07
$V_1 V_2$	1	-5.10996726E+00	JJJJ	1	1.33225114E-07	JJ	2	1.19818476E-04	JJ	5	1.93689217E-10
$V_1 V_3$	1	-1.66744690E+01	$V_1 V_1$ JJ	1	-1.63962886E-06	C	12	-1.39669292E-02	C	6	-3.54743786E-02
L L	1	-9.55670450E-01	$V_1 V_2$ JJ	1	1.14343729E-05				V_1	6	2.71173660E-04
$V_2 V_2$	1	1.49001026E+00	$V_1 V_3$ JJ	1	1.43563311E-05				V_2	6	4.98523936E-04
$V_2 V_3$	1	-1.17563066E+01	L L JJ	1	5.41271129E-06				V_3	6	1.18399184E-04
$V_3 V_3$	1	-1.17358589E+01	$V_2 V_2$ JJ	1	-9.40911923E-06				JJ	6	-4.44104842E-09
$V_1 V_1 V_1$	1	2.42212862E-02	$V_2 V_3$ JJ	1	9.70381097E-06				L	6	5.14849555E-04
$V_1 V_1 V_2$	1	6.81840815E-04	$V_3 V_3$ JJ	1	8.31453508E-06				C	8	-9.99099848E-06
$V_1 V_1 V_3$	1	-7.75144100E-02	V_1 JJJJ	1	9.66982605E-11						
V_1 L L	1	5.56888692E-02	V_2 JJJJ	1	2.21335705E-09						
$V_1 V_2 V_2$	1	-4.48305979E-02	V_3 JJJJ	1	3.21788374E-10						
$V_1 V_2 V_3$	1	8.88499618E-02	JJJJJJ	1	8.98219487E-15						
$V_1 V_3 V_3$	1	6.22764491E-02	L K	1	-2.20200862E-04						
V_2 L L	1	6.34629279E-03	V_1 L K	1	0.0						
$V_2 V_2 V_2$	1	-4.22560513E-03	V_3 L K	1	-4.40401724E-04						
$V_2 V_2 V_3$	1	2.47777030E-02	L K JJ	1	1.12720784E-07						
$V_2 V_3 V_3$	1	-2.02638209E-02									
$V_3 V_3 V_3$	1	1.70341805E-02									
	1	5.62079996E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{18}O^{12}C^{18}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.2597C726E+03	JJ	1	3.46817970E-01	C	2	-2.47745E19E+01	C	5	-1.22963873E-04
V_2	1	6.5712963 9E+02	V_1 JJ	1	1.00779464E-03	V_1	2	2.51752675E-01	V_1	5	5.06856168E-07
V_3	1	2.32627344E+03	V_2 JJ	1	6.63301442E-04	V_2	2	3.09540927E-01	V_2	5	-3.62374237E-07
$V_1 V_1$	1	-2.64822865E+00	V_3 JJ	1	2.75638537E-03	V_3	2	8.72752070E-02	V_3	5	-3.03775778E-07
$V_1 V_2$	1	-4.91946125E+C0	JJJJ	1	1.05275888E-07	JJ	2	1.09340122E-04	JJ	5	1.33806952E-10
$V_1 V_3$	1	-1.76662445E+01	$V_1 V_1$ JJ	1	-1.29480577E-06	C	12	-2.43191607E-02	C	6	-3.32556963E-02
L L	1	-9.8841 9354E-01	$V_1 V_2$ JJ	1	9.98265841E-06				V_1	6	3.23991291E-04
$V_2 V_2$	1	1.53667545E+00	$V_1 V_3$ JJ	1	1.20652367E-05				V_1	6	4.57057264E-04
$V_2 V_3$	1	-1.22801752E+01	L L JJ	1	5.19505375E-06				V_2	6	9.45989595E-05
$V_3 V_3$	1	-1.22202263E+01	$V_2 V_2$ JJ	1	-8.63783316E-06				V_3	6	-1.70421410E-09
$V_1 V_1 V_1$	1	2.0290441 8E-02	$V_2 V_3$ JJ	1	8.92632397E-06				L	6	4.78134258E-04
$V_1 V_1 V_2$	1	1.834731 92E-02	$V_3 V_3$ JJ	1	7.71781015E-06				C	8	-7.68426253E-06
$V_1 V_1 V_3$	1	-6.492942 57E-02	V_1 JJJJ	1	7.19850846E-11						
V_1 L L	1	6.42254353E-02	V_2 JJJJ	1	1.81398385E-09						
$V_1 V_2 V_2$	1	-5.375C5411E-02	V_3 JJJJ	1	2.44507081E-10						
$V_1 V_2 V_3$	1	8.965373 04E-02	JJJJJJ	1	6.30334475E-15						
$V_1 V_3 V_3$	1	5.07543981E-02	L K	1	-1.71568143E-04						
V_2 L L	1	6.85001910E-03	V_1 L K	1	0.0						
$V_2 V_2 V_2$	1	-4.64221835E-03	V_3 L K	1	-3.43136257E-04						
$V_2 V_2 V_3$	1	2.3174C326E-02	L K JJ	1	7.70033353E-08						
$V_2 V_3 V_3$	1	-1.76066160E-02									
$V_3 V_3 V_3$	1	1.7172657 0E-02									
	1	6.2550418 1E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{18}O^{13}C^{18}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.2601CC83E+03	JJ	1	3.46834779E-01	C	2	-2.40586853E+01	C	5	-1.26681392E-04
V_2	1	6.37956787E+02	V_1 JJ	1	1.00834272E-03	V_1	2	2.39053965E-01	V	5	4.48309834E-07
V_3	1	2.25E7644CE+03	V_2 JJ	1	6.29528193E-04	V_2	2	2.88358033E-01	V_1	5	-3.50002608E-07
$V_1 V_1$	1	-2.65632248E+00	V_3 JJ	1	2.65550474E-03	V_3	2	1.09814644E-01	V_2	5	-2.80347479E-07
$V_1 V_2$	1	-4.75883961E+00	JJJJ	1	1.05207278E-07	JJ	2	1.04326537E-04	V_3 JJ	5	1.37977102E-10
$V_1 V_3$	1	-1.72162476E+01	$V_1 V_1$ JJ	1	-1.29480577E-06	C	12	-1.84269510E-02	C	6	-3.21605653E-02
L L	1	-9.29932714E-01	$V_1 V_2$ JJ	1	9.58851342E-06				V_1	6	2.78670341E-04
$V_2 V_2$	1	1.44661808E+00	$V_1 V_3$ JJ	1	1.17573054E-05				V_2	6	4.35311114E-04
$V_2 V_3$	1	-1.14946766E+01	L L JJ	1	4.79963182E-06				V_3 JJ	6	9.47532535E-05
$V_3 V_3$	1	-1.14536555E+01	$V_2 V_2$ JJ	1	-8.12087092E-06				L	6	-2.40380893E-09
$V_1 V_1 V_1$	1	2.0290441BE-02	$V_2 V_3$ JJ	1	8.38514643E-06				C	8	4.52724053E-04
$V_1 V_1 V_2$	1	1.01900548E-02	$V_3 V_3$ JJ	1	7.22520963E-06						-7.86090732E-06
$V_1 V_1 V_3$	1	-6.48028255E-02	V_1 JJJJ	1	7.19850846E-11						
V_1 L L	1	5.62257357E-02	V_2 JJJJ	1	1.74451498E-09						
$V_1 V_2 V_2$	1	-4.63361256E-02	V_3 JJJJ	1	2.42754927E-10						
$V_1 V_2 V_3$	1	8.29777122E-02	JJJJJJ	1	6.30334475E-15						
$V_1 V_3 V_3$	1	5.11868857E-02	L K	1	-1.76758869E-04						
V_2 L L	1	6.17253406E-03	V_1 L K	1	0.0						
$V_2 V_2 V_2$	1	-4.15357947E-03	V_3 L K	1	-3.53517709E-04						
$V_2 V_2 V_3$	1	2.21336943E-02	L K JJ	1	8.17332761E-08						
$V_2 V_3 V_3$	1	-1.73536725E-02									
V_3 L L	1	1.59106627E-02									
$V_3 V_3 V_3$	1	5.58249652E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{14}C^{16}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.33664307E+03	JJ	1	3.90255690E-01	C	2	-2.45283356E+01	C	5	-1.61990829E-04
V_2	1	6.31778809E+02	V_1 JJ	1	1.20421639E-03	V_1	2	2.46778071E-01	V_1	5	3.91377228E-07
V_3	1	2.23690552E+03	V_2 JJ	1	6.60420628E-04	V_2	2	2.83924758E-01	V_2	5	-4.02612045E-07
$V_1 V_1$	1	-3.00616074E+00	V_3 JJ	1	2.90127471E-03	V_3	2	1.76554024E-01	V_3	5	-2.85288934E-07
$V_1 V_2$	1	-4.96467686E+00	JJJJ	1	1.33149285E-07	JJ	2	1.14864524E-04	JJ	5	1.98978112E-10
$V_1 V_3$	1	-1.82676392E+01	$V_1 V_1$ JJ	1	-1.63962886E-06	C	12	-1.00532956E-02	C	6	-3.44274826E-02
L L	1	-9.05893662E-01	$V_1 V_2$ JJ	1	1.10058454E-05				V_1	6	2.31736631E-04
$V_2 V_2$	1	1.41301441E+00	$V_1 V_3$ JJ	1	1.40460879E-05				V_2	6	4.76974528E-04
$V_2 V_3$	1	-1.10852871E+01	L L JJ	1	5.03438969E-06				V_3	6	1.20555982E-04
$V_3 V_3$	1	-1.10796719E+01	$V_2 V_2$ JJ	1	-8.90880801E-06				JJ	6	-5.24028110E-09
$V_1 V_1 V_1$	1	2.42212862E-02	$V_2 V_3$ JJ	1	9.18194110E-06				L	6	4.90945764E-04
$V_1 V_1 V_2$	1	-5.13432920E-03	$V_3 V_3$ JJ	1	7.93607993E-06				C	8	-1.01994465E-05
$V_1 V_1 V_3$	1	-7.78579712E-02	V_1 JJJJ	1	9.66982605E-11						
V_1 L L	1	4.97918986E-02	V_2 JJJJ	1	2.13589857E-09						
$V_1 V_2 V_2$	1	-3.94816287E-02	V_3 JJJJ	1	3.18527871E-10						
$V_1 V_2 V_3$	1	8.32527280E-02	JJJJJJ	1	8.98219487E-15						
$V_1 V_3 V_3$	1	6.32750392E-02	L K	1	-2.26013013E-04						
V_1 L L	1	5.79255917E-03	V_1 L K	1	0.0						
$V_2 V_2 V_2$	1	-3.83153185E-03	V_2 L K	1	-4.52026026E-04						
$V_2 V_2 V_3$	1	2.38456650E-02	V_3 L K JJ	1	1.18749767E-07						
$V_2 V_3 V_3$	1	-1.99799612E-02									
$V_3 V_3 V_3$	1	1.59421526E-02									
	1	5.03674570E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{12}C^{18}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.29752368E+03	JJ	1	3.68186355E-01	C	2	-2.53078156E+01	C	5	-1.37539973E-04
V_2	1	6.62186279E+02	V_1 JJ	1	1.10346475E-03	V_1	2	2.60749221E-01	V_1	5	5.25220003E-07
V_3	1	2.34445927E+03	V_2 JJ	1	6.97528711E-04	V_2	2	3.16247523E-01	V_2	5	-3.98213274E-07
V_1V_1	1	-2.81461906E+00	V_3 JJ	1	2.93027516E-03	V_3	2	1.11058474E-01	V_3	5	-3.26298732E-07
V_1V_2	1	-5.09194469E+00	JJJJ	1	1.18695652E-07	JJ	2	1.17235919E-04	JJ	5	1.58987656E-10
V_1V_3	1	-1.83853760E+01	V_1V_1 JJ	1	-1.47060928E-06	C	12	-2.15591416E-02	C	6	-3.49470936E-02
L L	1	-9.96319664E-01	V_1V_2 JJ	1	1.08752765E-05				V_1	6	3.23986635E-04
V_2V_2	1	1.55563164E+00	V_1V_3 JJ	1	1.34369220E-05				V_2	6	4.89340629E-04
V_2V_3	1	-1.24049444E+01	L L JJ	1	5.51487483E-06				V_3	6	1.05046638E-04
V_3V_3	1	-1.23646302E+01	V_2V_2 JJ	1	-9.29427733E-06				JJ	6	-2.46951615E-09
$V_1V_1V_1$	1	2.19703987E-02	V_2V_3 JJ	1	9.63599268E-06				L	6	5.09137986E-04
$V_1V_1V_2$	1	1.27382129E-02	V_3V_3 JJ	1	8.22903439E-06				C	8	-8.67156268E-06
$V_1V_1V_3$	1	-6.97281361E-02	V_1 JJJJ	1	8.63532440E-11						
V_1 L L	1	6.35179281E-02	V_2 JJJJ	1	2.04616768E-09						
$V_1V_2V_2$	1	-5.24969958E-02	V_3 JJJJ	1	2.86762836E-10						
$V_1V_2V_3$	1	9.36879516E-02	JJJJJJ	1	7.50756133E-15						
$V_1V_3V_3$	1	5.56059652E-02	L K	1	-1.91890416E-04						
V_2 L L	1	6.95434585E-03	V_1 L K	1	-1.60707089E-07						
$V_2V_2V_2$	1	-4.69372794E-03	V_3 L K	1	-3.83620150E-04						
V_2 L L	1	2.41217689E-02	L K JJ	1	9.07623416E-08						
$V_2V_2V_3$	1	-1.86538696E-02									
$V_2V_3V_3$	1	1.72453783E-02									
$V_3V_3V_3$	1	6.47703186E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{12}C^{17}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
ν_1	1	1.31567842E+03	JJ	1	3.78615558E-01	C	2	-2.65904999E+01	C	5	-1.44928446E-04
ν_2	1	6.64546875E+02	ν_1 JJ	1	1.14995847E-03	ν_1	2	2.65464246E-01	ν_1	5	5.28712405E-07
ν_3	1	2.35244995E+03	ν_2 JJ	1	7.14215217E-04	ν_2	2	3.19678426E-01	ν_2	5	-4.15818192E-07
$\nu_1\nu_1$	1	-2.85993858E+00	ν_3 JJ	1	3.01733729E-03	ν_3	2	1.24304593E-01	ν_3	5	-3.32546676E-07
$\nu_1\nu_2$	1	-5.18055056E+00	JJJJ	1	1.25499412E-07	JJ	2	1.21127479E-04	JJ	5	1.72337603E-10
$\nu_1\nu_3$	1	-1.87447205E+01	$\nu_1\nu_1$ JJ	1	-1.54626559E-06	C	12	-2.04343721E-02	C	6	-3.57557200E-02
LL	1	-1.00656223E+00	$\nu_1\nu_1$ JJ	1	1.13736642E-05				ν_1	6	3.20821768E-04
$\nu_2\nu_2$	1	1.56775551E+00	$\nu_1\nu_2$ JJ	1	1.40090133E-05				ν_2	6	5.05334465E-04
$\nu_2\nu_3$	1	-1.24707298E+01	$\nu_1\nu_3$ JJ	1	5.67765983E-06				ν_3	6	1.10553214E-04
$\nu_3\nu_3$	1	-1.24346555E+01	$\nu_2\nu_2$ JJ	1	-9.62403556E-06				JJ	6	-2.98366820E-09
$\nu_1\nu_1\nu_1$	1	2.30912343E-02	$\nu_2\nu_3$ JJ	1	9.94732000E-06				L	6	5.25146723E-04
$\nu_1\nu_1\nu_2$	1	1.07016191E-02	$\nu_3\nu_3$ JJ	1	8.54425434E-06				C	8	-9.18422575E-06
$\nu_1\nu_1\nu_3$	1	-7.36294386E-02	ν_1 JJJJ	1	9.04890329E-11						
ν_1LL	1	6.33456495E-02	ν_2 JJJJ	1	2.16487983E-09						
$\nu_1\nu_2\nu_2$	1	-5.21143340E-02	ν_3 JJJJ	1	3.03405745E-10						
$\nu_1\nu_2\nu_3$	1	9.42778567E-02	JJJJJJ	1	8.18133877E-15						
$\nu_1\nu_3\nu_3$	1	5.83572946E-02	LK	1	-2.02198629E-04						
ν_1LL	1	6.57962567E-03	ν_1LK	1	-4.79701860E-08						
$\nu_2\nu_2\nu_2$	1	-4.69494584E-03	ν_2LK	1	-4.04349295E-04						
ν_2LL	1	2.50407755E-02	ν_3LK JJ	1	9.79702008E-08						
$\nu_2\nu_2\nu_3$	1	-1.96578614E-02									
$\nu_2\nu_3\nu_3$	1	1.76757878E-02									
$\nu_3\nu_3\nu_3$	1	6.35736808E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{13}C^{18}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.29785229E+03	JJ	1	3.68183672E-01	C	2	-2.45834351E+01	C	5	-1.41615135E-04
V_2	1	6.43170410E+02	V_1 JJ	1	1.10413181E-03	V_1	2	2.48011172E-01	V_1	5	4.59175737E-07
V_3	1	2.27760962E+03	V_2 JJ	1	6.61555445E-04	V_2	2	2.94938982E-01	V_2	5	-3.83774875E-07
$V_1 V_1$	1	-2.82470608E+00	V_3 JJ	1	2.82317726E-03	V_3	2	1.32450223E-01	V_3	5	-2.99715452E-07
$V_1 V_2$	1	-4.92785549E+00	JJJJ	1	1.18609648E-07	JJ	2	1.11888468E-04	JJ	5	1.63845867E-10
$V_1 V_3$	1	-1.79146729E+01	$V_1 V_1$ JJ	1	-1.46943239E-06	C	12	-1.60839073E-02	C	6	-3.37990203E-02
L L	1	-9.39555347E-01	$V_1 V_2$ JJ	1	1.04372948E-05				V_1	6	2.76753912E-04
$V_2 V_2$	1	1.46529484E+00	$V_1 V_3$ JJ	1	1.31092283E-05				V_2	6	4.66362806E-04
$V_2 V_3$	1	-1.16191034E+01	L L JJ	1	5.09578058E-06				V_3	6	1.05854502E-04
$V_3 V_3$	1	-1.16205482E+01	$V_2 V_2$ JJ	1	-8.74437410E-06				JJ	6	-3.23563065E-09
$V_1 V_1 V_1$	1	2.20120884E-02	$V_2 V_3$ JJ	1	9.06450350E-06				L	6	4.82689124E-04
$V_1 V_1 V_2$	1	5.04381582E-03	$V_3 V_3$ JJ	1	7.69546932E-06				C	8	-8.86793444E-06
$V_1 V_1 V_3$	1	-6.99086785E-02	V_1 JJJJ	1	8.67960148E-11						
V_1 L L	1	5.58316870E-02	V_2 JJJJ	1	1.96780414E-09						
$V_1 V_2 V_2$	1	-4.54317927E-02	V_3 JJJJ	1	2.84952728E-10						
$V_1 V_2 V_3$	1	8.67207050E-02	JJJJJJ	1	7.50493553E-15						
$V_1 V_3 V_3$	1	5.59392311E-02	L K	1	-1.97580157E-04						
V_2 L L	1	6.27385825E-03	V_1 L K	1	-1.97353074E-07						
$V_2 V_2 V_2$	1	-4.20356169E-03	V_3 L K	1	-3.94962961E-04						
V_3 L L	1	2.30547339E-02	L K JJ	1	9.62312470E-08						
$V_2 V_2 V_3$	1	-1.83718018E-02									
$V_2 V_3 V_3$	1	1.59103945E-02									
$V_3 V_3 V_3$	1	5.83770126E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{16}O^{13}C^{17}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
V_1	1	1.3162624SE+03	JJ	1	3.78627896E-01	C	2	-2.48666840E+01	C	5	-1.49201762E-04
V_2	1	6.456C3516E+02	V_1 JJ	1	1.15060550E-03	V_1	2	2.52647638E-01	V	5	4.59211037E-07
V_3	1	2.28577222E+03	V_2 JJ	1	6.77221688E-04	V_2	2	2.98316360E-01	V_1	5	-4.00280214E-07
$V_1 V_1$	1	-2.9C81E787E+0C	V_3 JJ	1	2.90764938E-03	V_3	2	1.45665315E-01	V_2	5	-3.04919070E-07
$V_1 V_2$	1	-5.C1548004E+C0	JJJJ	1	1.25415625E-07	JJ	2	1.15630173E-04	V_3	5	1.77585585E-10
$V_1 V_3$	1	-1.827S4952E+01	$V_1 V_1$ JJ	1	-1.54650570E-06	C	12	-1.50793567E-02	JJ	6	-3.45915519E-02
$L_1 L_1$	1	-9.48C38220E-01	$V_1 V_1$ JJ	1	1.09175335E-05				C	6	2.73855170E-04
$V_2 V_2$	1	1.47764111E+0C	$V_1 V_2$ JJ	1	1.36689641E-05				V_1	6	4.81525203E-04
$V_2 V_3$	1	-1.168511C1E+01	$L_1 L_1$ JJ	1	5.24691404E-06				V_2	6	1.11650559E-04
$V_3 V_3$	1	-1.166852CCE+01	$V_2 V_2$ JJ	1	-9.05806792E-06				JJ	6	-3.80944343E-09
$V_1 V_1 V_1$	1	2.3C967775E-02	$V_2 V_3$ JJ	1	9.35649677E-06				L	6	4.97951405E-04
$V_1 V_1 V_2$	1	3.C8221S51E-03	$V_3 V_3$ JJ	1	8.00139151E-06				C	8	-9.39347137E-06
$V_1 V_1 V_3$	1	-7.37C97859E-02	V_1 JJJJ	1	9.06228564E-11						
$V_1 L_1 L_1$	1	5.5754S968E-02	V_2 JJJJ	1	2.08228190E-09						
$V_1 V_2 V_2$	1	-4.515C1980E-02	V_3 JJJJ	1	3.00910630E-10						
$V_1 V_2 V_3$	1	8.74837C41E-02	JJJJJJ	1	8.18032233E-15						
$V_1 V_3 V_3$	1	5.8S9143S6E-02	L K	1	-2.08165948E-04						
$V_1 L_1 L_1$	1	6.3030C805E-03	V_1 L K	1	-5.83677214E-08						
$V_2 V_2 V_2$	1	-4.2C9464S1E-03	V_1 L K	1	-4.16273018E-04						
$V_2 L_1 L_1$	1	2.3S6C43C2E-02	L_3 K JJ	1	1.03840136E-07						
$V_3 V_3 V_3$	1	-1.S3645246E-02									
$V_2 V_3 V_3$	1	1.65733EC7E-02									
$V_2 V_3 V_3$	1	5.6S3C3466E-03									

TABLE IX—*Continued*

Spectroscopic Constants of the Molecule $^{17}O^{12}C^{18}O$ (cm^{-1})

VIBRATIONAL DIAGONAL (1)	(2)	
V_1	1	1.27781543E+03
V_2	1	6.595C8545E+02
V_3	1	2.33472998E+03
V_1V_1	1	-2.72751617E+00
V_1V_2	1	-5.CC246C48E+00
V_1V_3	1	-1.8011C168E+01
LL	1	-5.93824959E-C1
V_2V_2	1	1.54625893E+00
V_2V_3	1	-1.234C2443E+01
V_3V_3	1	-1.22917757E+01
$V_1V_1V_1$	1	2.11256184E-02
$V_1V_1V_2$	1	1.57921277E-02
$V_1V_1V_3$	1	-6.74C593915-02
V_1LL	1	6.3677C461E-02
$V_1V_2V_2$	1	-5.31514846E-02
$V_1V_2V_3$	1	C.12991166E-02
$V_1V_3V_3$	1	5.3CE64820E-02
V_2LL	1	6.894C2966E-03
$V_2V_2V_2$	1	-4.661668C9E-03
V_3LL	1	2.3714E590E-02
$V_2V_2V_3$	1	-1.82C14480E-02
$V_2V_3V_3$	1	1.7335C647E-02
$V_3V_3V_3$	1	6.31C4E658E-03

ROTATIONAL DIAGONAL (1)	(2)	
JJ	1	3.56932938E-01
V_1JJ	1	1.05245947E-03
V_2JJ	1	6.79624500E-04
V_3JJ	1	2.83902045E-03
JJJJ	1	1.11519910E-07
V_1V_1JJ	1	-1.37461393E-06
V_1V_2JJ	1	1.04133696E-05
V_1V_3JJ	1	1.26842669E-05
LLJJ	1	5.34754781E-06
V_2V_2JJ	1	-8.94758705E-06
V_2V_3JJ	1	9.25159748E-06
V_3V_3JJ	1	7.97253779E-06
V_1JJJJ	1	7.79185466E-11
V_2JJJJ	1	1.92215399E-09
V_3JJJJ	1	2.62998512E-10
JJJJJJ	1	6.85285230E-15
LK	1	-1.81069088E-04
V_1LK	1	-3.32845431E-08
V_3LK	1	-3.62104969E-04
LKJJ	1	8.33417175E-08

VIBRATIONAL COUPLINGS (1)	(2)	
C	2	-2.50343170E+01
V_1	2	2.5E0665E1E-01
V_2	2	3.12744677E-01
V_3	2	9.89308357E-02
JJ	2	1.13064467E-04
C	12	-2.30046138E-02

ROTATIONAL COUPLINGS (1)	(2)	
C	5	-1.29777225E-04
V_1	5	5.15355737E-07
V_2	5	-3.79237633E-07
V_3	5	-3.13688020E-07
JJ	5	1.45384330E-10
C	6	-3.40558589E-02
V_1	6	3.23670916E-04
V_2	6	4.72281594E-04
V_3	6	9.94203001E-05
JJ	6	-2.07176898E-09
L	6	4.92861727E-04
C	8	-8.14589839E-06

TABLE X

16 12 16 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(0 0 0 0)1	0.0	0	0.390221	-1	1.33	0
(0 1 0 1)1 C	667.381	-1	0.390642	-1	1.35	0
(0 1 0 1)1 D	667.391	-1	0.391257	-2	1.36	0
(1 0 0 0)1	1388.188	0	0.390181	9	1.15	0
(1 0 0 0)2	1285.414	0	0.390482	0	1.57	0
(0 2 0 2)1 C	1335.123	5	0.391670	10	1.37	1
(0 2 0 2)1 D	1335.123	5	0.391670	10	1.38	0
(0 0 1 0)1	2349.150	-2	0.387132	9	1.33	-3
(1 1 0 1)1 C	2076.875	-19	0.390402	9	1.26	0
(1 1 0 1)2 C	1932.482	-4	0.390749	-3	1.49	-4
(0 3 0 3)1 C	2003.234	2	0.392383	-2	1.40	0
(1 1 0 1)1 D	2076.875	-19	0.391322	12	1.22	0
(1 1 0 1)2 D	1932.482	-4	0.391695	-4	1.56	-1
(0 3 0 3)1 D	2003.234	2	0.392383	2	1.40	0
(2 0 0 0)1	2797.135	19	0.390587	-23	0.99	0
(2 0 0 0)2	2671.122	-9	0.389543	12	1.35	-1
(2 0 0 0)3	2543.380	-5	0.391119	-11	1.81	-2
(0 1 1 1)1 C	3004.016	-4	0.387581	12	1.35	0
(1 2 0 2)1 C	2760.778	-42	0.391536	-1	1.42	2
(1 2 0 2)2 C	2585.049	-9	0.391952		1.39	
(0 4 0 4)1 C	2671.725		0.393097		1.42	
(0 1 1 1)1 D	3004.016	-4	0.388180	10	1.35	0
(1 2 0 2)1 D	2760.778	-42	0.391536	-1	1.29	-3
(1 2 0 2)2 D	2585.049	-9	0.391952		1.52	
(0 4 0 4)1 D	2671.725		0.393087		1.42	
(1 0 1 0)1	3714.779	10	0.387055	0	1.15	0
(1 0 1 0)2	3612.837	8	0.387501	1	1.57	0
(2 1 0 1)1 C	3500.677	-37	0.390370	10	1.18	0
(2 1 0 1)2 C	3339.306	34	0.390024	6	1.39	-1
(2 1 0 1)3 C	3181.472	-21	0.391042	-21	1.63	0
(0 2 1 2)1 C	3659.257	12	0.388622	10	1.36	-3
(1 3 0 3)1 C	3442.342		0.392204		1.37	
(1 3 0 3)2 C	3240.588		0.392672		1.49	
(0 5 0 5)1 C	3340.605		0.393783		1.45	
(2 1 0 1)1 D	3500.677	-37	0.391687	-27	1.10	-8
(2 1 0 1)2 D	3339.306	34	0.391157	3	1.39	-1
(2 1 0 1)3 D	3181.472	-21	0.392369		1.74	
(0 2 1 2)1 D	3659.257	12	0.388622	10	1.37	4
(1 3 0 3)1 D	3442.342		0.392204		1.37	
(1 3 0 3)2 D	3240.588		0.392672		1.49	
(0 5 0 5)1 D	3340.605		0.393783		1.45	
(3 0 0 0)1	4225.043		0.390959		0.75	
(3 0 0 0)2	4054.197	-95	0.389556		1.01	
(3 0 0 0)3	3942.561	-91	0.389579	5	1.70	-4
(3 0 0 0)4	3792.673	5	0.391790	-9	2.10	0
(0 0 2 0)1	4673.332	0	0.384050	4	1.32	0
(1 1 1 1)1 C	4390.637	-7	0.387347	19	1.25	-1
(1 1 1 1)2 C	4247.715	-2	0.387774	7	1.49	-3
(2 2 0 2)1 C	4197.426		0.391571		1.60	
(2 2 0 2)2 C	4007.873		0.391426		1.44	
(2 2 0 2)3 C	3822.039		0.392375		1.30	
(0 3 1 3)1 C	4314.983	21	0.389357	28	1.39	
(1 4 0 4)1 C	4122.527		0.392864		1.40	
(1 4 0 4)2 C	3893.460		0.393381		1.51	
(0 6 0 6)1 C	4009.984		0.394471		1.47	

TABLE X—*Continued*

16 12 16 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 1 1)1 D	4390.637	-7	0.388220	15	1.21	1
(1 1 1 1)2 D	4247.715	-7	0.388708	- 1	1.55	1
(2 2 0 2)1 D	4197.426		0.391571		1.20	
(2 2 0 2)2 D	4007.873		0.391426		1.41	
(2 2 0 2)3 D	3822.038		0.392374		1.67	
(0 3 1 3)1 D	4314.883	21	0.389357	28	1.39	
(1 4 0 4)1 D	4122.527		0.392864		1.40	
(1 4 0 4)2 D	3898.460		0.393381		1.51	
(0 6 0 6)1 D	4009.884		0.394471		1.47	
(2 0 1 0)1	5099.669	0	0.387474	16	0.98	-5
(2 0 1 0)2	4977.809	19	0.386830	0	1.36	-1
(2 0 1 0)3	4853.625	-3	0.388207	-19	1.80	-3
(3 1 0 1)1 C	4938.324	62	0.390361	9	1.11	-3
(3 1 0 1)2 C	4753.270		0.389590		1.24	
(3 1 0 1)3 C	4591.027	89	0.389927	-2	1.54	-1
(3 1 0 1)4 C	4415.121	27	0.391411	-51	1.77	0
(0 1 2 1)1 C	5315.719	-3	0.384534	14	1.34	0
(1 2 1 2)1 C	5061.812	-35	0.388496	14	1.38	0
(1 2 1 2)2 C	* 4888.027	-27	0.388941	-1	1.13	20
(2 3 0 3)1 C	* 4890.277		0.392181		1.62	
(2 3 0 3)2 C	4676.812		0.392185		1.45	
(2 3 0 3)3 C	4467.223		0.393062		1.55	
(0 4 1 4)1 C	4970.305		0.390087		1.41	
(1 5 0 5)1 C	4801.836		0.393516		1.43	
(1 5 0 5)2 C	4557.883		0.394081		1.53	
(0 7 0 7)1 C	4679.574		0.395152		1.49	
(3 1 0 1)1 D	4938.324		0.392136		0.99	
(3 1 0 1)2 D	4753.270	179	0.391147	31	1.18	33
(3 1 0 1)3 D	4591.027	89	0.391349	-19	1.63	-14
(3 1 0 1)4 D	4415.121	27	0.393143	-54	1.93	-16
(0 1 2 1)1 D	5315.719	-3	0.385119	7	1.35	0
(1 2 1 2)1 D	5061.812	-35	0.388496	14	1.28	-15
(1 2 1 2)2 D	* 4888.027	-27	0.388941	-1	1.28	15
(2 3 0 3)1 D	* 4890.277		0.392181		1.62	
(2 3 0 3)2 D	4676.812		0.392185		1.45	
(2 3 0 3)3 D	4467.223		0.393062		1.55	
(0 4 1 4)1 D	4970.305		0.390087		1.41	
(1 5 0 5)1 D	4801.836		0.393516		1.43	
(1 5 0 5)2 D	4557.883		0.394081		1.53	
(0 7 0 7)1 D	4679.574		0.395152		1.49	
(4 0 0 0)1	5667.488		0.391244		0.32	
(4 0 0 0)2	* 5475.285		0.390201		2.47	
(4 0 0 0)3	5329.746		0.388668		1.26	
(4 0 0 0)4	5197.266	-15	0.390129	-11	2.03	-1
(4 0 0 0)5	5022.273		0.392454		2.52	
(1 0 2 0)1	5015.576	11	0.393943	18	1.15	2
(1 0 2 0)2	4915.215	0	0.394537	5	1.56	0
(2 1 1 1)1 C	5790.594	-15	0.387359	30	1.16	-4
(2 1 1 1)2 C	5632.734	27	0.387017	-7	1.32	0
(2 1 1 1)3 C	* 5475.047		0.387952		-0.09	
(3 2 0 2)1 C	5644.867		0.391682		2.03	
(3 2 0 2)2 C	5435.288		0.391177		1.58	
(3 2 0 2)3 C	5245.316	144	0.391510	1	1.41	4
(3 2 0 2)4 C	5047.238		0.392876		1.10	
(0 2 2 2)1 C	5958.496	42	0.385588	32	1.34	0
(1 3 1 3)1 C	5730.635		0.389203		1.34	
(1 3 1 3)2 C	5531.391		0.389712		1.47	
(2 4 0 4)1 C	5590.574		0.392725		1.41	
(2 4 0 4)2 C	5346.129		0.392904		1.48	
(2 4 0 4)3 C	5115.711		0.393752		1.58	
(0 5 1 5)1 C	5627.332		0.390814		1.43	
(1 6 0 6)1 C	5480.574		0.394150		1.45	
(1 6 0 6)2 C	5213.672		0.394771		1.55	
(0 8 0 8)1 C	5349.580		0.395824		1.51	

TABLE X—*Continued*

16 12 16 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(2 1 1 1)1 D	5790.594	-15	0.388583	6	1.09	2
(2 1 1 1)2 D	5632.734	27	0.388124	-3	1.34	0
(2 1 1 1)3 D	5475.086	-15	0.389457	-37	1.74	-3
(3 2 0 2)1 D	5644.857		0.391682		1.16	
(3 2 0 2)2 D	5436.983		0.391177		1.29	
(3 2 0 2)3 D	5245.316	144	0.391510	1	1.57	3
(3 2 0 2)4 D	5047.238		0.392875		1.83	
(0 2 2 2)1 D	5958.496	42	0.385588	32	1.37	-5
(1 3 1 3)1 D	5730.695		0.389203		1.34	
(1 3 1 3)2 D	5531.391		0.389712		1.47	
(2 4 0 4)1 D	5580.574		0.392725		1.41	
(2 4 0 4)2 D	5346.129		0.392904		1.48	
(2 4 0 4)3 D	5115.711		0.393752		1.58	
(0 5 1 5)1 D	5627.332		0.390814		1.43	
(1 6 0 6)1 D	5480.574		0.394160		1.45	
(1 6 0 6)2 D	5219.672		0.394771		1.55	
(0 8 0 8)1 D	5349.580		0.395824		1.51	
(3 0 1 0)1	6503.078	3	0.387913	51	0.78	-6
(3 0 1 0)2	6347.871	-15	0.386447	3	1.01	-5
(3 0 1 0)3	6227.930	-15	0.386715	2	1.72	3
(3 0 1 0)4	6075.969	15	0.388935	-30	2.11	1
(0 0 3 0)1	6972.582	3	0.391005	-12	1.32	0
(4 1 0 1)1 C	6387.910	175	0.390307	-7	1.01	0
(4 1 0 1)2 C	6178.570		0.389580		1.14	
(4 1 0 1)3 C	6000.297		0.389158		1.37	
(4 1 0 1)4 C	5830.645		0.390060		1.72	
(4 1 0 1)5 C	5638.402		0.391802		1.94	
(1 1 2 1)1 C	6679.691	19	0.394301	14	1.24	0
(1 1 2 1)2 C	6537.988	-23	0.384810	7	1.48	1
(2 2 1 2)1 C	6474.602	-70	0.388570	79	1.50	0
(2 2 1 2)2 C	6238.492	3	0.388453	-1	1.41	-2
(2 2 1 2)3 C	6103.738	-54	0.389451	-26	1.26	3
(3 3 0 3)1 C	6346.926		0.392122		1.46	
(3 3 0 3)2 C	6113.344		0.391901		1.45	
(3 3 0 3)3 C	5902.582		0.392299		1.53	
(3 3 0 3)4 C	5683.687		0.393518		1.59	
(0 3 2 3)1 C	6601.676		0.386344		1.38	
(1 4 1 4)1 C	6398.238		0.389915		1.36	
(1 4 1 4)2 C	6175.879		0.390446		1.48	
(4 1 0 1)1 D	6387.910		0.392561		0.85	
(4 1 0 1)2 D	6178.570		0.391478		1.01	
(4 1 0 1)3 D	6000.297		0.390800		1.36	
(4 1 0 1)4 D	5830.645		0.391836		1.86	
(4 1 0 1)5 D	5638.402		0.393948		2.13	
(1 1 2 1)1 D	6679.691	19	0.385131	-3	1.21	-9
(1 1 2 1)2 D	6537.988	-23	0.385738	-16	1.55	3
(2 2 1 2)1 D	6474.602	-70	0.388570	79	1.18	3
(2 2 1 2)2 D	6238.492	3	0.388453	-1	1.41	-14
(2 2 1 2)3 D	6103.738	-54	0.389451	-26	1.55	2
(3 3 0 3)1 D	6346.926		0.392122		1.46	
(3 3 0 3)2 D	6113.344		0.391901		1.45	
(3 3 0 3)3 D	5902.582		0.392299		1.53	
(3 3 0 3)4 D	5683.687		0.393518		1.59	
(0 3 2 3)1 D	6601.676		0.386344		1.38	
(1 4 1 4)1 D	6398.238		0.389915		1.36	
(1 4 1 4)2 D	6175.879		0.390446		1.49	

TABLE X—*Continued*

16 12 16 O C O ATTRIBUTION		BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(5 0 0 0)1		7121.445		0.391301		-0.49	
(5 0 0 0)2		6903.207		0.390425		0.39	
(5 0 0 0)3		6725.102		0.388893		0.80	
(5 0 0 0)4		6588.730		0.388414		1.73	
(5 0 0 0)5		6435.398		0.390803		2.37	
(5 0 0 0)6		6239.852		0.393088		3.11	
(2 0 2 0)1		7377.687	-7	0.394353	39	0.98	
(2 0 2 0)2		7259.723		0.393537		1.39	
(2 0 2 0)3		7133.840	-19	0.395308	-18	1.80	-4
(3 1 1 1)1	C	7203.836	-7	0.387470	136	1.06	-1
(3 1 1 1)2	C	7023.629	50	0.396711	39	1.23	-3
(3 1 1 1)3	C	6863.508	50	0.386973	-41	1.43	0
(3 1 1 1)4	C	6698.164	15	0.388561	-16	1.75	0
(0 1 3 1)1	C	7602.527	-15	0.381504	-1	1.33	-1
(4 2 0 2)1	C	7102.074		0.391717		2.74	
(4 2 0 2)2	C	6874.547		0.391157		2.00	
(4 2 0 2)3	C	6670.113		0.390911		1.56	
(4 2 0 2)4	C	6472.738		0.391758		1.30	
(4 2 0 2)5	C	6261.898		0.393415		0.74	
(1 2 2 2)1	C	7338.152	7	0.385463	27	1.35	-1
(1 2 2 2)2	C	7166.099		0.386040		1.39	
(2 3 1 3)1	C	7154.703		0.389199		1.27	
(2 3 1 3)2	C	6944.781		0.389251		1.43	
(2 3 1 3)3	C	6736.906		0.390166		1.54	
(3 1 1 1)1	D	7203.836	-7	0.389070	50	0.99	-8
(3 1 1 1)2	D	7023.629	50	0.388093	24	1.18	-1
(3 1 1 1)3	D	6863.508	50	0.388388	-19	1.54	0
(3 1 1 1)4	D	6688.164	15	0.390299	-44	1.91	0
(0 1 3 1)1	D	7602.527	-15	0.382076	-7	1.35	0
(4 2 0 2)1	D	7102.078		0.391714		1.02	
(4 2 0 2)2	D	6874.547		0.391158		1.26	
(4 2 0 2)3	D	6670.113		0.390912		1.43	
(4 2 0 2)4	D	6472.738		0.391758		1.74	
(4 2 0 2)5	D	6261.898		0.393415		1.99	
(1 2 2 2)1	D	7338.152	7	0.385463	27	1.27	1
(1 2 2 2)2	D	7166.099		0.386040		1.56	
(2 3 1 3)1	D	7154.703		0.389199		1.26	
(2 3 1 3)2	D	6944.781		0.389251		1.43	
(2 3 1 3)3	D	6736.906		0.390166		1.54	
(4 0 1 0)1		7920.801	39	0.388358	191	0.53	2
(4 0 1 0)2		7734.410	50	0.396926	23	0.81	2
(4 0 1 0)3		7593.965	-265	0.395625	-42	1.20	-3
(4 0 1 0)4		7460.539	-7	0.387364	-15	2.03	-3
(4 0 1 0)5		7293.918	62	0.389640	-54	2.54	-3
(1 0 3 0)1		8293.922	35	0.390845	-45	1.15	0
(1 0 3 0)2		8192.578	-19	0.381590	-12	1.56	4
(5 1 0 1)1	C	7947.520		0.390040		0.86	
(5 1 0 1)2	C	7615.687		0.389572		1.12	
(5 1 0 1)3	C	7415.633		0.388796		1.20	
(5 1 0 1)4	C	7239.805		0.388871		1.56	
(5 1 0 1)5	C	7058.184		0.390323		1.89	
(5 1 0 1)6	C	6850.012		0.392194		2.15	
(2 1 2 1)1	C	8055.989	31	0.394341	59	1.15	-3
(2 1 2 1)2	C	* 7901.254		0.396225		45.14	
(2 1 2 1)3	C	7743.750	-50	0.385278	2	1.57	-31
(3 2 1 2)1	C	* 7897.932		0.386641		-42.00	
(3 2 1 2)2	C	7694.359	19	0.388227	-27	1.50	11
(3 2 1 2)3	C	7505.184	35	0.388609	23	1.38	-15
(3 2 1 2)4	C	7307.684	-35	0.390016	-14	1.05	-2
(0 2 3 2)1	C	8232.879	0	0.392570	29	1.33	-2
(4 3 0 3)1	C	7811.645		0.391952		1.77	
(4 3 0 3)2	C	7565.543		0.391703		1.49	
(4 3 0 3)3	C	7339.914		0.391740		1.50	
(4 3 0 3)4	C	7119.637		0.392527		1.59	
(4 3 0 3)5	C	6890.973		0.394012		1.60	
(1 3 2 3)1	C	7994.383		0.386205		1.32	
(1 3 2 3)2	C	7797.234		0.386765		1.46	

TABLE X—*Continued*

16 12 16 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(5 1 0 1)1 D	7847.520		0.392901		0.66	
(5 1 0 1)2 D	7615.687		0.391992		0.86	
(5 1 0 1)3 D	7415.633		0.390824		1.09	
(5 1 0 1)4 D	7239.805		0.390772		1.64	
(5 1 0 1)5 D	7058.184		0.392490		2.08	
(5 1 0 1)6 D	6850.012		0.394756		2.35	
(2 1 2 1)1 D	8055.988	31	0.385485	39	1.08	0
(2 1 2 1)2 D	* 7901.516		0.384174		-27.84	
(2 1 2 1)3 D	7743.750	-50	0.386560	-70	1.73	-19
(3 2 1 2)1 D	* 7897.570		0.389749		30.14	
(3 2 1 2)2 D	7694.359	19	0.388227	-27	1.27	-12
(3 2 1 2)3 D	7505.184	35	0.388609	23	1.57	-22
(3 2 1 2)4 D	7307.684	-35	0.390016	-14	1.82	0
(0 2 3 2)1 D	8232.879	0	0.382570	29	1.36	3
(4 3 0 3)1 D	7811.645		0.391952		1.78	
(4 3 0 3)2 D	7566.543		0.391703		1.49	
(4 3 0 3)3 D	7339.914		0.391740		1.50	
(4 3 0 3)4 D	7119.637		0.392527		1.59	
(4 3 0 3)5 D	6890.973		0.394012		1.60	
(1 3 2 3)1 D	7994.383		0.386205		1.32	
(1 3 2 3)2 D	7797.234		0.386765		1.46	
(6 0 0 0)1	+ 8584.980		0.390586		-11.45	
(6 0 0 0)2	8344.168		0.390632		-0.34	
(6 0 0 0)3	8137.539		0.389432		0.45	
(6 0 0 0)4	7974.918		0.387758		1.05	
(6 0 0 0)5	7835.324		0.388807		2.20	
(6 0 0 0)6	7659.559		0.391445		2.82	
(6 0 0 0)7	7447.215		0.393705		3.87	
(3 0 2 0)1	8756.707		0.384954		0.80	
(3 0 2 0)2	8607.133		0.393327		0.96	
(3 0 2 0)3	8483.387		0.393867		1.73	
(3 0 2 0)4	8334.297		0.386083		2.12	
(0 0 4 0)1	9246.937	3	0.377966	-41	1.32	1
(4 1 1 1)1 C	8628.660		0.397908		0.92	
(4 1 1 1)2 C	8424.867	132	0.396632	120	1.09	-8
(4 1 1 1)3 C	8250.637	7	0.396235	15	1.36	2
(4 1 1 1)4 C	8081.793		0.397185		1.66	
(4 1 1 1)5 C	7889.031		0.388939		1.82	
(1 1 3 1)1 C	8944.086	54	0.391255	-2	1.24	-1
(1 1 3 1)2 C	8803.332	-62	0.391862	-3	1.47	0
(5 2 0 2)1 C	8567.820		0.391464		4.52	
(5 2 0 2)2 C	8321.020		0.391107		2.59	
(5 2 0 2)3 C	8099.820		0.390672		1.85	
(5 2 0 2)4 C	7895.926		0.390837		1.59	
(5 2 0 2)5 C	7690.289		0.392120		1.06	
(5 2 0 2)6 C	7467.160		0.393968		0.22	
(2 2 2 2)1 C	8727.270		0.385561		1.43	
(2 2 2 2)2 C	8544.414		0.385487		1.37	
(2 2 2 2)3 C	* 8360.547		0.386441		0.62	
(3 3 1 3)1 C	+ 8586.909		0.390243		10.39	
(3 3 1 3)2 C	* 8362.887		0.389159		1.98	
(3 3 1 3)3 C	8149.961		0.389438		1.51	
(3 3 1 3)4 C	7932.375		0.390684		1.58	
(0 3 3 3)1 C	8863.648		0.383347		1.37	
(4 1 1 1)1 D	8628.660	11	0.389598	111	0.85	7
(4 1 1 1)2 D	8424.867	132	0.388435	82	1.00	-3
(4 1 1 1)3 D	8250.637	7	0.387825	-1	1.38	-2
(4 1 1 1)4 D	8081.793		0.388951		1.83	
(4 1 1 1)5 D	7889.031		0.391080		2.17	
(1 1 3 1)1 D	8944.086	54	0.382056	-24	1.21	-2
(1 1 3 1)2 D	8803.332	-62	0.382785	-41	1.54	1
(5 2 0 2)1 D	8567.820		0.391451		1.16	
(5 2 0 2)2 D	8321.020		0.391105		1.07	
(5 2 0 2)3 D	8099.820		0.390673		1.28	
(5 2 0 2)4 D	7895.926		0.390839		1.54	
(5 2 0 2)5 D	7690.289		0.392120		1.92	
(5 2 0 2)6 D	7467.160		0.393970		2.18	
(2 2 2 2)1 D	8727.270		0.385561		1.16	
(2 2 2 2)2 D	8544.414		0.385487		1.40	
(2 2 2 2)3 D	* 8360.547		0.386441		1.03	
(3 3 1 3)1 D	8586.930		0.389555		1.01	
(3 3 1 3)2 D	* 8362.887		0.389159		1.99	
(3 3 1 3)3 D	8149.961		0.389438		1.51	
(3 3 1 3)4 D	7932.375		0.390684		1.57	
(0 3 3 3)1 D	8863.648		0.383347		1.37	

TABLE X—*Continued*

16 12 16 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(5 0 1 0)1	9349.797		0.388943		0.06	
(5 0 1 0)2	9137.574	226	0.387534	252	0.55	10
(5 0 1 0)3	8965.578	347	0.385926	97	0.81	-6
(5 0 1 0)4	8832.098		0.385362		1.09	
(5 0 1 0)5	8676.545	70	0.388113	69	2.35	4
(5 0 1 0)6	8480.094		0.390385		3.19	
(2 0 3 0)1	9631.258	93	0.381255	-9	0.97	3
(2 0 3 0)2	9516.879	89	0.380568	-73	1.41	-1
(2 0 3 0)3	9389.055	-62	0.382422	-41	1.80	5
(6 1 0 1)1 C	9315.586		0.388930		0.97	
(6 1 0 1)2 C	9063.273		0.389284		0.90	
(6 1 0 1)3 C	8840.373		0.388920		1.74	
(6 1 0 1)4 C	8648.801		0.388090		1.34	
(6 1 0 1)5 C	8470.004		0.388770		1.70	
(6 1 0 1)6 C	8274.547		0.390638		2.08	
(6 1 0 1)7 C	8052.305		0.392578		2.38	
(3 1 2 1)1 C	9444.965		0.384533		1.04	
(3 1 2 1)2 C	9269.496		0.383726		1.21	
(3 1 2 1)3 C	* 9110.086		0.391048		83.63	
(3 1 2 1)4 C	8935.301		0.385712		1.75	
(0 1 4 1)1 C	9864.477	-31	0.378490	-25	1.33	0
(4 2 1 2)1 C	9330.086		0.389539		1.46	
(4 2 1 2)2 C	* 9109.039		0.381397		-80.30	
(4 2 1 2)3 C	8907.352		0.388026		1.50	
(4 2 1 2)4 C	8711.633		0.388930		1.27	
(4 2 1 2)5 C	8501.180		0.390617		0.71	
(1 2 3 2)1 C	9589.844	85	0.382441	9	1.32	6
(1 2 3 2)2 C	9419.305	-113	0.383084	-5	1.31	-10
(5 3 0 3)1 C	9283.590		0.392643		3.49	
(5 3 0 3)2 C	9021.766		0.391200		2.40	
(5 3 0 3)3 C	8781.285		0.391291		1.56	
(5 3 0 3)4 C	8554.502		0.391629		1.58	
(5 3 0 3)5 C	8328.195		0.392841		1.53	
(5 3 0 3)6 C	8089.930		0.394526		1.59	
(2 3 2 3)1 C	9394.637		0.386253		1.27	
(2 3 2 3)2 C	9188.090		0.386321		1.44	
(2 3 2 3)3 C	8981.762		0.387281		1.52	
(6 1 0 1)1 D	9315.590		0.392728		0.86	
(6 1 0 1)2 D	9063.273		0.392240		0.68	
(6 1 0 1)3 D	8840.369		0.391156		0.89	
(6 1 0 1)4 D	8648.801		0.390300		1.29	
(6 1 0 1)5 D	8470.004		0.391064		1.94	
(6 1 0 1)6 D	8274.547		0.393214		2.30	
(6 1 0 1)7 D	8052.305		0.395539		2.62	
(3 1 2 1)1 D	9444.965		0.385996		0.98	
(3 1 2 1)2 D	9269.496		0.385047		1.18	
(3 1 2 1)3 D	* 9110.203		0.391923		84.51	
(3 1 2 1)4 D	8935.301		0.387459		1.91	
(0 1 4 1)1 D	9864.477	-31	0.379049	-37	1.34	0
(4 2 1 2)1 D	9330.082		0.389552		0.19	
(4 2 1 2)2 D	* 9108.922		0.391953		-81.57	
(4 2 1 2)3 D	8907.352		0.388026		1.41	
(4 2 1 2)4 D	8711.633		0.388930		1.74	
(4 2 1 2)5 D	8501.190		0.390617		1.99	
(1 2 3 2)1 D	9589.844	85	0.382441	9	1.26	1
(1 2 3 2)2 D	9419.305	-113	0.383084	-5	1.49	13
(5 3 0 3)1 D	9283.586		0.392646		3.54	
(5 3 0 3)2 D	9021.766		0.391199		2.38	
(5 3 0 3)3 D	8781.295		0.391292		1.57	
(5 3 0 3)4 D	8554.502		0.391629		1.58	
(5 3 0 3)5 D	8328.195		0.392841		1.63	
(5 3 0 3)6 D	8089.930		0.394525		1.58	
(2 3 2 3)1 D	9394.637		0.386253		1.27	
(2 3 2 3)2 D	9188.090		0.386321		1.44	
(2 3 2 3)3 D	8981.762		0.387281		1.52	

TABLE X—*Continued*

16 13 16 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0	0	0.390240	1	1.33	2
(0 1 0 1)1 C	648.487	-3	0.390614	-9	1.35	0
(0 1 0 1)1 D	648.487	-3	0.391247	-10	1.36	0
(1 0 0 0)1	1370.088	-27	0.389707	14	1.20	1
(1 0 0 0)2	1265.834	-7	0.390918	2	1.58	2
(0 2 0 2)1 C	1297.270	0	0.391613	-2	1.30	-2
(0 2 0 2)1 D	1297.270	0	0.391613	-2	1.38	1
(0 0 1 0)1	2283.509	-23	0.387264	13	1.33	1
(1 1 0 1)1 C	2037.137	-37	0.390042	55	1.28	3
(1 1 0 1)2 C	1896.557	-17	0.390969		1.49	
(0 3 0 3)1 C	1946.358		0.392288	13	1.37	-7
(1 1 0 1)1 D	2037.137	-37	0.390904	56	1.24	0
(1 1 0 1)2 D	1896.557	-17	0.392028	13	1.56	1
(0 3 0 3)1 D	1946.358	-15	0.392288	13	1.37	-7
(2 0 0 0)1	2750.589		0.389701		1.01	
(2 0 0 0)2	2645.080		0.389692		1.52	
(2 0 0 0)3	2507.559		0.391689		1.89	
(0 1 1 1)1 C	2920.263	-19	0.387667	3	1.34	0
(1 2 0 2)1 C	2700.333		0.391169		1.32	
(1 2 0 2)2 C	2531.720	8	0.392132	-2	1.22	
(0 4 0 4)1 C	2595.759		0.392955		1.39	
(0 1 1 1)1 D	2920.263	-19	0.388282	3	1.35	1
(1 2 0 2)1 D	2700.333		0.391169		1.30	
(1 2 0 2)2 D	2531.720	8	0.392132		1.53	
(0 4 0 4)1 D	2595.759		0.392955		1.39	
(1 0 1 0)1	3632.939	-30	0.386720	7	1.21	0
(1 0 1 0)2	3527.750	-13	0.388029	12	1.57	2
(2 1 0 1)1 C	3433.733		0.389707		1.20	
(2 1 0 1)2 C	3289.700		0.389984		1.44	
(2 1 0 1)3 C	3127.371		0.391408		1.64	
(0 2 1 2)1 C	3557.328	- 9	0.388680	16	1.29	-3
(1 3 0 3)1 C	3361.527		0.391837		1.34	
(1 3 0 3)2 C	3169.493		0.392778		1.43	
(0 5 0 5)1 C	3245.482		0.393615		1.42	
(2 1 0 1)1 D	3433.733		0.390904		1.10	
(2 1 0 1)2 D	3289.700		0.391167		1.47	
(2 1 0 1)3 D	3127.371		0.392920		1.75	
(0 2 1 2)1 D	3557.328	-9	0.388680	16	1.37	1
(1 3 0 3)1 D	3361.527		0.391837		1.34	
(1 3 0 3)2 D	3169.493		0.392778		1.43	
(0 5 0 5)1 D	3245.482		0.393615		1.42	
(3 0 0 0)1	* 4145.898		0.388565		-26.07	
(3 0 0 0)2	4020.188		0.388843		1.25	
(3 0 0 0)3	3898.823		0.390319		1.84	
(3 0 0 0)4	3732.171		0.392418		2.37	
(0 0 2 0)1	4543.578	-28	0.384304	10	1.32	2
(1 1 1 1)1 C	4287.746	-36	0.387117	61	1.27	3
(1 1 1 1)2 C	* 4147.152		0.389477		28.33	
(2 2 0 2)1 C	4109.680		0.390913		1.43	
(2 2 0 2)2 C	3937.120		0.391336		1.29	
(2 2 0 2)3 C	3752.458		0.392730		0.97	
(0 3 1 3)1 C	4194.711		0.389379	11	1.36	-6
(1 4 0 4)1 C	4021.499		0.392486		1.37	
(1 4 0 4)2 C	3809.110		0.393425		1.46	
(0 6 0 6)1 C	3895.537		0.394266		1.44	

TABLE X—*Continued*

16 13 16 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 1 1)1 D	4287.746	-50	0.387929	55	1.24	0
(1 1 1 1)2 D	4147.258	-25	0.389143	21	1.55	2
(2 2 0 2)1 D	4109.680		0.390913		1.21	
(2 2 0 2)2 D	3937.120		0.391336		1.46	
(2 2 0 2)3 D	3752.458		0.392730		1.69	
(0 3 1 3)1 D	4194.711	-11	0.389379	11	1.36	-6
(1 4 0 4)1 D	4021.499		0.392486		1.37	
(1 4 0 4)2 D	3809.110		0.393425		1.46	
(0 6 0 6)1 D	3895.537		0.394266		1.44	
(2 0 1 0)1	4991.379	-27	0.386700	11	1.01	-4
(2 0 1 0)2	4887.387	3	0.386849	-1	1.52	-4
(2 0 1 0)3	4748.102	-39	0.388854	-5	1.88	-1
(3 1 0 1)1 C	4840.148		0.389491		1.13	
(3 1 0 1)2 C	4683.008	109	0.389213	22	1.33	0
(3 1 0 1)3 C	4527.285	-3	0.390200	19	1.61	
(3 1 0 1)4 C	4344.152		0.391856		1.83	
(0 1 2 1)1 C	5168.637	-35	0.384735	-39	1.34	0
(1 2 1 2)1 C	4938.898		0.388258		1.29	
(1 2 1 2)2 C	4771.035	-8	0.389251	-1	1.17	
(2 3 0 3)1 C	4781.738		0.391531		1.36	
(2 3 0 3)2 C	4585.715		0.392042		1.42	
(2 3 0 3)3 C	4381.160		0.393330		1.46	
(0 4 1 4)1 C	4832.422		0.390077		1.38	
(1 5 0 5)1 C	4680.672		0.393121		1.40	
(1 5 0 5)2 C	4450.168		0.394069		1.49	
(0 7 0 7)1 C	4545.934		0.394910		1.46	
(3 1 0 1)1 D	4840.148		0.391119		0.98	
(3 1 0 1)2 D	4683.008	109	0.390619		1.29	
(3 1 0 1)3 D	4527.285	-3	0.391787		1.71	
(3 1 0 1)4 D	4344.152		0.393829		1.96	
(0 1 2 1)1 D	5168.637	-35	0.385333	-28	1.35	-1
(1 2 1 2)1 D	4938.898		0.388258		1.28	
(1 2 1 2)2 D	4771.035	-8	0.389251	-1	1.50	
(2 3 0 3)1 D	4781.738		0.391531		1.36	
(2 3 0 3)2 D	4585.715		0.392042		1.42	
(2 3 0 3)3 D	4381.160		0.393330		1.46	
(0 4 1 4)1 D	4832.422		0.390077		1.38	
(1 5 0 5)1 D	4680.672		0.393121		1.40	
(1 5 0 5)2 D	4450.168		0.394069		1.49	
(0 7 0 7)1 D	4545.934		0.394910		1.46	
(4 0 0 0)1	5554.363		0.390152		0.53	
(4 0 0 0)2	5400.586		0.388776		0.89	
(4 0 0 0)3	5280.562		0.388803		1.70	
(4 0 0 0)4	5132.730		0.391037		2.23	
(4 0 0 0)5	4943.547		0.393117		3.06	
(1 0 2 0)1	5872.570	-11	0.383746	-10	1.21	-5
(1 0 2 0)2	5766.203	-19	0.385153	-3	1.56	-3
(2 1 1 1)1 C	5662.262	3	0.386829	7	1.17	-7
(2 1 1 1)2 C	5519.922	7	0.387071	-3	1.20	8
(2 1 1 1)3 C	5357.086	-74	0.388611	2	1.60	-1
(3 2 0 2)1 C	5526.367		0.390837		1.91	
(3 2 0 2)2 C	5343.734		0.390707		1.43	
(3 2 0 2)3 C	5161.859		0.391694		1.13	
(3 2 0 2)4 C	4961.406		0.393355		0.52	
(0 2 2 2)1 C	5794.016	-39	0.385759	14	1.28	0
(1 3 1 3)1 C	5588.137		0.388973		1.31	
(1 3 1 3)2 C	5397.363		0.389932		1.42	
(2 4 0 4)1 C	5451.293		0.392087		1.39	
(2 4 0 4)2 C	5235.031		0.392715		1.44	
(2 4 0 4)3 C	5012.578		0.393945		1.51	
(0 5 1 5)1 C	5470.469		0.390790		1.38	
(1 6 0 6)1 C	5339.309		0.393742		1.43	
(1 6 0 6)2 C	5092.422		0.394708		1.51	
(0 8 0 8)1 C	5196.676		0.395546		1.48	

TABLE X—*Continued*

16 13 16 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(2 1 1 1)1 D	5662.262	3	0.387924	11	1.09	-9
(2 1 1 1)2 D	5519.922	7	0.388228	-19	1.27	2
(2 1 1 1)3 D	5357.086	-74	0.390110	-15	1.72	-4
(3 2 0 2)1 D	5526.367		0.390839		1.32	
(3 2 0 2)2 D	5343.734		0.390708		1.37	
(3 2 0 2)3 D	5161.859		0.391694		1.63	
(3 2 0 2)4 D	4961.406		0.393355		1.87	
(0 2 2 2)1 D	5794.016	-39	0.385759	14	1.36	0
(1 3 1 3)1 D	5588.137		0.388973		1.31	
(1 3 1 3)2 D	5397.363		0.389933		1.42	
(2 4 0 4)1 D	5451.293		0.392087		1.39	
(2 4 0 4)2 D	5235.031		0.392715		1.44	
(2 4 0 4)3 D	5012.578		0.393945		1.51	
(0 5 1 5)1 D	5470.469		0.390790		1.38	
(1 6 0 6)1 D	5339.309		0.393742		1.43	
(1 6 0 6)2 D	5092.422		0.394708		1.51	
(0 8 0 8)1 D	5196.676		0.395546		1.48	
(3 0 1 0)1	6363.555	70	0.386982	40	0.85	3
(3 0 1 0)2	6242.047	-74	0.385865	-9	1.09	0
(3 0 1 0)3	6119.641	-19	0.387579	12	1.82	2
(3 0 1 0)4	5951.711	-101	0.389646	-6	2.36	-3
(0 0 3 0)1	6780.246	-27	0.381360	-12	1.32	0
(4 1 0 1)1 C	6256.285		0.389357		1.22	
(4 1 0 1)2 C	6080.617		0.388710		1.21	
(4 1 0 1)3 C	5922.648		0.389031		1.52	
(4 1 0 1)4 C	5750.551		0.390534		1.79	
(4 1 0 1)5 C	5549.301		0.392295		2.07	
(1 1 2 1)1 C	6515.145		0.384199		1.26	
(1 1 2 1)2 C	6374.555		0.385264		1.45	
(2 2 1 2)1 C	6326.051	-7	0.388054	10	1.35	-30
(2 2 1 2)2 C	6155.441	-19	0.388495	24	1.27	9
(2 2 1 2)3 C	5971.094	-132	0.389916	30	0.94	62
(3 3 0 3)1 C	6207.668		0.391247		1.41	
(3 3 0 3)2 C	6003.324		0.391436		1.41	
(3 3 0 3)3 C	5799.680		0.392376		1.47	
(3 3 0 3)4 C	5582.676		0.393912		1.45	
(0 3 2 3)1 C	6419.730		0.386481		1.35	
(1 4 1 4)1 C	6236.203		0.389704		1.31	
(1 4 1 4)2 C	6025.508		0.390623		1.44	
(4 1 0 1)1 D	6256.285		0.391390		0.86	
(4 1 0 1)2 D	6080.617		0.390488		1.08	
(4 1 0 1)3 D	5922.648		0.390761		1.58	
(4 1 0 1)4 D	5750.551		0.392564		1.93	
(4 1 0 1)5 D	5549.301		0.394729		2.22	
(1 1 2 1)1 D	6515.145		0.384968		1.23	
(1 1 2 1)2 D	6374.555		0.386270		1.54	
(2 2 1 2)1 D	6326.051	-7	0.388055	9	1.18	17
(2 2 1 2)2 D	6155.441	-19	0.388495	24	1.45	-14
(2 2 1 2)3 D	5971.094	-132	0.389916	30	1.68	-34
(3 3 0 3)1 D	6207.668		0.391247		1.41	
(3 3 0 3)2 D	6003.324		0.391436		1.41	
(3 3 0 3)3 D	5799.680		0.392376		1.47	
(3 3 0 3)4 D	5582.676		0.393912		1.45	
(0 3 2 3)1 D	6419.730		0.386481		1.35	
(1 4 1 4)1 D	6236.203		0.389704		1.31	
(1 4 1 4)2 D	6025.508		0.390623		1.44	

TABLE X—*Continued*

16 13 16 O C O ATTRIBUTION		BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(5 0 0 0)1		6973.535		0.390197		-0.09	
(5 0 0 0)2		6793.664		0.389083		0.58	
(5 0 0 0)3		6655.559		0.387806		1.24	
(5 0 0 0)4		6523.973		0.389376		2.13	
(5 0 0 0)5		6351.164		0.391712		2.80	
(5 0 0 0)6		6144.004		0.393795		4.02	
(2 0 2 0)1		7209.180		0.383704		1.01	
(2 0 2 0)2		7106.359		0.384015		1.52	
(2 0 2 0)3		6965.230		0.386029		1.88	
(3 1 1 1)1 C		7045.879	148	0.386750	84	1.06	-7
(3 1 1 1)2 C		6892.020	35	0.386383	15	1.33	-6
(3 1 1 1)3 C		6736.695	0	0.387379	-10	1.55	7
(3 1 1 1)4 C		6553.094	-140	0.389083	-30	1.79	-15
(0 1 3 1)1 C		7393.637	-31	0.381816	-10	1.33	-1
(4 2 0 2)1 C		6950.484		0.390628		2.31	
(4 2 0 2)2 C		6753.887		0.390365		1.65	
(4 2 0 2)3 C		6568.828		0.390779		1.31	
(4 2 0 2)4 C		6375.105		0.392162		0.79	
(4 2 0 2)5 C		6160.176		0.393982		-0.18	
(1 2 2 2)1 C		7154.277		0.385352		1.27	
(1 2 2 2)2 C	*	6987.613		0.383876		-21.32	
(2 3 1 3)1 C	*	6985.395		0.391287		23.65	
(2 3 1 3)2 C		6792.230		0.389259		1.38	
(2 3 1 3)3 C		6588.605		0.390561		1.45	
(3 1 1 1)1 D		7045.879	148	0.388170	65	0.96	-7
(3 1 1 1)2 D		6892.020	35	0.387709	18	1.24	1
(3 1 1 1)3 D		6736.695	0	0.388940	-33	1.68	-9
(3 1 1 1)4 D		6553.094	-140	0.391016	-61	1.98	-14
(0 1 3 1)1 D		7393.637	-31	0.382400	-9	1.34	1
(4 2 0 2)1 D		6950.488		0.390627		1.08	
(4 2 0 2)2 D		6753.887		0.390366		1.25	
(4 2 0 2)3 D		6568.828		0.390779		1.50	
(4 2 0 2)4 D		6375.105		0.392162		1.82	
(4 2 0 2)5 D		6160.172		0.393985		2.08	
(1 2 2 2)1 D		7154.277		0.385352		1.28	
(1 2 2 2)2 D	*	6987.613		0.383892		-20.86	
(2 3 1 3)1 D	*	6985.398		0.391271		23.52	
(2 3 1 3)2 D		6792.230		0.389258		1.37	
(2 3 1 3)3 D		6588.605		0.390561		1.45	
(4 0 1 0)1		7748.781		0.387406		0.67	
(4 0 1 0)2		7600.219	-89	0.385822	55	0.89	-5
(4 0 1 0)3	*	7481.836		0.385427		-1.31	
(4 0 1 0)4		7332.234		0.388442		2.17	
(4 0 1 0)5		7142.367		0.390228		3.48	
(1 0 3 0)1		8089.012	15	0.380788	-49	1.21	1
(1 0 3 0)2		7981.230	-42	0.382289	-10	1.56	1
(5 1 0 1)1 C		7681.043		0.388868		0.98	
(5 1 0 1)2 C	*	7485.680		0.389055		4.09	
(5 1 0 1)3 C		7316.316		0.388025		1.38	
(5 1 0 1)4 C		7150.367		0.389248		1.09	
(5 1 0 1)5 C		6961.258		0.390899		2.00	
(5 1 0 1)6 C		6744.551		0.392719		2.35	
(2 1 2 1)1 C		7867.777		0.383940		1.15	
(2 1 2 1)2 C	*	7726.887		0.384474		3.19	
(2 1 2 1)3 C		7563.426		0.385804		1.61	
(3 2 1 2)1 C	*	7719.961		0.387891		-0.42	
(3 2 1 2)2 C		7540.453		0.387917		1.37	
(3 2 1 2)3 C		7359.723		0.388932		1.14	
(3 2 1 2)4 C		7159.512		0.390649		0.71	
(0 2 3 2)1 C		8007.367	-58	0.382853	6	1.26	3
(4 3 0 3)1 C	*	7639.441		0.389984		3.25	
(4 3 0 3)2 C		7423.793		0.390665		1.81	
(4 3 0 3)3 C		7216.707		0.391645		1.55	
(4 3 0 3)4 C		7003.910		0.392796		1.49	
(4 3 0 3)5 C		6775.199		0.394505		1.41	
(1 3 2 3)1 C		7791.586		0.386103		1.29	
(1 3 2 3)2 C		7601.914		0.387095		1.40	

TABLE X—*Continued*

16 13 16 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(5 1 0 1)1 D	7681.043		0.391580		0.74	
(5 1 0 1)2 D	7485.676		0.390631		0.90	
(5 1 0 1)3 D	7316.316		0.390094		1.32	
(5 1 0 1)4 D	7150.367		0.391208		1.91	
(5 1 0 1)5 D	6961.258		0.393385		2.18	
(5 1 0 1)6 D	6744.551		0.395614		2.54	
(2 1 2 1)1 D	7867.777		0.384949		1.09	
(2 1 2 1)2 D	* 7726.895		0.385557		1.54	
(2 1 2 1)3 D	7563.426		0.387293		1.72	
(3 2 1 2)1 D	* 7719.957		0.387961		0.92	
(3 2 1 2)2 D	7540.449		0.387917		1.33	
(3 2 1 2)3 D	7359.723		0.388932		1.62	
(3 2 1 2)4 D	7159.512		0.390649		1.81	
(0 2 3 2)1 D	8007.367	-58	0.382853	6	1.36	-1
(4 3 0 3)1 D	* 7639.441		0.389984		3.23	
(4 3 0 3)2 D	7423.793		0.390665		1.81	
(4 3 0 3)3 D	7216.707		0.391645		1.55	
(4 3 0 3)4 D	7003.910		0.392796		1.48	
(4 3 0 3)5 D	6775.199		0.394505		1.40	
(1 3 2 3)1 D	7791.586		0.386103		1.29	
(1 3 2 3)2 D	7601.914		0.387095		1.40	
(6 0 0 0)1	8401.227		0.389929		-1.26	
(6 0 0 0)2	8199.273		0.389268		0.11	
(6 0 0 0)3	8034.437		0.387771		0.71	
(6 0 0 0)4	7903.734		0.387618		1.82	
(6 0 0 0)5	7750.125		0.390217		2.57	
(6 0 0 0)6	7557.187		0.392352		3.61	
(6 0 0 0)7	7335.141		0.394455		5.27	
(3 0 2 0)1	8558.480		0.384026		0.84	
(3 0 2 0)2	* 8442.914		0.374144		-77.40	
(3 0 2 0)3	8317.117		0.384830		1.80	
(3 0 2 0)4	8147.895		0.386874		2.37	
(0 0 4 0)1	8993.539	-125	0.378431	-39	1.31	
(4 1 1 1)1 C	* 8436.695		0.395900		79.40	
(4 1 1 1)2 C	8267.465		0.385970		1.14	
(4 1 1 1)3 C	8111.441		0.386261		1.51	
(4 1 1 1)4 C	7939.129		0.387834		1.78	
(4 1 1 1)5 C	7737.687		0.389448		2.06	
(1 1 3 1)1 C	8719.375	35	0.381290	-46	1.25	-24
(1 1 3 1)2 C	8578.473	-74	0.382411	-21	1.46	
(5 2 0 2)1 C	* 8381.500		0.389621		4.74	
(5 2 0 2)2 C	8169.281		0.389955		2.16	
(5 2 0 2)3 C	7975.441		0.389937		1.49	
(5 2 0 2)4 C	7784.129		0.391084		1.09	
(5 2 0 2)5 C	7577.859		0.392683		0.24	
(5 2 0 2)6 C	7350.039		0.394603		-1.11	
(2 2 2 2)1 C	8519.414		0.385178		1.29	
(2 2 2 2)2 C	8350.570		0.385654		1.24	
(2 2 2 2)3 C	8166.406		0.387103		0.91	
(3 3 1 3)1 C	* 8389.102		0.389478		0.09	
(3 3 1 3)2 C	8187.949		0.388865		1.26	
(3 3 1 3)3 C	7985.977		0.389774		1.11	
(3 3 1 3)4 C	7769.812		0.391108		1.41	
(0 3 3 3)1 C	8621.445		0.383596		1.34	
(4 1 1 1)1 D	8438.762		0.388557		0.84	
(4 1 1 1)2 D	8267.465		0.387577		1.07	
(4 1 1 1)3 D	8111.441		0.387940		1.58	
(4 1 1 1)4 D	7939.129		0.389789		1.94	
(4 1 1 1)5 D	7737.687		0.391975		2.24	
(1 1 3 1)1 D	8719.375	35	0.382020	-30	1.23	8
(1 1 3 1)2 D	8578.473	-74	0.383411	43	1.53	
(5 2 0 2)1 D	* 8381.500		0.389624		2.28	
(5 2 0 2)2 D	8169.281		0.389954		1.19	
(5 2 0 2)3 D	7975.441		0.389937		1.36	
(5 2 0 2)4 D	7784.129		0.391084		1.74	
(5 2 0 2)5 D	7577.855		0.392684		2.02	
(5 2 0 2)6 D	7350.035		0.394609		2.33	
(2 2 2 2)1 D	8519.414		0.385178		1.16	
(2 2 2 2)2 D	8350.570		0.385654		1.43	
(2 2 2 2)3 D	8166.406		0.387103		1.66	
(3 3 1 3)1 D	* 8389.102		0.389467		-0.11	
(3 3 1 3)2 D	8187.949		0.388866		1.27	
(3 3 1 3)3 D	7985.977		0.389773		1.10	
(3 3 1 3)4 D	7769.812		0.391108		1.40	
(0 3 3 3)1 D	8621.445		0.383596		1.34	

TABLE X—*Continued*

18 12 18 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(0 0 0 0)1	0.0	0	0.346818	0	1.05	0
(0 1 0 1)1 C	657.333	-2	0.347231	-10	1.07	-2
(0 1 0 1)1 D	657.333	-2	0.347725	-9	1.07	-2
(1 0 0 0)1	1347.093	6	0.347399	7	0.92	3
(1 0 0 0)2	1230.356	-24	0.346519	9	1.17	3
(0 2 0 2)1 C	1315.077		0.348131		1.14	
(0 2 0 2)1 D	1315.077		0.348131		1.09	
(0 0 1 0)1	2314.059	-5	0.344080	14	1.05	4
(1 1 0 1)1 C	2024.552		0.347458		1.00	
(1 1 0 1)2 C	1868.747		0.347004		1.15	
(0 3 0 3)1 C	1973.243		0.348778		1.12	
(1 1 0 1)1 D	2024.552		0.348284		0.98	
(1 1 0 1)2 D	1868.747		0.347672		1.19	
(0 3 0 3)1 D	1973.243		0.348778		1.12	
(2 0 0 0)1	2721.962		0.347970		0.78	
(2 0 0 0)2	2560.174		0.346678		0.93	
(2 0 0 0)3	2447.087		0.346592		1.35	
(0 1 1 1)1 C	2959.137		0.344514		1.06	
(1 2 0 2)1 C	2698.097		0.348412		1.22	
(1 2 0 2)2 C	2511.869		0.348072		1.19	
(0 4 0 4)1 C	2631.839		0.349417		1.14	
(0 1 1 1)1 D	2959.137		0.344997		1.07	
(1 2 0 2)1 D	2698.097		0.348413		1.02	
(1 2 0 2)2 D	2511.869		0.348072		1.18	
(0 4 0 4)1 D	2631.839		0.349417		1.14	
(1 0 1 0)1	3638.062	4	0.344639	-15	0.92	0
(1 0 1 0)2	3525.228	-22	0.343859	0	1.18	0
(2 1 0 1)1 C	3411.584		0.347712		0.95	
(2 1 0 1)2 C	3224.134		0.346944		1.04	
(2 1 0 1)3 C	3068.772		0.346962		1.25	
(0 2 1 2)1 C	3604.642		0.345425		1.12	
(1 3 0 3)1 C	3369.625		0.348976		1.11	
(1 3 0 3)2 C	3157.843		0.348770		1.20	
(0 5 0 5)1 C	3290.875		0.350050		1.16	
(2 1 0 1)1 D	3411.584		0.348907		0.90	
(2 1 0 1)2 D	3224.134		0.347884		1.03	
(2 1 0 1)3 D	3068.772		0.347848		1.34	
(0 2 1 2)1 D	3604.642		0.345425		1.08	
(1 3 0 3)1 D	3369.625		0.348976		1.11	
(1 3 0 3)2 D	3157.843		0.348770		1.20	
(0 5 0 5)1 D	3290.875		0.350050		1.16	
(3 0 0 0)1	4116.266		0.348430		0.51	
(3 0 0 0)2	3917.165		0.347317		0.78	
(3 0 0 0)3	3769.624		0.345999		1.03	
(3 0 0 0)4	3648.692		0.346966		1.54	
(0 0 2 0)1	4603.715		0.341358		1.05	
(1 1 1 1)1 C	4303.383	7	0.344741	14	0.99	-1
(1 1 1 1)2 C	4151.266	-39	0.344351	-9	1.15	-2
(2 2 0 2)1 C	4096.637		0.348741		1.43	
(2 2 0 2)2 C	3885.483		0.348102		1.24	
(2 2 0 2)3 C	3698.847		0.348151		1.23	
(0 3 1 3)1 C	4250.586		0.346088		1.12	
(1 4 0 4)1 C	4039.952		0.349548		1.13	
(1 4 0 4)2 C	3805.872		0.349446		1.21	
(0 6 0 6)1 C	3950.362		0.350677		1.18	

TABLE X—*Continued*

18 12 18 O C O ATTRIBUTION		BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 1 1)1	D	4303.383	7	0.345535	12	0.97	-1
(1 1 1 1)2	D	4151.266	-39	0.345016	-6	1.19	2
(2 2 0 2)1	D	4096.637		0.348741		0.96	
(2 2 0 2)2	D	3885.483		0.348102		1.08	
(2 2 0 2)3	D	3698.847		0.348151		1.28	
(0 3 1 3)1	D	4250.586		0.346088		1.12	
(1 4 0 4)1	D	4039.952		0.349548		1.13	
(1 4 0 4)2	D	3805.872		0.349446		1.21	
(0 6 0 6)1	D	3950.362		0.350677		1.18	
(2 0 1 0)1		4989.281	-46	0.345233	-23	0.79	-3
(2 0 1 0)2		4833.270	-7	0.343940	-5	0.92	-7
(2 0 1 0)3		4721.930	-7	0.344026	-8	1.36	-2
(3 1 0 1)1	C	4814.637		0.347904		0.87	
(3 1 0 1)2	C	4597.465		0.347126		0.96	
(3 1 0 1)3	C	4417.687		0.346516		1.11	
(3 1 0 1)4	C	4257.246		0.347071		1.36	
(0 1 2 1)1	C	5236.574		0.341811		1.06	
(1 2 1 2)1	C	4964.672		0.345705		1.19	
(1 2 1 2)2	C	* 4782.145		0.345913		12.33	
(2 3 0 3)1	C	* 4778.766		0.348740		-9.93	
(2 3 0 3)2	C	4546.160		0.348760		1.16	
(2 3 0 3)3	C	4333.801		0.348863		1.27	
(3 1 0 1)1	D	4814.637		0.349491		0.81	
(3 1 0 1)2	D	4597.465		0.348423		0.92	
(3 1 0 1)3	D	4417.687		0.347590		1.13	
(3 1 0 1)4	D	4257.246		0.348211		1.48	
(0 1 2 1)1	D	5236.574		0.342284		1.07	
(1 2 1 2)1	D	4964.672	-58	0.345705	28	1.01	
(1 2 1 2)2	D	* 4782.145		0.345912		12.32	
(2 3 0 3)1	D	* 4778.766		0.348740		-9.92	
(2 3 0 3)2	D	4546.160		0.348760		1.16	
(2 3 0 3)3	D	4333.801		0.348863		1.27	
(4 0 0 0)1		5525.477		0.348810		0.07	
(4 0 0 0)2		5296.687		0.347835		0.53	
(4 0 0 0)3		5109.762		0.346535		0.76	
(4 0 0 0)4		4973.250		0.345632		1.27	
(4 0 0 0)5		4835.805		0.347459		1.71	
(1 0 2 0)1		5904.855		0.341887		0.91	
(1 0 2 0)2		5795.641		0.341219		1.18	
(2 1 1 1)1	C	5666.859	-109	0.345027	36	0.94	-9
(2 1 1 1)2	C	5484.602	-11	0.344260	-12	1.03	-4
(2 1 1 1)3	C	5331.574	-15	0.344377	-21	1.25	0
(3 2 0 2)1	C	5508.523		0.349050		1.82	
(3 2 0 2)2	C	5272.777		0.348318		1.43	
(3 2 0 2)3	C	5066.457		0.347879		1.26	
(3 2 0 2)4	C	4876.160		0.348345		1.23	
(0 2 2 2)1	C	5869.871		0.342733		1.11	
(2 1 1 1)1	D	5666.859	-109	0.346163	21	0.90	-11
(2 1 1 1)2	D	5484.602	-11	0.345175	0	1.02	-4
(2 1 1 1)3	D	5331.574	-15	0.345268	-28	1.34	1
(3 2 0 2)1	D	5508.523		0.349050		0.89	
(3 2 0 2)2	D	5272.777		0.348318		0.99	
(3 2 0 2)3	D	5066.457		0.347879		1.16	
(3 2 0 2)4	D	4876.160		0.348345		1.39	
(0 2 2 2)1	D	5869.871		0.342733		1.08	

TABLE X—*Continued*

18 13 18 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0		0.346835		1.05	
(0 1 0 1)1 C	638.128		0.347207		1.07	
(0 1 0 1)1 D	638.128		0.347715		1.07	
(1 0 0 0)1	1318.718	-11	0.347016	5	0.91	-1
(1 0 0 0)2	1220.414	-20	0.346870	1	1.22	0
(0 2 0 2)1 C	1276.603		0.348081		1.10	
(0 2 0 2)1 D	1276.603		0.348081		1.09	
(0 0 1 0)1	2247.316	-15	0.344198	9	1.05	0
(1 1 0 1)1 C	1976.963		0.347150		1.00	
(1 1 0 1)2 C	1839.410		0.347190		1.17	
(0 3 0 3)1 C	1915.435		0.348694		1.11	
(1 1 0 1)1 D	1976.963		0.347942		0.96	
(1 1 0 1)2 D	1839.410		0.347938		1.22	
(0 3 0 3)1 D	1915.435		0.348694		1.11	
(2 0 0 0)1	2660.248		0.347454		0.78	
(2 0 0 0)2	2531.475		0.346343		1.00	
(2 0 0 0)3	2422.172		0.347297		1.40	
(0 1 1 1)1 C	2873.972	-53	0.344592		1.06	
(1 2 0 2)1 C	2630.763		0.348102		1.15	
(1 2 0 2)2 C	2463.540		0.348218		1.13	
(0 4 0 4)1 C	2554.632		0.349301		1.13	
(0 1 1 1)1 D	2873.972		0.345088		1.07	
(1 2 0 2)1 D	2630.763		0.348102		1.01	
(1 2 0 2)2 D	2463.540		0.348218		1.19	
(0 4 0 4)1 D	2554.632		0.349301		1.13	
(1 0 1 0)1	3545.058	-29	0.344346	9	0.90	-1
(1 0 1 0)2	3448.449		0.344322		1.22	

15 14 16 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0		0.390256		1.33	
(0 1 0 1)1 C	631.897		0.390588		1.35	
(0 1 0 1)1 D	631.897		0.391237		1.36	
(1 0 0 0)1	1359.711		0.389415		1.26	
(1 0 0 0)2	1243.129		0.391176		1.60	
(0 2 0 2)1 C	1264.035		0.391562		1.21	
(0 2 0 2)1 D	1264.035		0.391561		1.37	
(0 0 1 0)1	2225.831	-15	0.387381	- 9	1.33	0
(1 1 0 1)1 C	2006.671		0.389771		1.30	
(1 1 0 1)2 C	1860.599		0.391113		1.50	
(0 3 0 3)1 C	1896.422		0.392203		1.33	
(1 1 0 1)1 D	2006.671		0.390587		1.27	
(1 1 0 1)2 D	1860.599		0.392269		1.55	
(0 3 0 3)1 D	1896.422		0.392203		1.33	

TABLE XI

16 12 18 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(0 0 0 0)1	0.0	0	0.368186	-7	1.19	-2
(0 1 0 1)1 C	662.377	-9	0.368604	-5	1.20	4
(0 1 0 1)1 D	662.377	-9	0.369156	15	1.21	3
(1 0 0 0)1	1365.845		0.368507		1.02	
(1 0 0 0)2	1259.444		0.368119		1.36	
(0 2 0 2)1 C	1325.141		0.369567		1.26	
(0 2 0 2)1 D	1325.141		0.369567		1.23	
(0 0 1 0)1	2332.121	-9	0.365277	7	1.18	-1
(1 1 0 1)1 C	2049.362		0.368622		1.12	
(1 1 0 1)2 C	1901.764		0.368522		1.31	
(0 3 0 3)1 C	1988.302		0.370246		1.26	
(1 1 0 1)1 D	2049.362		0.369500		1.09	
(1 1 0 1)2 D	1901.764		0.369316		1.37	
(0 3 0 3)1 D	1988.302		0.370246		1.26	
(2 0 0 0)1	2757.211	18	0.369024	15	0.88	-11
(2 0 0 0)2	2614.227	8	0.367727	79	1.10	-5
(2 0 0 0)3	2500.782	-52	0.368497	-34	1.58	-13
(0 1 1 1)1 C	2982.116	-11	0.365719	-12	1.20	-6
(1 2 0 2)1 C	2728.309		0.369661		1.32	
(1 2 0 2)2 C	2549.422		0.369658		1.30	
(0 4 0 4)1 C	2651.869		0.370917		1.28	
(0 1 1 1)1 D	2982.116		0.366258		1.21	
(1 2 0 2)1 D	2728.309		0.369661		1.14	
(1 2 0 2)2 D	2549.422		0.369658		1.34	
(0 4 0 4)1 D	2651.869		0.370917		1.28	
(1 0 1 0)1	3675.131	-1	0.365563	2	1.02	-3
(1 0 1 0)2	3571.151	-7	0.365307	1	1.37	0
(2 1 0 1)1 C	3453.970		0.368758		1.06	
(2 1 0 1)2 C	3280.977		0.368134		1.19	
(2 1 0 1)3 C	3127.357		0.368646		1.43	
(0 2 1 2)1 C	3632.519		0.366693		1.24	
(1 3 0 3)1 C	3405.000		0.370273		1.24	
(1 3 0 3)2 C	3200.123		0.370371		1.34	
(0 5 0 5)1 C	3315.853		0.371582		1.30	
(2 1 0 1)1 D	3453.970		0.370024		1.00	
(2 1 0 1)2 D	3280.977		0.369162		1.18	
(2 1 0 1)3 D	3127.357		0.369735		1.54	
(0 2 1 2)1 D	3632.519		0.366693		1.22	
(1 3 0 3)1 D	3405.000		0.370273		1.24	
(1 3 0 3)2 D	3200.123		0.370371		1.34	
(0 5 0 5)1 D	3315.853		0.371582		1.30	
(3 0 0 0)1	4167.996	-89	0.369440	32	0.62	5
(3 0 0 0)2	3987.509	85	0.368195	36	0.87	9
(3 0 0 0)3	3856.779	-122	0.367282	123	1.34	-9
(3 0 0 0)4	3724.401		0.369062		1.78	
(0 0 2 0)1	4639.512	-7	0.362384	6	1.18	1
(1 1 1 1)1 C	4346.199		0.365737		1.11	
(1 1 1 1)2 C	4201.180		0.365712		1.31	
(2 2 0 2)1 C	4145.066		0.369866		1.53	
(2 2 0 2)2 C	3946.170		0.369420		1.34	
(2 2 0 2)3 C	3762.294		0.369909		1.30	
(0 3 1 3)1 C	4283.332		0.367392		1.25	
(1 4 0 4)1 C	4080.363		0.370887		1.26	
(1 4 0 4)2 C	3852.958		0.371065		1.36	
(0 6 0 6)1 C	3980.262		0.372239		1.32	

TABLE XI—*Continued*

16 12 18 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 1 1)1 D	4346.199		0.366576		1.08	
(1 1 1 1)2 D	4201.180		0.366502		1.37	
(2 2 0 2)1 D	4145.066		0.369866		1.07	
(2 2 0 2)2 D	3946.170		0.369420		1.23	
(2 2 0 2)3 D	3762.294		0.369900		1.47	
(0 3 1 3)1 D	4283.332		0.367392		1.25	
(1 4 0 4)1 D	4080.363		0.370887		1.26	
(1 4 0 4)2 D	3852.958		0.371065		1.36	
(0 6 0 6)1 D	3980.262		0.372239		1.32	
(2 0 1 0)1	5042.613	-31	0.366105	27	0.88	-2
(2 0 1 0)2	4904.855	-3	0.364841	3	1.10	0
(2 0 1 0)3	4791.266	-3	0.365773	-15	1.58	0
(3 1 0 1)1 C	4873.812		0.368863		0.98	
(3 1 0 1)2 C	4673.160		0.368108		1.08	
(3 1 0 1)3 C	4504.797		0.367828		1.31	
(3 1 0 1)4 C	4339.734		0.368895		1.56	
(0 1 2 1)1 C	5277.164	-7	0.362849	23	1.19	5
(1 2 1 2)1 C	5012.668		0.366789		1.28	
(1 2 1 2)2 C	* 4836.621		0.366678		-3.67	
(2 3 0 3)1 C	* 4832.684		0.370553		6.18	
(2 3 0 3)2 C	4611.094		0.370132		1.30	
(2 3 0 3)3 C	4402.133		0.370603		1.41	
(3 1 0 1)1 D	4873.812		0.370554		0.90	
(3 1 0 1)2 D	4673.160		0.369489		1.02	
(3 1 0 1)3 D	4504.797		0.369051		1.36	
(3 1 0 1)4 D	4339.734		0.370312		1.70	
(0 1 2 1)1 D	5277.164	-7	0.363376	18	1.20	0
(1 2 1 2)1 D	5012.668		0.366789		1.14	
(1 2 1 2)2 D	* 4836.621		0.366677		-3.64	
(2 3 0 3)1 D	* 4832.684		0.370554		6.20	
(2 3 0 3)2 D	4611.094		0.370132		1.30	
(2 3 0 3)3 D	4402.133		0.370603		1.41	
(4 0 0 0)1	5593.484		0.369772		0.17	
(4 0 0 0)2	5382.629		0.368709		0.64	
(4 0 0 0)3	5216.410		0.367236		0.91	
(4 0 0 0)4	5088.555		0.367386		1.67	
(4 0 0 0)5	4933.035		0.369641		2.04	
(1 0 2 0)1	5959.949	7	0.362628	2	1.01	0
(1 0 2 0)2	5858.047	-19	0.362515	-13	1.37	1
(2 1 1 1)1 C	5727.102	-54	0.365911	43	1.04	-5
(2 1 1 1)2 C	5558.562	7	0.365292	25	1.17	6
(2 1 1 1)3 C	5406.070	3	0.365909	-9	1.43	-8
(3 2 0 2)1 C	5574.133		0.370090		1.93	
(3 2 0 2)2 C	5353.137		0.369437		1.51	
(3 2 0 2)3 C	5156.258		0.369319		1.36	
(3 2 0 2)4 C	4964.309		0.370248		1.22	
(0 2 2 2)1 C	5915.234		0.363834		1.23	
(2 1 1 1)1 D	5727.102	-54	0.367103	57	0.99	4
(2 1 1 1)2 D	5558.562	7	0.366293	16	1.17	6
(2 1 1 1)3 D	5406.070	3	0.366998	-6	1.54	5
(3 2 0 2)1 D	5574.133		0.370090		1.01	
(3 2 0 2)2 D	5353.137		0.369437		1.12	
(3 2 0 2)3 D	5156.258		0.369319		1.35	
(3 2 0 2)4 D	4964.309		0.370247		1.60	
(0 2 2 2)1 D	5915.234		0.363834		1.22	
(3 0 1 0)1	6429.281	-105	0.366570	63	0.65	-2
(3 0 1 0)2	6254.617	-23	0.365270	15	0.86	0
(3 0 1 0)3	6127.832	-50	0.364520	-1	1.37	3
(3 0 1 0)4	5993.535	46	0.366390	-34	1.79	3
(0 0 3 0)1	6922.211	0	0.359507	-14	1.18	-1
(4 1 0 1)1 C	6306.273		0.368904		0.86	
(4 1 0 1)2 C	6080.066		0.368182		0.99	
(4 1 0 1)3 C	5888.066		0.367494		1.15	
(4 1 0 1)4 C	5719.332		0.367740		1.46	
(4 1 0 1)5 C	5540.121		0.369203		1.68	
(1 1 2 1)1 C	6618.559		0.362862		1.11	
(1 1 2 1)2 C	6475.863	-85	0.362918	93	1.31	

TABLE XI—*Continued*

16 12 18 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(4 1 0 1)1 D	6306.273		0.371038		0.76	
(4 1 0 1)2 D	6080.066		0.369985		0.91	
(4 1 0 1)3 D	5888.066		0.369000		1.11	
(4 1 0 1)4 D	5719.332		0.369216		1.57	
(4 1 0 1)5 D	5540.121		0.370964		1.85	
(1 1 2 1)1 D	6618.559		0.363664		1.07	
(1 1 2 1)2 D	6475.863	-85	0.363704	38	1.37	
(5 0 0 0)1	7030.805		0.369996		-0.62	
(5 0 0 0)2	6794.527		0.369107		0.23	
(5 0 0 0)3	* 6595.094		0.368868		19.04	
(5 0 0 0)4	6443.242		0.366432		1.16	
(5 0 0 0)5	6306.340		0.367876		1.96	
(5 0 0 0)6	6129.289		0.370200		2.39	
(2 0 2 0)1	7303.680		0.363189		0.88	
(2 0 2 0)2	7171.031		0.361971		1.12	
(2 0 2 0)3	7056.887		0.363065		1.58	
(3 1 1 1)1 C	7122.836		0.366105		0.96	
(3 1 1 1)2 C	6927.586		0.365284		1.07	
(3 1 1 1)3 C	6762.641		0.365030		1.21	
(3 1 1 1)4 C	* 6597.797		0.365188		-16.82	
(0 1 3 1)1 C	7547.551	-15	0.359993	25	1.19	
(3 1 1 1)1 D	7122.836		0.367664		0.90	
(3 1 1 1)2 D	6927.586		0.366600		1.02	
(3 1 1 1)3 D	6762.645		0.366246		1.28	
(3 1 1 1)4 D	6597.621	82	0.367639	-69	1.69	-24
(0 1 3 1)1 D	7547.551	-15	0.360509	11	1.20	
(4 0 1 0)1	7830.445		0.367010		0.35	
(4 0 1 0)2	7625.559		0.365839		0.66	
(4 0 1 0)3	7465.297		0.364324		0.89	
(4 0 1 0)4	7338.871		0.364760		1.69	
(4 0 1 0)5	7181.039		0.367008		2.05	
(1 0 3 0)1	8220.352	11	0.359705	-22	1.01	0
(1 0 3 0)2	8120.152	-42	0.359743	-38	1.38	2

16 12 17 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0	0	0.378616	21	1.25	7
(0 1 0 1)1 C	664.731		0.379035		1.27	
(0 1 0 1)1 D	664.731		0.379617		1.28	
(1 0 0 0)1	1376.030		0.378772		1.08	
(1 0 0 0)2	1272.299		0.378698		1.46	
(0 2 0 2)1 C	1329.836		0.380029		1.31	
(0 2 0 2)1 D	1329.836		0.380029		1.30	
(0 0 1 0)1	2340.021	- 8	0.375621	34	1.25	7
(1 1 0 1)1 C	2062.120		0.378934		1.19	
(1 1 0 1)2 C	1916.712		0.379041		1.40	
(0 3 0 3)1 C	1995.325		0.380724		1.32	
(1 1 0 1)1 D	2062.120		0.379833		1.15	
(1 1 0 1)2 D	1916.712		0.379905		1.46	
(0 3 0 3)1 D	1995.325		0.380724		1.32	

TABLE XI—*Continued*

15 12 17 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(2 0 0 0)1	2775.610		0.379248		0.93	
(2 0 0 0)2	2641.218		0.378029		1.21	
(2 0 0 0)3	2524.266		0.379212		1.68	
(0 1 1 1)1 C	2992.307	3	0.376066	- 27	1.27	
(1 2 0 2)1 C	2743.463		0.380018		1.37	
(1 2 0 2)2 C	2566.615		0.380209		1.35	
(0 4 0 4)1 C	2661.207		0.381411		1.35	
(0 1 1 1)1 D	2992.307	3	0.376633	6	1.28	
(1 2 0 2)1 D	2743.463		0.380018		1.21	
(1 2 0 2)2 D	2566.615		0.380208		1.43	
(0 4 0 4)1 D	2661.207		0.381411		1.35	
(1 0 1 0)1	3693.354		0.375739		1.08	
(1 0 1 0)2	3591.273		0.375807		1.46	
(2 1 0 1)1 C	3475.671		0.378996		1.12	
(2 1 0 1)2 C	3308.708		0.378488		1.28	
(2 1 0 1)3 C	3153.733		0.379248		1.53	
(0 2 1 2)1 C	3644.983		0.377072		1.30	
(1 3 0 3)1 C	3422.485		0.380656		1.30	
(1 3 0 3)2 C	3219.600		0.380925		1.41	
(0 5 0 5)1 C	3327.493		0.382090		1.37	
(2 1 0 1)1 D	3475.671		0.380289		1.04	
(2 1 0 1)2 D	3308.708		0.379563		1.28	
(2 1 0 1)3 D	3153.733		0.380447		1.64	
(0 2 1 2)1 D	3644.983		0.377072		1.29	
(1 3 0 3)1 D	3422.485		0.380656		1.30	
(1 3 0 3)2 D	3219.600		0.380925		1.41	
(0 5 0 5)1 D	3327.493		0.382090		1.37	
(3 0 0 0)1	4194.500		0.379643		0.68	
(3 0 0 0)2	4023.333	85	0.378317	33	0.93	10
(3 0 0 0)3	3898.265		0.377796		1.51	
(3 0 0 0)4	3757.844		0.379839		1.92	
(0 0 2 0)1	4655.211	-7	0.372643	7	1.25	0
(1 1 1 1)1 C	4366.812		0.375967		1.18	
(1 1 1 1)2 C	4223.441		0.376152		1.39	
(2 2 0 2)1 C	4169.508		0.390147		1.56	
(2 2 0 2)2 C	3975.506		0.379832		1.39	
(2 2 0 2)3 C	3791.221		0.380537		1.31	
(0 3 1 3)1 C	4298.059		0.377788		1.31	
(1 4 0 4)1 C	4100.145		0.381291		1.32	
(1 4 0 4)2 C	3874.730		0.381627		1.43	
(0 6 0 6)1 C	3994.191		0.382762		1.39	
(1 1 1 1)1 D	4366.812		0.376823		1.14	
(1 1 1 1)2 D	4223.441		0.377009		1.45	
(2 2 0 2)1 D	4169.508		0.380147		1.13	
(2 2 0 2)2 D	3975.506		0.379832		1.31	
(2 2 0 2)3 D	3791.221		0.380537		1.56	
(0 3 1 3)1 D	4298.059		0.377788		1.31	
(1 4 0 4)1 D	4100.145		0.381291		1.32	
(1 4 0 4)2 D	3874.730		0.381627		1.43	
(0 6 0 6)1 D	3994.191		0.382762		1.39	
(2 0 1 0)1	5068.945	-15	0.376237	26	0.93	-2
(2 0 1 0)2	4939.348	3	0.375081	37	1.22	-1
(2 0 1 0)3	4821.520	-3	0.376400	-5	1.68	3
(3 1 0 1)1 C	4903.891		0.379051		1.04	
(3 1 0 1)2 C	4710.891		0.378325		1.15	
(3 1 0 1)3 C	4546.227		0.378274		1.42	
(3 1 0 1)4 C	4376.910		0.379558		1.66	
(0 1 2 1)1 C	5295.090	11	0.373112	12	1.26	
(1 2 1 2)1 C	5035.555		0.377065		1.33	
(1 2 1 2)2 C	* 4860.965		0.378881		25.75	
(2 3 0 3)1 C	* 4859.789		0.379156		-22.97	
(2 3 0 3)2 C	4642.324		0.380568		1.37	
(2 3 0 3)3 C	4433.512		0.381232		1.48	

TABLE XI—*Continued*

16 12 17 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(3 1 0 1)1 D	4903.891		0.380784		0.94	
(3 1 0 1)2 D	4710.891		0.379743		1.09	
(3 1 0 1)3 D	4546.227		0.379586		1.48	
(3 1 0 1)4 D	4376.910		0.381122		1.80	
(0 1 2 1)1 D	5295.090	11	0.373666	1	1.27	
(1 2 1 2)1 D	5035.555		0.377065		1.20	
(1 2 1 2)2 D	* 4860.965		0.378887		25.93	
(2 3 0 3)1 D	* 4859.789		0.379150		-23.05	
(2 3 0 3)2 D	4642.324		0.380568		1.37	
(2 3 0 3)3 D	4433.512		0.381232		1.48	
(4 0 0 0)1	5628.016		0.379952		0.24	
(4 0 0 0)2	5425.977		0.378818		0.71	
(4 0 0 0)3	5269.969		0.377342		1.05	
(4 0 0 0)4	5141.570		0.378139		1.85	
(4 0 0 0)5	4976.523		0.380453		2.25	
(1 0 2 0)1	5986.117	15	0.372718	0	1.07	0
(1 0 2 0)2	5885.332	-7	0.372936	-12	1.46	0
(2 1 1 1)1 C	5756.582		0.376070		1.10	
(2 1 1 1)2 C	5593.648	-3	0.375567	32	1.24	-1
(2 1 1 1)3 C	5439.305		0.376439		1.51	
(3 2 0 2)1 C	5607.203		0.380321		1.98	
(3 2 0 2)2 C	5392.746		0.379729		1.54	
(3 2 0 2)3 C	5198.926		0.379814		1.39	
(3 2 0 2)4 C	5004.473		0.380960		1.18	
(0 2 2 2)1 C	5935.371		0.374130		1.28	
(2 1 1 1)1 D	5756.582		0.377281		1.03	
(2 1 1 1)2 D	5593.648	-3	0.376616	24	1.25	-3
(2 1 1 1)3 D	5439.305		0.377628		1.63	
(3 2 0 2)1 D	5607.203		0.380321		1.08	
(3 2 0 2)2 D	5392.746		0.379729		1.20	
(3 2 0 2)3 D	5198.926		0.379814		1.45	
(3 2 0 2)4 D	5004.473		0.380959		1.70	
(0 2 2 2)1 D	5935.371		0.374130		1.29	
(3 0 1 0)1	6463.535	-50	0.376689	68	0.71	8
(3 0 1 0)2	6298.133	-19	0.375301	11	0.92	-2
(3 0 1 0)3	6175.960	-27	0.374989	-6	1.54	-2
(3 0 1 0)4	6033.445	35	0.377074	-32	1.93	0
(0 0 3 0)1	6945.609	-7	0.369681	-7	1.24	1
(4 1 0 1)1 C	6344.437		0.379048		0.93	
(4 1 0 1)2 C	6126.363		0.378323		1.06	
(4 1 0 1)3 C	5941.375		0.377735		1.25	
(4 1 0 1)4 C	5773.102		0.378295		1.58	
(4 1 0 1)5 C	5587.852		0.379908		1.80	
(1 1 2 1)1 C	6646.934		0.373010		1.17	
(1 1 2 1)2 C	6505.344		0.373278		1.39	
(4 1 0 1)1 D	6344.437		0.381243		0.80	
(4 1 0 1)2 D	6126.363		0.380173		0.95	
(4 1 0 1)3 D	5941.375		0.379300		1.22	
(4 1 0 1)4 D	5773.102		0.379907		1.71	
(4 1 0 1)5 D	5587.852		0.381850		1.98	
(1 1 2 1)1 D	6646.934		0.373826		1.14	
(1 1 2 1)2 D	6505.344		0.374130		1.45	
(4 0 1 0)1	7872.539		0.377129		0.43	
(4 0 1 0)2	7676.402		0.375858		0.72	
(4 0 1 0)3	7526.059		0.374366		1.04	
(4 0 1 0)4	7397.812		0.375453		1.86	
(4 0 1 0)5	7230.746		0.377731		2.27	
(1 0 3 0)1	8254.371	23	0.369710	22	1.07	
(1 0 3 0)2	8154.504	-35	0.370082	-39	1.46	

TABLE XI—*Continued*

$^{16}O\ ^{13}C\ ^{18}O$	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0	0	0.368184	9	1.19	-2
(0 1 0 1)1 C	643.329		0.368558		1.20	
(0 1 0 1)1 D	643.329		0.369126		1.21	
(1 0 0 0)1	1342.301		0.368016		1.04	
(1 0 0 0)2	1244.917		0.368534		1.40	
(0 2 0 2)1 C	1286.982		0.369493		1.21	
(0 2 0 2)1 D	1286.982		0.369493		1.23	
(0 0 1 0)1	2265.995	-23	0.365382	21	1.18	-1
(1 1 0 1)1 C	2005.491		0.368254		1.13	
(1 1 0 1)2 C	1869.271		0.368719		1.32	
(0 3 0 3)1 C	1930.966		0.370136		1.24	
(1 1 0 1)1 D	2005.491		0.369083		1.09	
(1 1 0 1)2 D	1869.271		0.369613		1.38	
(0 3 0 3)1 D	1930.966		0.370136		1.24	
(2 0 0 0)1	2701.968		0.368312		0.88	
(2 0 0 0)2	2588.195		0.367556		1.25	
(2 0 0 0)3	2467.467		0.369163		1.61	
(0 1 1 1)1 C	2897.728		0.365781		1.20	
(1 2 0 2)1 C	2664.250		0.369294		1.24	
(1 2 0 2)2 C	2498.698		0.369810		1.20	
(0 4 0 4)1 C	2575.291		0.370773		1.26	
(0 1 1 1)1 D	2897.728		0.366335		1.20	
(1 2 0 2)1 D	2664.250		0.369294		1.15	
(1 2 0 2)2 D	2498.698		0.369810		1.35	
(0 4 0 4)1 D	2575.291		0.370773		1.26	
(1 0 1 0)1	3587.580	-40	0.365185	64	1.03	-2
(1 0 1 0)2	3490.419	-29	0.365826	54	1.39	-13
(2 1 0 1)1 C	3379.814		0.368160		1.05	
(2 1 0 1)2 C	3230.898		0.367976		1.24	
(2 1 0 1)3 C	3079.290		0.369030		1.45	
(0 2 1 2)1 C	3529.798		0.366727		1.19	
(1 3 0 3)1 C	3320.824		0.369908		1.21	
(1 3 0 3)2 C	3130.969		0.370450		1.31	
(0 5 0 5)1 C	3219.965		0.371403		1.28	
(2 1 0 1)1 D	3379.814		0.369338		0.98	
(2 1 0 1)2 D	3230.898		0.369021		1.26	
(2 1 0 1)3 D	3079.290		0.370292		1.55	
(0 2 1 2)1 D	3529.798		0.366727		1.22	
(1 3 0 3)1 D	3320.824		0.369908		1.21	
(1 3 0 3)2 D	3130.969		0.370450		1.31	
(0 5 0 5)1 D	3219.965		0.371403		1.28	
(3 0 0 0)1	4079.120		0.368634		0.69	
(3 0 0 0)2	3934.768		0.367350		0.94	
(3 0 0 0)3	3820.545		0.367769		1.56	
(3 0 0 0)4	3672.367		0.369798		1.91	
(0 0 2 0)1	4508.785	-39	0.362596	12	1.18	2
(1 1 1 1)1 C	4238.812		0.365484		1.12	
(1 1 1 1)2 C	4103.551		0.366018		1.32	
(2 2 0 2)1 C	4050.924		0.369263		1.38	
(2 2 0 2)2 C	3874.711		0.369254		1.25	
(2 2 0 2)3 C	3697.695		0.370244		1.09	
(0 3 1 3)1 C	4162.215		0.367391		1.23	
(1 4 0 4)1 C	3976.101		0.370512		1.23	
(1 4 0 4)2 C	3765.213		0.371085		1.33	
(0 6 0 6)1 C	3864.997		0.372025		1.30	

TABLE XI—*Continued*

16 13 18 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 1 1)1 D	4238.812		0.366271		1.08	
(1 1 1 1)2 D	4103.551		0.366901		1.38	
(2 2 0 2)1 D	4050.924		0.369263		1.07	
(2 2 0 2)2 D	3874.711		0.369254		1.27	
(2 2 0 2)3 D	3697.695		0.370243		1.49	
(0 3 1 3)1 D	4162.215		0.367391		1.23	
(1 4 0 4)1 D	3976.101		0.370512		1.23	
(1 4 0 4)2 D	3765.213		0.371085		1.33	
(0 6 0 6)1 D	3864.997		0.372025		1.30	
(2 0 1 0)1	4925.070	-58	0.365484	73	0.88	
(2 0 1 0)2	4814.578	-7	0.364838	-23	1.26	-14
(2 0 1 0)3	4692.227	-46	0.366511	0	1.61	3
(3 1 0 1)1 C	4766.574		0.368116		0.99	
(3 1 0 1)2 C	4597.344		0.367562		1.12	
(3 1 0 1)3 C	4444.926		0.367970		1.39	
(3 1 0 1)4 C	4275.496		0.369392		1.58	
(0 1 2 1)1 C	5128.953		0.363018		1.19	
(3 1 0 1)1 D	4766.574		0.369705		0.88	
(3 1 0 1)2 D	4597.344		0.368878		1.07	
(3 1 0 1)3 D	4444.926		0.369309		1.47	
(3 1 0 1)4 D	4275.496		0.371039		1.71	
(0 1 2 1)1 D	5128.953		0.363558		1.20	
(4 0 0 0)1	5470.008		0.368883		0.36	
(4 0 0 0)2	5295.828		0.367786		1.02	
(4 0 0 0)3	5165.141		0.366695		1.23	
(4 0 0 0)4	5035.379		0.368354		1.84	
(4 0 0 0)5	4863.520		0.370407		2.33	
(1 0 2 0)1	5809.895	-42	0.362366		1.04	
(1 0 2 0)2	5712.652		0.363132		1.39	
(2 1 1 1)1 C	5591.141		0.365426		1.04	
(2 1 1 1)2 C	* 5445.039		0.365191		0.71	
(2 1 1 1)3 C	5293.312		0.366315		1.12	
(2 1 1 1)1 D	5591.141		0.366522		0.97	
(2 1 1 1)2 D	* 5445.039		0.366211		0.70	
(2 1 1 1)3 D	5293.312		0.367644		1.54	
(3 0 1 0)1	6279.527		0.365862		0.71	
(3 0 1 0)2	6140.203		0.364531		0.92	
(3 0 1 0)3	6026.684	-54	0.365175	-7	1.56	-15
(3 0 1 0)4	5876.664		0.367188		1.91	
(0 0 3 0)1	6728.402	-42	0.359824	-2	1.17	3

16 13 17 O C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(0 0 0 0)1	0.0		0.378628	13	1.25	-17
(0 1 0 1)1 C	645.756		0.379002		1.27	
(0 1 0 1)1 D	645.756		0.379601		1.28	
(1 0 0 0)1	1355.128		0.378291		1.11	
(1 0 0 0)2	1255.333		0.379141		1.48	
(0 2 0 2)1 C	1291.821		0.379967		1.25	
(0 2 0 2)1 D	1291.821		0.379967		1.30	
(0 0 1 0)1	2274.109	-23	0.375744	29	1.25	-17
(1 1 0 1)1 C	2020.239		0.378570		1.20	
(1 1 0 1)2 C	1882.580		0.379258		1.40	
(0 3 0 3)1 C	1938.206		0.380626		1.30	

TABLE XI—*Continued*

16 13 17 D C O ATTRIBUTION	BAND CENTER (CM-1) CALC.	O-C	INERTIA CONSTANT (CM-1) CALC.	O-C	CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1) CALC.	O-C
(1 1 0 1)1 D	2020.239		0.379414		1.16	
(1 1 0 1)2 D	1882.580		0.380229		1.46	
(0 3 0 3)1 D	1938.206		0.380626		1.30	
(2 0 0 0)1	2724.325		0.378453		0.94	
(2 0 0 0)2	2615.628		0.378015		1.38	
(2 0 0 0)3	2487.167		0.379844		1.73	
(0 1 1 1)1 C	2908.204		0.376145		1.27	
(1 2 0 2)1 C	2681.139		0.379653		1.28	
(1 2 0 2)2 C	2514.652		0.380381		1.22	
(0 4 0 4)1 C	2584.918		0.381277		1.32	
(0 1 1 1)1 D	2908.204		0.376727		1.27	
(1 2 0 2)1 D	2681.139		0.379653		1.22	
(1 2 0 2)2 D	2514.652		0.380381		1.43	
(0 4 0 4)1 D	2584.918		0.381277		1.32	
(1 0 1 0)1	3608.580		0.375374		1.11	
(1 0 1 0)2	3508.391		0.376347		1.48	
(2 1 0 1)1 C	3404.888		0.378368		1.12	
(2 1 0 1)2 C	3259.043		0.378390		1.33	
(2 1 0 1)3 C	3102.717		0.379634		1.54	
(0 2 1 2)1 C	3542.622		0.377122		1.24	
(1 3 0 3)1 C	3339.920		0.380293		1.27	
(1 3 0 3)2 C	3149.477		0.381024		1.37	
(0 5 0 5)1 C	3231.966		0.381921		1.34	
(2 1 0 1)1 D	3404.888		0.379557		1.03	
(2 1 0 1)2 D	3259.043		0.379497		1.36	
(2 1 0 1)3 D	3102.717		0.381014		1.64	
(0 2 1 2)1 D	3542.622		0.377122		1.29	
(1 3 0 3)1 D	3339.920		0.380293		1.27	
(1 3 0 3)2 D	3149.477		0.381024		1.37	
(0 5 0 5)1 D	3231.966		0.381921		1.34	
(3 0 0 0)1	4109.988		0.378744		0.76	
(3 0 0 0)2	3975.011		0.377494		1.07	
(3 0 0 0)3	3858.759		0.378451		1.69	
(3 0 0 0)4	3701.541		0.380522		2.10	
(0 0 2 0)1	4524.914	-31	0.372875	3	1.25	
(1 1 1 1)1 C	4261.582		0.375726		1.19	
(1 1 1 1)2 C	4124.434		0.376483		1.39	
(2 2 0 2)1 C	4078.385		0.379520		1.40	
(2 2 0 2)2 C	3904.490		0.379706		1.27	
(2 2 0 2)3 C	3724.218		0.380896		1.05	
(0 3 1 3)1 C	4177.371		0.377803		1.29	
(1 4 0 4)1 C	3997.429		0.380919		1.30	
(1 4 0 4)2 C	3786.225		0.381664		1.39	
(0 6 0 6)1 C	3879.359		0.382558		1.36	
(1 1 1 1)1 D	4261.582		0.376524		1.15	
(1 1 1 1)2 D	4124.434		0.377435		1.46	
(2 2 0 2)1 D	4078.385		0.379520		1.13	
(2 2 0 2)2 D	3904.490		0.379706		1.36	
(2 2 0 2)3 D	3724.218		0.380895		1.58	
(0 3 1 3)1 D	4177.371		0.377803		1.29	
(1 4 0 4)1 D	3997.429		0.380919		1.30	
(1 4 0 4)2 D	3786.225		0.381664		1.39	
(0 6 0 6)1 D	3879.359		0.382558		1.36	
(2 0 1 0)1	4955.633		0.375540		0.93	
(2 0 1 0)2	4849.398		0.375239		1.39	
(2 0 1 0)3	4719.164	-3	0.377103	5	1.73	
(3 1 0 1)1 C	4800.844		0.378251		1.05	
(3 1 0 1)2 C	4637.879		0.377805		1.22	
(3 1 0 1)3 C	4484.707		0.378494		1.49	
(3 1 0 1)4 C	4308.859		0.380039		1.69	
(0 1 2 1)1 C	5147.383		0.373301		1.26	
(3 1 0 1)1 D	4800.844		0.379861		0.93	
(3 1 0 1)2 D	4637.879		0.379159		1.17	
(3 1 0 1)3 D	4484.707		0.379948		1.58	
(3 1 0 1)4 D	4308.859		0.381839		1.82	
(0 1 2 1)1 D	5147.383		0.373869		1.27	

TABLE XI—*Continued*

16 13 17 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(4 0 0 0)1	5509.285		0.378978		0.44	
(4 0 0 0)2	5344.703		0.377710		0.78	
(4 0 0 0)3	5220.695		0.377103		1.46	
(4 0 0 0)4	5082.820		0.379117		2.00	
(4 0 0 0)5	4902.406		0.381173		2.64	
(1 0 2 0)1	5838.961		0.372482		1.11	
(1 0 2 0)2	5738.102		0.373567		1.47	
(2 1 1 1)1 C	5624.191		0.375564		1.10	
(2 1 1 1)2 C	* 5480.629		0.375539		0.92	
(2 1 1 1)3 C	5323.949		0.376903		1.52	
(2 1 1 1)1 D	5624.191		0.376661		1.02	
(2 1 1 1)2 D	* 5480.629		0.376622		0.96	
(2 1 1 1)3 D	5323.949		0.378284		1.63	
(3 0 1 0)1	6318.418		0.375891		0.77	
(3 0 1 0)2	6188.129		0.374604		1.02	
(3 0 1 0)3	6071.609		0.375789		1.69	
(3 0 1 0)4	5912.816		0.377830		2.11	
(0 0 3 0)1	6752.453	11	0.370021	-23	1.24	-9

17 12 18 O C O	BAND CENTER (CM-1)		INERTIA CONSTANT (CM-1)		CENTRIFUGAL DIST. CONSTANT 10(-7) (CM-1)	
ATTRIBUTION	CALC.	O-C	CALC.	O-C	CALC.	O-C
(0 0 0 0)1	0.0		0.356933		1.11	
(0 1 0 1)1 C	659.705		0.357349		1.13	
(0 1 0 1)1 D	659.705		0.357869		1.14	
(1 0 0 0)1	1355.655		0.357402		0.97	
(1 0 0 0)2	1244.619		0.356734		1.26	
(0 2 0 2)1 C	1319.811		0.358278		1.19	
(0 2 0 2)1 D	1319.811		0.358278		1.15	
(0 0 1 0)1	2322.445		0.354114		1.11	
(1 1 0 1)1 C	2036.063		0.357481		1.06	
(1 1 0 1)2 C	1884.760		0.357186		1.23	
(0 3 0 3)1 C	1980.326		0.358940		1.19	
(1 1 0 1)1 D	2036.063		0.358333		1.03	
(1 1 0 1)2 D	1884.760		0.357912		1.27	
(0 3 0 3)1 D	1980.326		0.358940		1.19	
(2 0 0 0)1	2738.287		0.357950		0.83	
(2 0 0 0)2	2585.649		0.356637		1.00	
(2 0 0 0)3	2473.516		0.356954		1.46	
(1 0 1 0)1	3655.120	10	0.354553	-1	0.96	-12
(1 0 1 0)2	3547.299		0.354002		1.27	

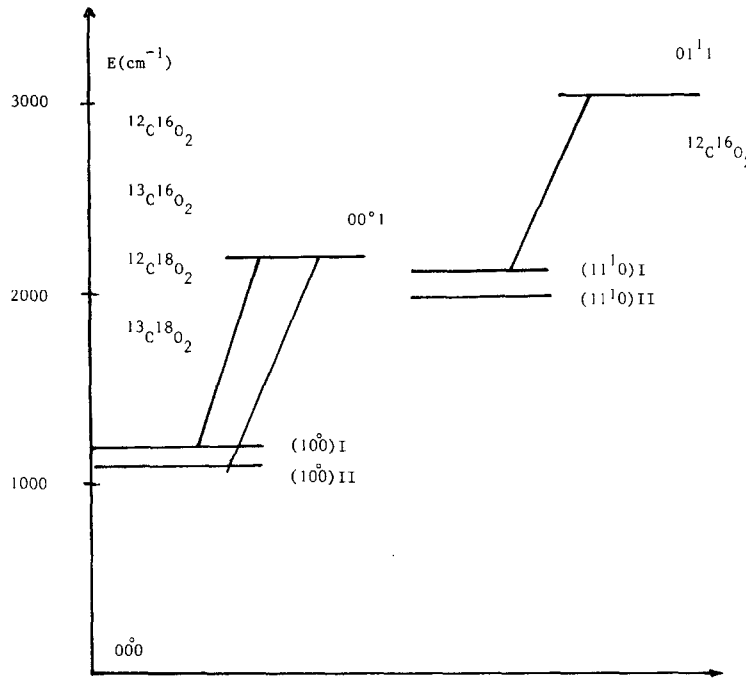


FIG. 5. Measured lasing transitions in CO_2.

knowns (28) is about 25 which is satisfactory. The mean deviations from theory to experiment are:

0.035 cm^{-1}	for the band centers (213 data);
2.2×10^{-5} cm^{-1}	for the inertia constants (299 data);
3.0×10^{-9} cm^{-1}	for the centrifugal distortion constants (276 data).

In addition to these global results some more local comparisons are interesting, mainly in two extreme cases: first, the lower states since a few of them have been very accurately measured using laser techniques and are ideally suited to the theory we have developed; second, the upper states which have also been accurately measured recently using emission techniques. The latter allow for a judgment on the behavior of the solution when it is extrapolated.

Figure 5 shows the measured lasing transitions in CO_2 (see Appendix) and Table XII presents some comparisons between this work (calculated values) and the laser data for four isotopic species: $^{12}C^{16}O_2$, $^{13}C^{16}O_2$, $^{12}C^{18}O_2$, and $^{13}C^{18}O_2$. This table shows an excellent agreement and points out that the mean errors on the band centers, the $\Delta\nu_0$'s (0.009 cm^{-1}), on the ΔB's (5.10^{-6} cm^{-1}) and on the ΔD's (6.10^{-10} cm^{-1}) are smaller than the mean errors on the ν_0's, the B's, and the D's (see Tables X and XI). This is an important and general remark. The agreement with the quantities directly related to the observed transitions is almost always better than the agreement with the ν_0's, the B's, and the D's. This demonstrates the quality of the present solution but also indicates that we should have iterated it

TABLE XII

Comparisons with Laser Data (in cm^{-1})

$^{16}O^{12}C^{16}O$

	EXP.	CALC.	O-C
E(001-I)	960.958	960.962	-0.004
E(001-II)	1063.734	1063.736	-0.002
B(001)	0.387141	0.387132	9.(E-6)
B(I-001)	3.047(E-3)	3.049(E-3)	2.(E-6)
B(II-001)	3.341(E-3)	3.350(E-3)	-9.(E-6)
D(001)	1.30(E-7)	1.33(E-7)	-3.(E-9)
D(I-001)	-1.81(E-8)	-1.76(E-8)	-5.(E-10)
D(II-001)	2.41(E-8)	2.39(E-8)	2.(E-10)
H(I-001)	1.74(E-13)	1.74(E-13)	0
H(II-001)	2.16(E-13)	2.10(E-13)	6.(E-15)

$^{16}O^{13}C^{16}O$

	EXP.	CALC.	O-C
E(001-I)	913.425	913.421	0.004
E(001-II)	1017.659	1017.675	-0.016
B(001)	0.387277	0.387264	1.3(E-5)
B(I-001)	2.444(E-3)	2.443(E-3)	1.(E-6)
B(II-001)	3.643(E-3)	3.654(E-3)	-1.1(E-5)
D(001)	1.34(E-7)	1.33(E-7)	1.(E-9)
D(I-001)	-1.31(E-8)	-1.22(E-8)	-9.(E-10)
D(II-001)	2.53(E-8)	2.52(E-8)	1.(E-10)

$^{18}O^{12}C^{18}O$

	EXP.	CALC.	O-C
E(001-I)	966.955	966.966	-0.011
E(001-II)	1083.723	1083.703	-0.020
B(001)	0.344094	0.344080	1.4(E-5)
B(I-001)	3.313(E-3)	3.319(E-3)	-6.(E-6)
B(II-001)	2.434(E-3)	2.439(E-3)	-5.(E-6)
D(001)	1.09(E-7)	1.05(E-7)	4.(E-9)
D(I-001)	-1.27(E-8)	-1.24(E-8)	-3.(E-10)
D(II-001)	1.15(E-8)	1.21(E-8)	-6.(E-10)

$^{18}O^{13}C^{18}O$

	EXP.	CALC.	O-C
E(001-I)	928.594	928.598	- 0.004
E(001-II)	1026.906	1026.902	- 0.004
B(001)	0.344207	0.344198	9.(E-6)
B(I-001)	2.814(E-3)	2.818(E-3)	- 4.(E-6)
B(II-001)	2.644(E-3)	2.663(E-3)	1.(E-6)
D(001)	1.04(E-7)	1.05(E-7)	1.(E-9)
D(I-001)	-1.46(E-8)	-1.41(E-8)	- 5.(E-10)
D(II-001)	1.64(E-8)	1.68(E-8)	- 4.(E-10)

$^{16}O^{12}C^{16}O$

	EXP.	CALC.	O-C
$E(01^{1}1\text{-I})$	927.156	927.141	0.015
$B(01^{1}1\text{-I})C$	-2.817(E-4)	-2.821(E-3)	4.(E-6)
$B(01^{1}1\text{-I})D$	-3.144(E-3)	-3.142(E-3)	-2.(E-6)
$D(01^{1}1\text{-I})C$	9.25(E-9)	8.30(E-9)	9.(E-10)
$D(01^{1}1\text{-I})D$	1.48(E-8)	1.38(E-8)	1.(E-9)

on the former quantities rather than on the latter. This choice has been made for the sake of simplicity and speed. However, as we have already said, the difference between the two solutions may be expected to be small.

On the other hand, the emission spectrum of CO_2 has recently been widely and accurately measured, mainly in the 4.3-μm region (see Appendix). Two kinds of transitions have been obtained up to relatively high values of the quantum numbers: $00^{0}n \rightarrow 00^{0}n - 1$ from $n = 1$ to 10 and $01^{1}n \rightarrow 01^{1}n - 1$ from $n = 1$ to 9. Table XIII shows the agreement between experiment and the present work for the band centers (the $\Delta\nu_0$'s). Only five states (four transitions) participated in the iterative process. The other calculated values have been extrapolated following a simplified way which consists in removing the Coriolis interaction from the once transformed

TABLE XIII

Comparisons between Theory and Experiment for Upper Levels. An Example: The Emission Spectrum of CO_2 in the 4.3-μm Region

n	n-1	CALC.	O-C	
1	0	2349.150	-0.007	ITERATED
2	1	2324.181	0.002	
3	2	2299.250	0.003	
4	3	2274.257	-0.001	
5	4	2249.502	0.000	EXTRAPOLATED
6	5	2224.685	0.013	
7	6	2199.905	0.046	
8	7	2175.164	0.105	
9	8	2150.461	0.199	
10	9	2125.795	0.345	
00°10		22373.158	0.708	

EMISSION SPECTRUM OF $^{12}C^{16}O_2$

$00^{\circ}n \rightarrow 00^{\circ}n-1$

in cm^{-1}

n	n-1	CALC.	O-C	
1	0	2336.635	-0.005	ITERATED
2	1	2311.702	-0.003	
3	2	2286.809	-0.009	
4	3	2261.952	-0.017	
5	4	2237.133	-0.017	EXTRAPOLATED
6	5	2212.354	-0.007	
7	6	2187.610	0.033	
8	7	2162.906	0.099	
9	8	2138.239	0.207	
$01^{1}9$		20802.720	0.283	

EMISSION SPECTRUM OF $^{12}C^{16}O_2$

$01^{1}n \rightarrow 01^{1}n-1$

in cm^{-1}

Hamiltonian. Indeed, if this coupling is not removed by the perturbation treatment, the states 00^0n or 01^1n for $n > 4$ are obtained owing to the diagonalization of very large subblocks. Moreover we have verified that, for such particular states, the influence of this change in the usual procedure leads to negligible changes in the molecular constants from $n = 1$ to $n = 4$. From Table XIII it may be seen that the absolute error on the ν_0 value of the state 00^010 is only 0.708 cm^{-1} although this corresponds to an extrapolation from 00^04 at about[5] 9250 cm^{-1} to more than 22 000 cm^{-1}. The same conclusion holds for the extrapolation from 01^14 to 01^19. Table XIV gives the same kind of results for the ΔB's and the ΔD's and for the transitions $00^0n \rightarrow 00^0n - 1$; the conclusions are identical.

[5] The state 00^04 is the highest of the ν_3 series to be included in the iterative process for the adjustment of the potential and, consequently, has been assigned a small statistical weight (the same for 01^14).

TABLE XIV

Emission Spectrum of $^{12}C^{16}O_2$ (in cm^{-1}) $00^0n \rightarrow 00^0n - 1$

		B'-B" (E+5)		D'-D" (E+9)		
n	n-1	CALC.	O-C	CALC.	O-C	
1	0	-308.9	0.7	0	0	INTERATED
2	1	-307.2	-0.3	0	1	
3	2	-305.5	-1.6	0	0	
4	3	-303.9	-2.9	0	0	
5	4	-301.6	-5.0	0	0	EXTRAPOLATED
6	5	-300.1	-6.3	-1	0	
7	6	-298.3	-7.6	0	0	
8	7	-296.5	-9.3	0	0	
9	8	-294.8	-10.5	-1	0	
10	9	-293.0	-13.0	0	0	

D. The Vibration–Rotation Levels

The vibration–rotation levels of all the isotopic species may be computed either from the molecular constants or from the direct diagonalization of the energy matrix for all the values of J which are required. This last possibility becomes essential when a vibrational state is strongly involved in a Coriolis interaction with another vibrational state.

We have computed a number of transitions from the molecular constants of the two states (upper and lower) involved and we have found that the error due to the rotational part of the energy is almost always less than 0.05 cm^{-1} for $J = 50$ and very rarely larger. Moreover, we have observed that the errors on the molecular constants of the upper state often compensate (at least partly) for the errors on those of the lower state in such a way that the agreement for the observed transition is, as already said, better than the agreement for the molecular constants presented in Tables X and XI.

On the other hand, we have studied transitions strongly perturbed by the Coriolis interactions. As is well known, the molecular constants calculated for the two states coupled are meaningless. However, it is possible to ascribe to each state a set of molecular constants which reproduce correctly the lines corresponding to the values of J where the interaction is weak. This may simply be achieved by modifying the second canonical transformation of the Hamiltonian so as to remove the Coriolis coupling from the once transformed Hamiltonian. The results we have obtained for several such transitions are in agreement with the experimental errors. If the Coriolis interaction is not removed from the Hamiltonian by the perturbation treatment, the only way to compute the line positions is to diagonalize the energy

matrix for all the values of J in the range of the observations. Here again, the agreement is good.

E. Discussion

The agreement between theory and experiment which comes out from the preceding results is satisfactory and, in any case, much better than we had expected when this work was started. It must be kept in mind that all these results are deduced from only 28 constants. Even if it is still too much it is less than the 58 spectroscopic constants per isotopic species used in the usual vibrorotational analysis approach. This is certainly a good example of data reduction. However, it is also clear that the present solution is not completely stabilized yet. If its behavior when extrapolating to high values of the quantum number v_3 is very satisfactory, it is not so good when the value of the quantum number v_1 increases. Some slight but significant difficulties are noticed from $v_1 = 3$ and increase[6] for $v_1 = 4$, and so on. For these states, it must be remarked that they are strongly coupled to other states by the first order Fermi interaction. If we take the example of 40^01, it is coupled to 32^01, 24^01, 16^01, and 08^01. If the maximum value of v_1 is 4, the maximum value of v_2 is 8 like the "good" quantum number $2v_1 + v_2$. Such a value is quite large if one remembers that the usual scheme for the expansion in orders of magnitude implies that the vibrational part of the matrix elements is not too much different from 1. In other words, such states are at the limit or already beyond the scope of the theory. This is the reason for which the states with a value of $2v_1 + v_2$ equal or larger to six have been given smaller and smaller statistical weights up to zero for the states like 50^00 or 50^01 where $2v_1 + v_2$ equals 10.

As for the states with large v_3 values, the errors for the states with large $2v_1 + v_2$ values, although noticeable, remain relatively small and, certainly sufficiently small to leave a good hope to adjust the calculated values to experiment without significantly adulterating the physical content of the potential. A new cycle of iterations will be initiated in the future with larger statistical weights for these "bad" levels as soon as a sufficiently large number of new and accurate data become available. In this connection, we must say that the compilation of the best experimental data is not an easy task and this is due to the frequent difficulty in choosing among several experimental determinations. This is mainly the case of the fundamentals which are very highly weighted and for which errors of a few thousands of cm^{-1} for the band centers or a few 10^{-6} cm^{-1} for the inertia constants may have some consequences in the fourth order potential constants. For example, the relatively large error on the inertia constant of ν_3 of $^{12}C^{16}O_2$ (7 or 8 10^{-6} cm^{-1}) is due to the fact that the iterations have been carried out with an "old" value precisely equal to the present calculated value (0.387132) instead of the new value obtained by laser techniques. This is also the case for ν_3 of $^{13}C^{16}O_2$. Many local remarks of that kind could be done by carefully examining the present solution. The new cycle of iterations will take care of these minor points.

All the results we have presented in this section have been obtained in the mathe-

[6] No more, however, than an error of 0.1 to 0.3 cm^{-1} on the band center and 1×10^{-4} to 2×10^{-4} on the inertia constant.

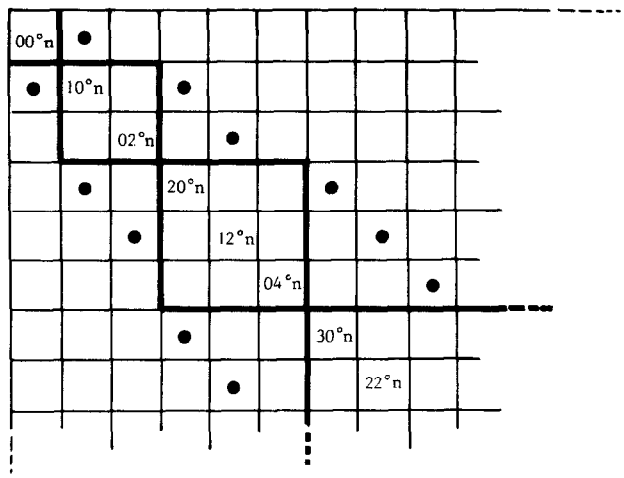


FIG. 6. Direct diagonalization of the couplings $\Delta v_1 = \pm 1$. The heavy lines encompass the Fermi subblocks. The matrix elements: $\langle v_1, v_2, l_2, v_3 | H | v_1 \pm 1, v_2, l_2, v_3 \rangle$ are indicated by the dots.

matical and physical frame of the algebraized contact transformation and within the limits of the Born–Oppenheimer approximation. The effects of this last approximation can hardly be detected from the present results and are probably slightly smaller than the deviation between theory and experiment we have achieved. On the contrary, the effects of the assumption of harmonicity of the zeroth order Hamiltonian are clearly seen.

F. Modifications of the Coupling Scheme: Flexibility of the Algebraized Contact Transformation Formalism; the Upper Spectrum of CO_2

The physical meaning of the decoupling process (block diagonalization) depends upon the assumptions which lead us to remove or not to remove such or such coupling. If these assumptions are (or become) too weak, it may be important to enlarge the blocks by including the suspicious states. On the contrary, if the assumptions mentioned above were too strong, it is interesting to reduce the dimensions of the blocks. Such a study provides a hierarchy of approximations associated with computations the complexity of which may be very different.

One of the most important characteristics of the algebraized contact transformation is its great flexibility to changes in the set of couplings which are not removed by the canonical transformations. The results we have presented in the preceding paragraphs included a Fermi-type interaction, an l-type interaction, and a Coriolis-type interaction. It is immediately possible to subtract (or to add) any coupling in view of simplifying (or sophisticating) the set of couplings which will not be removed from the Hamiltonian by the perturbation treatment. As an example of simplification, we have carried out the computation of the molecular constants of the 10 isotopic species of CO_2 already considered by removing from the Hamiltonian the couplings corresponding to the Coriolis interaction. This simplification allows for a systematic study of the influence of this interaction on the molecular constants.

TABLE XV

State	Energy value (cm^{-1}) (1)[a]	(2)[b]	(2)-(1) (cm^{-1})
00°0	2530.29	2530.88	0.59
00°1	4878.82	4880.03	1.21
00°2	7201.88	7204.21	2.33
00°3	9499.32	9503.46	4.14
00°4	11770.97	11777.82	6.85
00°5	14016.53	14027.32	10.79
00°6	16235.65	16252.02	16.37
00°7	18427.89	18451.97	24.08
00°8	20592.74	20627.24	34.50
00°9	22729.73	22777.90	48.17
00°10	24838.72	24904.04	65.32

[a] In the "non-diagonal" approach, from the minimum of the potential.

[b] In the "diagonal" approach, from the minimum of the potential (in fact: the observed value for the transition 00°n ← 00°0 plus 2530.88).

It is as simple to sophisticate the "coupling scheme" and we found it interesting to try to clarify the influence of the next important vibrational coupling after the Fermi interaction, namely, the $\Delta v_1 = \pm 1$ coupling whose magnitude is due to that of H_1 (first order Hamiltonian). The question is: What happens if such a coupling is not removed by the perturbation treatment and directly diagonalized at the end of this process?

The first consequence of this approach is obviously to give rise to an energy matrix with infinite subblocks. Then, these subblocks must be truncated before the diagonalization. However, if we focus our interest on the series 00^0n, $n = 1$ to 10, recently accurately measured in emission (see Appendix) it is clear that these states may be, within the subblocks, "almost" isolated in the upper left corner. This convenient situation for truncation is shown in Fig. 6. One may remark that this figure has been drawn omitting the Coriolis couplings which have been supposed removed by the Hamiltonian transformation process. In favor of this simplification is the fact that the states 00^0n are not significantly affected by this interaction.[7] In addition, we shall see that for high v_3 values the sizes of the truncated

[7] Verification of this fact has been made up to $v_3 = 4$. This may be questionable for higher v_3 values (see the following).

subblocks may become so big that we may not be able to take account of this interaction. The second consequence of this approach is to completely change the values of the spectroscopic constants.

To simplify the problem one step further we have carried out the calculation of the energies of the only vibrational states. The purpose is once more to reduce the sizes of the truncated subblocks.

Even under these assumptions, and this is the third consequence, the ground level 00^00 is not diagonal and its energy value is obtained by diagonalizing the corresponding subblock (see Fig. 6 with $n = 0$). A slight difference is found between the ground level energy in the conventional approach (henceforth, the "diagonal" approach) and the approach where the couplings $\Delta v_1 = \pm 1$ are not removed by the perturbation treatment (henceforth, the "nondiagonal" approach). The complete results, from $v_3 = 0$ to $v_3 = 10$, are given in Table XV. In this table, the origin of the energies (in cm^{-1}) is the minimum of the potential. For $v_3 = 9$ or 10, a converged value for the energy is obtained when diagonalizing subblocks of dimension 55. This size decreases with v_3.

Two important ideas are apparent from an examination of this table:

1. It is easily verified that the deviations between the "diagonal" and the "nondiagonal" approaches may be fitted by a polynomial of degree 3 in the quantum number v_3. Since it is the case for the energies in the "diagonal" approach, this means that it is also the case in the "nondiagonal" approach. In other words, an adjustment procedure of the potential to the experimental energies cannot make the difference between the two approaches. The fit will not be distorted by the inclusion or the noninclusion of the $\Delta v_1 = \pm 1$ couplings. The main problem is then to have an idea of the order of magnitude of the difference between the two approaches.

2. If we divide the three or four last (the largest) deviations by $(v_3 + \frac{1}{2})^3$, we find a constant ratio of approximately 0.056 cm^{-1} which will be the change to bring to the spectroscopic constant y_{333} so as to go from one approach to the other. This is typically a fourth order contribution. In fact, when fitting the deviations of Table XV one has to account also for those of the lower levels. Changing ω_3^0 by less than 0.1% and x_{33}^0 by 1 to 2%, together with the change of 0.056 for y_{333} allows for a perfect fit of these differences. Once more, all these effects are typically of the fourth order of approximation in the ordering scheme we have used.

From the present study one may conclude that, since by using one approach or the other the differences are of the fourth order of approximation, this means that the two corresponding effective Hamiltonians are equivalent to within this order. Indeed, an effective Hamiltonian expanded up to the nth order of approximation is in error in the same order (sometimes even in a lower order) and not in the next higher order as often stated.

The preceding conclusions must be restrained by the following remark: in the upper spectrum, even for a molecule like CO_2, the density of levels is so high that isolating the couplings $\Delta v_1 = \pm 1$ jointly with the Fermi-type couplings is not enough. Many other interactions become strong and only the generalized Van Vleck perturbation treatment [see Ref. (*11*)] may take care of such a complexity, leading (there is no other choice) to very large subblocks in the energy matrix. So,

the best way to carry out the computations is, in this upper part of the spectrum, to abandon or partly abandon the assumption of harmonicity of the zeroth order potential. The greater complexity in the numerical developments may be at least partly compensated by the new ordering scheme implied by that kind of approach. It is, however, our personal feeling that important amounts of computer time will be required to bring such a work to a successful conclusion.

One last comment about the v_3 series: Both the high density of levels and the important variety of possible relatively strong couplings in the upper spectrum may, in a certain way, explain the amazingly good behavior of the corresponding theoretical energy values obtained by extrapolating our solution: there are most probably as many interactions which tend to shift the energy downwards as there are interactions which tend to shift the energy upwards. They probably compensate for one another to a large extent.

The example we have just considered is important since it demonstrates the ability of the algebraized contact transformation formalism to isolate not only clusters of levels of finite sizes (as was the case for the previous approaches with the conventional contact transformation approach) but also clusters of levels of infinite sizes. In this case, a rule for truncating must be defined. In a certain way, this rule is a correction of the assumption of harmonicity. Between the two cases only a very few changes in the models are required to orient the fully automatized process. This flexibility of the method allows for a large variety of tests which may be very helpful in defining the main points in which the formalism has eventually to be sophisticated in order to answer such questions more accurately (for example, the problem of the states with high $2v_1 + v_2$ values in CO_2).

V. CONCLUSION

This study of the intramolecular potential of the carbon dioxide molecule has led us to the following conclusions.

1. A comparison between our previous determination of this potential (*17*, *18*) from the experimental vibrational energies and the present determination from experimental vibration–rotation energies clearly demonstrates the large influence of rotation. This influence was expected in principle but its magnitude was not evident. More quantitatively, the use of the pure vibration data allows for a reasonable determination of the potential up to the second order only while the use of all the data makes it possible to improve the determination up to the fourth order (the error being of the same order) with a convergence strictly compatible with that theoretically presupposed in the expansion of the Hamiltonian.

2. The agreement between theory and experiment achieved with this new determination of the potential is satisfactory since the mean errors on the vibrational energies, the inertia constants, and the centrifugal distortion constants are, respectively, 0.035 cm^{-1} (mean over 213 vibrational energies), 2.2×10^{-5} cm^{-1} (mean over 299 inertia constants), and 3.0×10^{-9} cm^{-1} (mean over 276 centrifugal distortion constants). The fit is much better for the lower spectrum (up to a factor of 10) and significantly less satisfactory in the upper spectrum (mainly for the large "Fermi-polyads") which is in accordance with the theory we have developed.

3. A dependence of the potential on the nuclear masses can hardly be detected and the agreement between theory and experiment is roughly the same for all the isotopic species we have studied. Correlated to the good convergence of this potential, this fact is in favor of the validity of the Born–Oppenheimer approximation at least within the limits of the errors mentioned above.

4. The deviations between theoretical and experimental energies increase significantly with the value of the energy. This is mainly the case for the large "Fermi-polyads" when the "good" quantum number $2v_1 + v_2$ increases. This limitation of the formalism is obviously due to the assumption of the harmonicity of the zeroth order Hamiltonian: the larger the displacements of the nuclei from equilibrium, the larger the errors. This limitation is all the more severe for molecules in which the nuclei undergo movements of large amplitude even in the lower spectrum (like HCN, H_2O, NH_3, etc.) and this is the reason for which many attempts have been made more or less recently to derive new Hamiltonians leaving partly or totally the zeroth order harmonic approximation and/or including a sophistication of the rigid rotor approximation. The generalized Van Vleck transformation is able to solve the resulting Schrödinger equations, but one of the conclusions of this study is that it would require very large amounts of computation time and very big computers. Iterating the *intra*molecular potential under such conditions is, in many respects, similar to iterating the *inter*molecular potential of colliding molecules from the observed cross sections. This has not been done yet. However, for a molecule like CO_2 a first interesting step could consist in a generalization of the infinite truncated matrices approach presented in Section III.F, for a better treatment of the upper spectrum where CO_2 resembles a "large amplitude molecule." The method consists in maintaining the assumption of the harmonicity of the zeroth order Hamiltonian but, as the energy increases, more and more interactions are considered as becoming too important to be treated by perturbation. The infinite series of systematically weak interactions are first treated via the algebraized contact transformation and the generalized Van Vleck transformation formalism is applied to the resulting Hamiltonian. Such an approach would allow for a variety of tests which could be very helpful in defining the main points in which the zeroth order approximation has to be sophisticated.

Additional improvements of the present solution may be obtained after a significant improvement of the *mean* accuracy of the experimental data set (see Section III.E). We intend, in the future, to carry out new iterations of the potential so as to take into account several new publications of accurate data and to eliminate, as far as possible, the discrepancies which exist on the vibrational energies, the inertia constants, or the centrifugal distortion constants when the agreement on their differences for transitions is almost always much better.

APPENDIX: REFERENCES FOR EXPERIMENTAL SPECTROSCOPY IN THE INFRARED FOR CO_2

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